Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM

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Cryptococcosis is a major worldwide disseminated invasive fungal infection. Cryptococcosis, particularly in its most lethal manifestation of cryptococcal meningitis, accounts for substantial mortality and morbidity. The breadth of the clinical cryptococcosis syndromes, the different patient types at-risk and affected, and the vastly disparate resource settings where clinicians practice pose a complex array of challenges. Expert contributors from diverse regions of the world have collated data, reviewed the evidence, and provided insightful guideline recommendations for health practitioners across the globe. This guideline offers updated practical guidance and implementable recommendations on the clinical approaches, screening, diagnosis, management, and follow-up care of a patient with cryptococcosis and serves as a comprehensive synthesis of current evidence on cryptococcosis. This Review seeks to facilitate optimal clinical decision making on cryptococcosis and addresses the myriad of clinical complications by incorporating data from historical and contemporary clinical trials. This guideline is grounded on a set of core management principles, while acknowledging the practical challenges of antifungal access and resource limitations faced by many clinicians and patients. More than 70 societies internationally have endorsed the content, structure, evidence, recommendation, and pragmatic wisdom of this global cryptococcosis guideline to inform clinicians about the past, present, and future of care for a patient with cryptococcosis.

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Introduction

Cryptococcosis accounts for substantial morbidity and mortality globally. In 2022, WHO listed *Cryptococcus neoformans* as a top fungal priority pathogen.[1] Cryptococcosis often involves the CNS or the lungs, but disseminated disease can affect any organ, yet appear localised. Despite the knowledge gained and improvements in clinical outcomes generated by multiple interventional trials[2–7] done primarily in low-income settings with insufficient resources, mortality from cryptococcal meningoencephalitis is high, ranging from 24 to 47% at 10 weeks.[2,4,7,8] The highest burden of disease is in low-income and middle-income countries, especially in sub-Saharan Africa,[9] where HIV and AIDS are the dominant risk factor, although new non-HIV immunocompromised risk groups and putatively immunocompetent individuals are increasingly reported in high-income settings with sufficient resources.

Complementary diagnostic and management guidelines for cryptococcosis exist.[10–21] This comprehensive management guideline serves primarily to facilitate clinical decision making while also providing an overview of the uncertainties in cryptococcosis management. With contributors across the globe, this guideline gives voice to expertise and challenges from diverse settings in a globally relevant Review. General principles and treatment recommendations are provided, and clinicians are urged to use careful clinical judgement when formulating

Key points

- Accurate delineation of the cryptococcosis clinical syndrome is important as it guides antifungal treatment choice and duration; cryptococcosis syndromes are divided into CNS, disseminated disease, isolated pulmonary disease, or direct skin inoculation (figure 1)
- Liposomal amphotericin B 3–4 mg/kg daily and flucytosine 25 mg/kg four times a day is the most optimal induction therapy option for cryptococcal meningitis, disseminated cryptococcosis, and severe isolated pulmonary cryptococcosis in high-income settings
- In low-income settings, patients with HIV-associated cryptococcal meningitis are best treated with liposomal amphotericin B 10 mg/kg as a single-dose, with 14 days of flucytosine 25 mg/kg four times a day and fluconazole 1200 mg daily as induction therapy; this induction therapy has not been trialled in non-HIV-associated cryptococcal meningitis or other non-CNS cryptococcosis syndromes
- Optimise outcomes by providing the most effective antifungal therapy while preventing, monitoring, and managing potential toxicity; do not stop or switch to an inferior regimen too early or unnecessarily
- Expect and monitor for clinical relapse and investigate thoroughly for causality; review adherence to antifungal therapy and consider drug–drug interactions; during treatment follow-up, do not escalate antifungal therapy for persistent blood antigenemia (blood cryptococcal antigen), persistently positive CSF cryptococcal antigen, visible cryptococci in CSF (without culture positivity), or abnormal CSF microscopy or biochemistry, as they are not necessarily indicators of microbiological failure
- Adapt and adopt these ECMM global guidelines to suit local practices, while constantly advocating for better antifungal access, scrutinising new trial data, and reviewing local data to improve patient outcomes

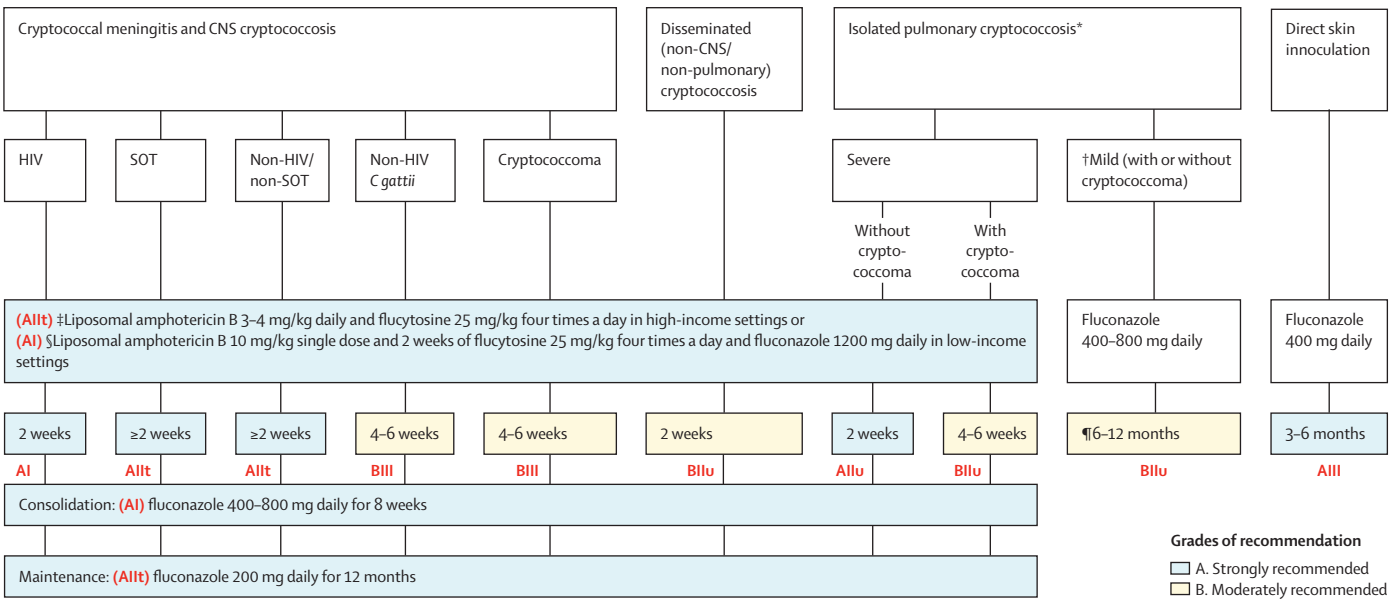


Figure 1: **Recommended first-line antifungal therapy by cryptococcosis syndrome**
Grading of recommendation and level of evidence in bolded red letters. Recommendation grading by shading: blue (strongly recommended; A) and yellow (moderately recommend; B). SOT=solid organ transplantation. *C gattii=Cryptococcal gattii*. w=weeks. *Isolated *Cryptococcal neoformans* or *Cryptococcal gattii* pulmonary cryptococcosis, mild is defined as asymptomatic or mildly symptomatic patients or with a solitary small nodule (<2 cm); whereas severe is defined as multiple lesions, large lesions (≥2 cm), lobar consolidation, cavitation, multi-lobar involvement, or hypoxaemic. †If the presence of *Cryptococcus* spp in respiratory specimens is deemed to be airway colonisation after careful evaluation and no treatment is elected, regular follow-up is recommended, especially in the setting of future immunosuppression. ‡Strongly preferred in cryptococcal meningitis and CNS cryptococcosis in SOT and non-HIV non-SOT patient populations, disseminated cryptococcosis, and severe pulmonary cryptococcosis. ‡Has not been directly compared against §. §Has only been trialled in people with cryptococcal meningitis and there are no supporting data of its use in SOT or non-HIV non-SOT patients or in other cryptococcosis syndromes. ¶Can consider a shorter duration (eg, 3 months) in immunocompetent individuals with mild isolated pulmonary cryptococcosis.

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treatment plans for the individual patient. See the appendix for more detailed text, tables, and panels relevant to each section. A summary of the first-line treatment for the different cryptococcosis syndromes is in figure 1.[3] An explanation of the evidence grading system used for the recommendations throughout is in panel 1.

Populations at high risk, clinical presentations, and outcomes

Primarily acquired via inhalation but occurring mainly upon reactivation after a period of latency, cryptococcosis has protean manifestations, with cryptococcal meningitis being the most common severe presentation. Pulmonary cryptococcosis is underdiagnosed and often subclinical. Disseminated cryptococcosis can involve any organ of the body, thus a thorough clinical assessment is required, even in individuals who appear asymptomatic.[24,25] Although classic patient populations at high risk include people living with HIV and solid organ transplant (SOT) recipients, individuals with other immunosuppressive conditions or receiving immunosuppressant drugs and people putatively immunocompetent are also affected by cryptococcosis (appendix pp 6, 79). Those who survive cryptococcosis report substantial morbidity, ranging from 10–70% depending on the disease syndrome and severity, underlying predisposing conditions of the host, and the health-care system in which the patient is managed[26–29] (panel 2; appendix pp 8, 39).

Yeasts causing cryptococcosis and diagnostic methods

C neoformans species complex is the predominant causative agent of cryptococcosis in people living with HIV, and *Cryptococcus gattii* species complex more commonly causes disease in people who appear immunocompetent. Although both can cause a similarly broad range of cryptococcosis syndromes, *C neoformans* has a predilection for CNS disease and *C gattii* is more often associated with pulmonary disease and large cryptococcomas.[30–32]

Diagnostic methods used to establish the diagnosis, extent, severity, and prognosis of cryptococcosis are constantly evolving (appendix pp 10, 41). Microscopy and culture of cerebrospinal fluid (CSF) pellet after centrifugation and blood culture, accompanied by CSF and blood (ie, serum, plasma, or whole blood) cryptococcal antigen testing (most commonly by lateral flow assay) and radiological studies, are central to the diagnosis of cryptococcosis (panel 3; appendix pp 10, 35, 41).[33,34]

Screening, primary prophylaxis, and pre-emptive therapy

Supportive evidence for cryptococcal screening is limited to people living with HIV and depends on blood cryptococcal antigen by lateral flow assay (panel 4; appendix p 49).

Panel 1: Grade of recommendation and level of evidence

This guideline follows the structure and definitions of previous European Confederation of Medical Mycology guidelines on invasive fungal infections,[22,23] which are in accordance with the Grading of Recommendations Assessment, Development and Evaluation and Appraisal of Guidelines for Research & Evaluation systems, as previously described. Strength of recommendation and quality of evidence are provided.

Grade of recommendation

- A: the guideline group strongly supports a recommendation for use
- B: the guideline group moderately supports a recommendation for use
- C: the guideline group marginally supports a recommendation for use
- D: the guideline group supports a recommendation against use

Level of evidence

- I: evidence from at least one well-designed randomised controlled trial (RCT)
- II: evidence from at least one well-designed clinical trial, without randomisation; from cohort or case-controlled analytic studies (preferably from >1 centre); from multiple time series; or from results of uncontrolled experiments
- III: evidence from opinions of respected authorities, based on clinical experience, descriptive case studies, or reports of expert committees

Added index for source of level II evidence

- r: meta-analysis or systematic review of RCT
- t: transferred evidence (ie, results from different patient cohorts or similar immune-status situations)
- h: historical control as control group
- u: uncontrolled trials
- a: for published abstract presented at an international symposium or meeting

Panel 2: Recommendations for populations at high risk, clinical presentations, and outcomes

- (AIII) Cryptococcosis should be considered in any patient presenting with compatible symptoms or microbiology, regardless of their immune status.
- (AIII) Among patients without known predisposition to cryptococcosis, exclusion of an underlying immunodeficiency (eg, performing HIV serology and CD4 T-cell count) is recommended

Panel 3: Recommendations for yeast causing cryptococcosis and diagnostic methods

(AIIt) All patients with suspected or confirmed cryptococcosis (including cryptococcal antigenemia) require clinical assessment for CNS, pulmonary, and other body site involvement.

Investigations for disseminated disease should include:

- Lumbar puncture with measurement of CSF opening pressure, glucose, protein, cell counts, microscopy, and culture and quantification of CSF cryptococcal antigen
- Quantification of blood cryptococcal antigen and cultures of blood, sputum (or other respiratory specimens), or other affected sites
- Brain imaging (preferably MRI) and chest imaging (preferably CT)

HIV-associated cryptococcal meningitis

Induction therapy

Multiple studies support the successful combination of amphotericin B plus flucytosine as the induction treatment of choice in HIV-associated cryptococcal meningitis. First trialled by van der Horst and colleagues, the addition of flucytosine to amphotericin B showed a trend towards improved CSF sterility at 2 weeks and reduced frequency of relapse.[35] In a subsequent trial, this combination cleared cryptococci (measured as early fungicidal activity [EFA]) more rapidly than either amphotericin B alone or amphotericin B plus fluconazole.[5] Importantly, the combination of amphotericin B 1 mg/kg daily plus flucytosine 25 mg/kg four times a day showed a survival advantage at day 70, compared with amphotericin B alone in the treatment of cryptococcal meningitis.[2] The nephroprotection of liposomal amphotericin B compared with amphotericin B is long recognised and the accessibility of liposomal amphotericin B in high-income settings led to the establishment of liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day for 2 weeks as the standard.

In low-income settings, challenges with antifungal access, adverse effects, and difficulty of monitoring and safely managing 2 weeks of amphotericin B induction treatment led to phase 2 studies exploring alternative regimens. Fluconazole monotherapy, even at doses up to 1200 mg daily, was associated with approximately 50% mortality at 10 weeks and up to 75% mortality at 1 year.[36–38] An oral combination of fluconazole 1200 mg daily plus flucytosine 25 mg/kg four times a day was associated with a significant improvement in EFA compared with fluconazole alone.[39] The addition of a short, 5–7 day course of amphotericin B at 1 mg/kg daily to oral fluconazole or combined oral fluconazole and flucytosine showed improved rates of cryptococcal clearance,[40,41] similar to rates observed with 14 days of amphotericin B.

In the phase 3 ACTA trial conducted in centres in Africa the oral combination of fluconazole 1200 mg daily and flucytosine 25 mg/kg four times a day for 2 weeks was compared with 1 week of amphotericin B 1 mg/kg daily and 2 weeks of amphotericin B 1 mg/kg daily as induction therapy, with the amphotericin B groups

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Panel 4: Recommendations for screening, primary prophylaxis, and pre-emptive therapy

Adults living with HIV who are antiretroviral therapy (ART)-naive or after a period of ART discontinuation with less than 200 CD4 cells per mm^3 must have:

- (AI) A lateral flow assay of blood cryptococcal antigen for the screening of cryptococcosis and the cryptococcal antigen titre should be measured if positive
- (AIIt) All patients with cryptococcal antigenaemia should be carefully assessed and investigated for cryptococcosis and treated as appropriate
- (AIIu) In people living with HIV who have asymptomatic cryptococcal antigenaemia but without clinical cryptococcosis after thorough investigation (including at least a lumbar puncture), fluconazole 1200 mg daily for 2 weeks (when ART can be initiated), followed by fluconazole 800 mg daily for 8 weeks, and 200 mg daily thereafter for 6 months is recommended (guidance might be updated contingent on results of prospective trials)
- (BI) In clinical settings where cryptococcal antigen lateral flow antigen screening is not available (despite WHO's strong recommendations), universal primary prophylaxis with fluconazole 100 mg daily in people living with HIV in high endemic areas with a CD4 count of less than 200 cells per mm^3 is recommended

In patients without HIV:

- (DIIu) Routine blood cryptococcal antigen screening, primary prophylaxis, and pre-emptive therapy are not recommended

being further randomly assigned to either fluconazole 1200 mg daily or flucytosine 25 mg/kg four times a day.[7] 1 week of amphotericin B 1 mg/kg daily plus flucytosine followed by fluconazole 1200 mg daily in the second week was the best-performing induction group, with a 24% 10-week mortality rate. This regimen was adopted as the preferred 10-week induction regimen by WHO and southern African guidelines until the AMBITION-cm study.[16,18]

In the AMBITION-cm phase 3 study, which had sites across Africa, a single initial 10 mg/kg dose of liposomal amphotericin B with oral fluconazole 1200 mg daily plus flucytosine 25 mg/kg four times a day for 2 weeks was compared with the WHO recommendation of 1 week of amphotericin B 1 mg/kg daily plus flucytosine followed by 1 week of fluconazole 1200 mg daily.[6] This new regimen met non-inferiority criteria (10-week mortality 24·8% *vs* 28·7%) with similar EFAs and was significantly better tolerated. The WHO guidelines now recommend the AMBITION-cm regimen as the preferred antifungal therapy in people living with HIV and cryptococcal meningitis.[10]

The applicability of the ACTA and AMBITION-cm trials to high-income settings and in non-HIV populations is contentious. The standard regimen is liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day for 2 weeks, which is different to the comparators used in the trials. Retrospective database reviews in the USA showed low rates of acute inpatient mortality from cryptococcal meningitis (10·5% in HIV-cryptococcal meningitis and 13·3% in non-HIV cryptococcal meningitis) and a remarkably low mortality rate at 1 year of 11·6% in the past two decades.[42,43] The reliance on high-dose fluconazole and flucytosine as the basis of induction therapy in the AMBITION-cm study might not be pragmatic in high-income settings, where more comorbidities occur, potential drug–drug interactions need to be carefully considered, and the risk of hepatotoxicity is less tolerated than in low-income settings. In the USA, only a third of patients completed the 14 days of flucytosine.[44] Although some experts support the inclusion of the AMBITION-cm triple regimen as a primary option in high-income settings, other experts call for further comparative trials in high-income settings to assess the regimen's effect in patients with HIV and patients without HIV (in whom no supporting data exist). Regardless of the induction antifungal regimen used, the complications of cryptococcal meningitis, such as increased intracranial pressure, require intense clinical monitoring, and most patients with cryptococcal meningitis require inpatient care for 1–2 weeks or more.

Mycological success, defined as cryptococcocal culture negativity (also termed CSF sterility) has been associated with improved outcomes and reduced clinical relapse.[45] In people living with HIV and cryptococcal meningitis, CSF sterility before ART commencement has been shown to be associated with reduced occurrence of neurological deterioration, microbiological relapse, and cryptococcosis-associated immune reconstitution inflammatory syndrome (C-IRIS).[45] Some treatment guidelines advocate performing a lumbar puncture after 2 weeks of induction therapy (before changing to consolidation therapy) to assess CSF culture sterility as a marker of successful induction.[11,15,18,20] Other guidelines—particularly those focused on low-income settings—do not.[10,16]

Consolidation and maintenance therapy

There have been no trials of consolidation and maintenance therapy in cryptococcal meningitis within the past two decades. Two early studies established 400 mg daily fluconazole for consolidation therapy.[35,46] With the accumulation of safety data of a 800 mg fluconazole daily dose and evidence of a fluconazole dose-response effect,[36,47] this regimen is the preferred consolidation dose in low-income settings, where suboptimal antifungal regimens are used.[16,18] A gradual rise in median fluconazole minimum inhibitory concentrations (MICs) in cryptococcal isolates collected

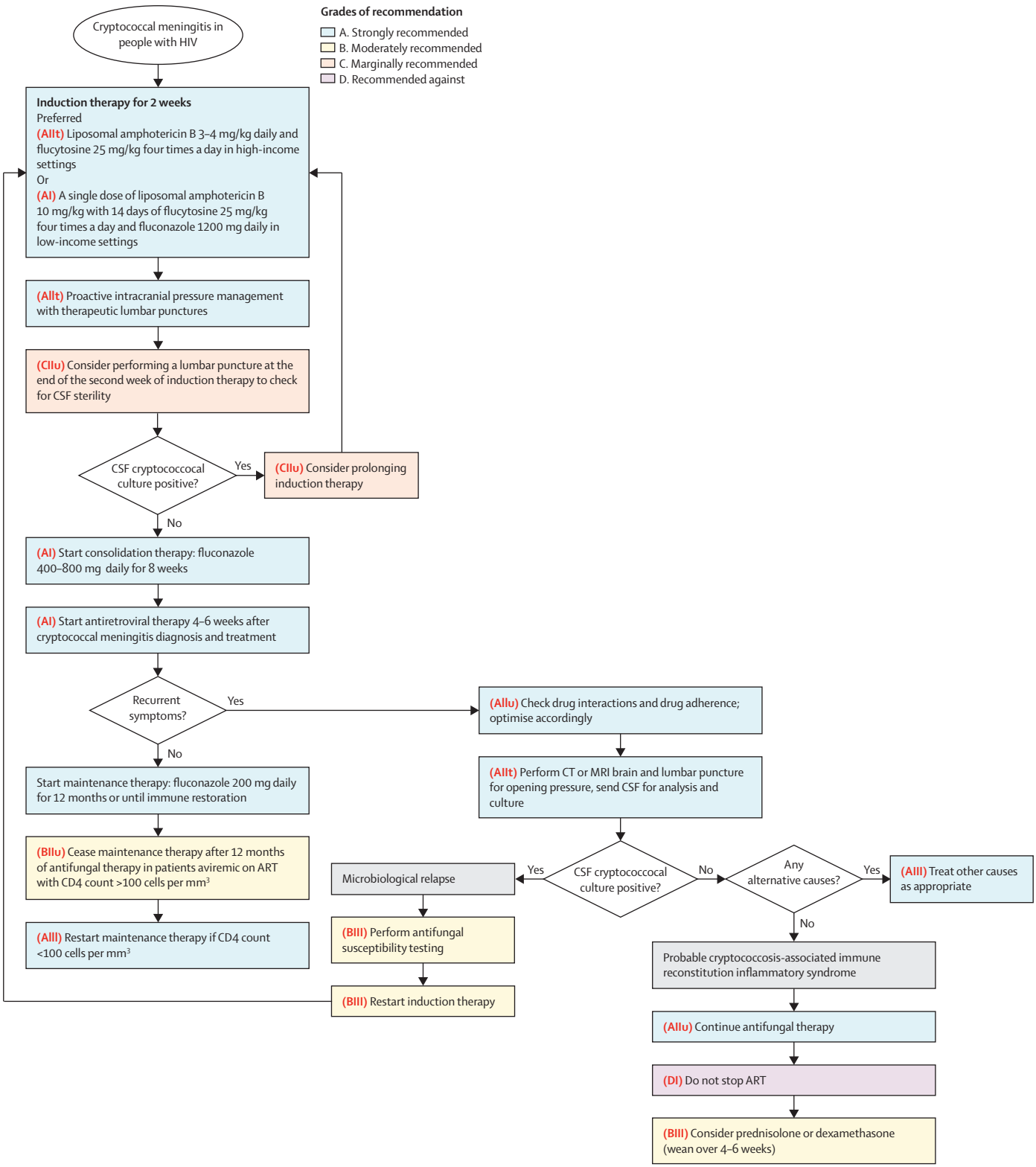


Figure 2: **Management algorithm for cryptococcal meningitis and cryptococcal meningoencephalitis in people with HIV**
ART=antiretroviral therapy. CSF=cerebrospinal fluid.

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Panel 5: Ten principles of cryptococcal meningitis management

The key principles in cryptococcal meningitis management are best read in context (see relevant sections in main text). Although most evidence and recommendations are derived from cryptococcal meningitis in people living with HIV, many of these principles are translatable to non-HIV settings.

1) Selectively screen, risk-stratify, and investigate for cryptococcosis

This principle is specific to people with HIV and cryptococcal meningitis (panel 4).

2) Provide best fungicidal induction regimen possible

- (AIIt) Liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day for 2 weeks (preferred in high-income settings and strongly recommended in SOT and non-HIV non-SOT settings); or (AI) a single dose of liposomal amphotericin B 10 mg/kg with 14 days of flucytosine 25 mg/kg four times a day and fluconazole 1200 mg daily (note: only trialled in people living with HIV in low-income settings).
- (CIIu) Consider performing a lumbar puncture at the end of the first or second week of induction therapy to check for CSF sterility before ART commencement.
- (CIIu) Consider prolonging induction therapy if CSF is persistently culture positive at 2 weeks.
- (BIII) In *Cryptococcus gattii* CNS infection occurring in non-HIV patients or CNS cryptococcoma consider extending induction therapy to 4–6 weeks.

3) Monitor for and minimise drug toxic effects

- (AIIu) In-hospital care for the first 1–2 weeks is encouraged to manage the major early complications seen with cryptococcal meningitis management.
- (AIIu) The use of amphotericin B and liposomal amphotericin B should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy.
- (AIIu) Frequent (at least every alternate day) complete blood counts, renal function tests, and electrolyte measurements are recommended to assess for therapy-related nephrotoxicity and bone marrow, fluid, and electrolyte changes. Liver function tests at baseline and at least weekly are recommended.
- See appendix pp 16, 30.

4) Manage raised intracranial pressure

- (AIIu) Opening pressure should be measured at every lumbar puncture in patients with cryptococcal meningitis.
- (AIIt) Acute symptomatic elevation of the intracranial pressure (≥20 cm of CSF) should be managed by daily therapeutic lumbar punctures (ie, removal of sufficient CSF, usually around 20–30 mL) to reduce the pressure to 50% of opening pressure or to a normal pressure of ≤20 cm of CSF (documented as a closing pressure).
- (BIIu) Perform a scheduled therapeutic lumbar puncture 48–72 h after initial lumbar puncture or 7 days, regardless of initial opening pressure.
- (AIIt) Persistent raised symptomatic intracranial pressure despite therapeutic lumbar punctures should be managed by surgical decompression via temporary lumbar drainage, shunting, or ventriculostomy, depending on local expertise and resources.

5) Look for an underlying immunosuppressive state

Exploring for an immunosuppressive state—particularly, but not limited to, HIV infection—is important in the management of cryptococcosis.

- (AIII) Among patients without known predisposition to cryptococcosis, exclusion of an underlying immunodeficiency (including performing HIV serology and CD4 T-cell count) is recommended in all patients with cryptococcosis.
- (BIII) Individuals without a known risk factor for disseminated cryptococcosis, particularly those with a history of other atypical fungal, mycobacterial, or bacterial infections, should be considered for evaluation of an undiagnosed immunodeficiency, preferably in consultation with a clinical immunologist (appendix pp 6, 79).

6) Provide and ensure adherence to consolidation and maintenance therapy

- Consolidation (8 weeks): (AI) Fluconazole 400–800 mg daily. 800 mg is preferred in low-income settings.
- Maintenance (12 months or until immune restoration): (AIIt) Fluconazole 200 mg daily.
- (AIIu) Check for drug–drug interactions and adjust the dose as necessary.
- (AIII) Close therapeutic drug monitoring of tacrolimus, cyclosporine, and sirolimus levels and dose reduction of these agents are recommended when azoles are co-administered.[73,74]

7) Optimal commencement of ART in people with HIV

This principle is specific to people with HIV and cryptococcal meningitis.

- (DI) Immediate or very early commencement of ART is not recommended.
- If there is inadequate access to antifungal induction therapy, (AI) delay ART for 4–6 weeks.
- If there is adequate access to antifungal induction therapy, (BIIu) consider further individualisation, taking into consideration resolution of symptoms and signs of cryptococcal meningitis and intracranial pressure (including normalisation of opening pressure), attainment of CSF cryptococcal sterility, successful identification and management of concurrent co-infections and other AIDS-defining illnesses, the patient's readiness for ART, and

(Continues on next page)

(Panel 5 continued from previous page)

local experience of cryptococcal meningitis and cryptococcosis-associated immune reconstitution inflammatory syndrome (C-IRIS) management (usual range is 4–6 weeks).

8) Monitor for clinical relapse and investigate causality

- (AIIt) Investigate thoroughly for causality (ie, CNS and non-CNS and infective and non-infective) in cases of apparent clinical relapse. Investigations should include CT or MRI of the brain, lumbar puncture for opening pressure, and CSF analyses including microscopy and culture.
- (AIIu) Review adherence to antifungal therapy, ART, immunosuppressants, and other medications and consider drug–drug interactions. Perform therapeutic drug monitoring if applicable. Optimise control of underlying diseases.
- (DIIu) The use of follow-up blood or CSF cryptococcal antigen (including monitoring of titres) for clinical decision making is discouraged.
- (DIIu) Do not escalate antifungal therapy for persistent blood antigenemia, persistently positive CSF cryptococcal antigen, visible cryptococci in CSF (without culture positivity), or abnormal CSF microscopy or biochemistry. These are not necessarily indicators of microbiological failure.

9) Evaluate for drug adherence, drug–drug interactions, and drug resistance

This principle is specific to people with culture-positive (microbiological) persistence or relapse.

- (BIII) Antifungal susceptibility testing should be done concurrently on all initial and relapse isolates (if stored and available). An increase in fluconazole minimum inhibitory concentration of >2 dilutions is concerning for the potential development of drug resistance.
- (BIII) Consider recommencing induction therapy with a more optimal regimen that is guided by antifungal susceptibility testing.

10) Carefully exclude alternative diagnoses before attributing clinical relapse to C-IRIS

- (AIIt) For patients with suspected paradoxical C-IRIS, carefully exclude recurrent cryptococcal disease or new infective or non-infective conditions before attributing symptoms and signs to C-IRIS. Perform a brain MRI and lumbar puncture to measure opening pressure and get CSF for microbiological, cellular, and biochemical analyses.
- (AIIu) Treatment of C-IRIS should include therapeutic lumbar puncture and symptomatic therapy, such as analgesia, antiemetics, and antiepileptics if appropriate.
- (AIII) Continue antifungal therapy.
- (BIII) High-dose prednisolone or prednisone (usually 0·5–1·0 mg/kg daily) or dexamethasone (usually 0·2–0·3 mg/kg daily), weaned over 4–6 weeks can be considered in those with persistent symptoms who are unresponsive to therapeutic lumbar punctures. Rarely, a second steroid course with taper is needed.
- (DIII) Do not stop ART.

during initial cryptococcal meningitis presentation have been reported in South Africa and Uganda.[48,49] Although this evidence could lend support for a higher consolidation dose of 800 mg daily of fluconazole in these settings, whether this regimen is required across all patient groups and settings is contentious. Widespread fluconazole use could also perpetuate further rises in MICs.

Maintenance therapy with fluconazole 200 mg daily has been shown to be highly effective at preventing relapse, superior to weekly amphotericin B and itraconazole capsules.[50–52] Rarely, triazoles, such as voriconazole,[53–60] posaconazole,[61–63] or isavuconazole,[64,65] are used as alternatives to fluconazole due to concerns of fluconazole resistance, drug toxicity, or drug–drug interactions. Notably, none of the newer triazoles have been formally trialled in cryptococcosis and none are readily available in low-income settings (appendix p 30).

A low incidence of cryptococcal meningitis relapse is observed after a minimum of 1 year of antifungal therapy in people living with HIV established on ART, who are virologically suppressed or have a CD4 count more than 100 cells/mm³.[66–72]

A management algorithm is described in figure 2 and key principles are discussed in panel 5. Recommendations for cryptococcal meningitis treatment in people living with HIV are based on the availability of antifungal drugs. Preferred and alternative strategies are offered in (figure 3 and figure 4A.

Adjunctive therapy

In the past decade, trials of adjunctive treatment in HIV-associated cryptococcal meningitis have all been shown to be ineffective, and in some cases harmful. These include high-dose dexamethasone,[75] sertraline,[76,77] and tamoxifen.[78] The debate regarding adjunctive exogenous interferon(IFN)-γ is unresolved. IFN-γ has been studied in two randomised trials of HIV-associated cryptococcal meningitis, which suggested faster clearance of yeasts in the CSF,[79,80] but further studies are needed. There is no trial evidence supporting its use in non-HIV-associated cryptococcal meningitis (appendix p 65).

Cryptococcal meningitis in SOT recipients

Cryptococcosis is the third most common invasive fungal infection in SOT recipients, with an incidence of 4·5–33·8%[26,28,29] and causing considerable mortality.[12] SOT recipients encompass a third of non-HIV-related cryptococcosis in the USA.[81] The majority of cryptococcosis

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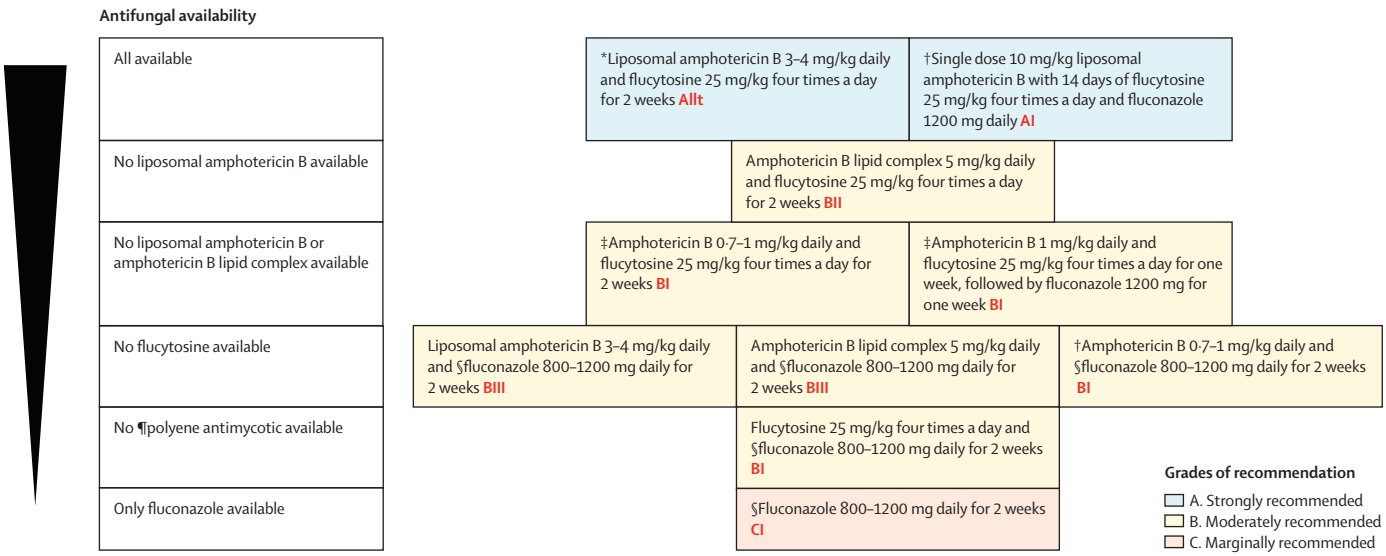


Figure 3: **HIV-cryptococcal meningitis antifungal induction treatment recommendations by antifungal drug availability**
Grading of recommendation and level of evidence in bolded red letters. *Has not been compared with †. †Has only been trialled in HIV-cryptococcal meningitis. ‡Amphotericin B: 1 mg/kg showed earlier fungicidal activity than 0·7 mg/kg, but some institutions use the low dose due to toxicity concerns. §Fluconazole induction doses of up to 1200 mg daily have been trialled but caution is advised; consider drug–drug interaction and liver toxicity. ¶Polyene antimycotic includes amphotericin B formulations such as conventional deoxycholate amphotericin B, liposomal amphotericin B, and amphotericin B lipid complex.

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in SOT occurs late (ie, months or years after transplantation) and is due to reactivated disease; however, acute donor-derived infections have been described.[14,82,83]

Anti-rejection drugs vary in their degree of immunosuppression and heart and small bowel transplant recipients are at the highest cryptococcal meningitis risk.[84] CNS and pulmonary cryptococcosis dominate but unusual manifestations, including cutaneous disease[85,86] and pericarditis,[87] have been reported. Notably, blood cryptococcal antigen can be negative in SOT recipients with cryptococcosis, particularly those with single pulmonary nodules or in lung transplant recipients.[88]

There are no randomised treatment trials targeted specifically at SOT recipients; hence, recommendations are extrapolated from evidence in people living with HIV. The use of lipid-formulations in SOT recipients with CNS cryptococcosis was independently associated with reduced mortality compared with amphotericin B.[89] The AMBITION-cm regimen has not been studied in non-HIV patients, and the evidence for high dose fluconazole (with the ensuant potential toxicity and drug–drug interactions) in this group is absent. A precipitous reduction in dosing of immunosuppressants, particularly calcineurin inhibitors, can lead to C-IRIS.[90] Figure 4B contains recommendations for treatment in SOT recipients (appendix p 62).

Cryptococcal meningitis in people without HIV or SOT

The group of people without HIV or SOT is heterogeneous, ranging from apparently healthy people to those with haematological malignancies or liver cirrhosis. There is no single therapeutic regimen or duration that meets all patients' needs, but the therapeutic principles mirror cryptococcal meningitis treatment in high-income settings, with liposomal amphotericin B 3–4 mg/kg daily and flucytosine 25 mg/kg four times a day as induction therapy. Induction therapy can be extended in those with persistently positive CSF cultures or persistent symptoms at 2 weeks. In 2022, the combination of liposomal amphotericin B and flucytosine was shown to have a low acute mortality of 6% in a nationwide observational study of non-HIV-associated cryptococcal meningitis in Japan.[91] Figure 4B contains recommendations for treatment in people without HIV or SOT (appendix pp 18, 62).

Pulmonary cryptococcosis

There are no randomised treatment studies in pulmonary cryptococcosis. Case series and clinical knowledge suggest that for patients with cryptococcaemia and evidence of CNS involvement, those with blood cryptococcal antigen titres more than 1:512 by latex agglutination (or ten-fold higher by lateral flow assay),[92] or severe pulmonary disease should be treated as cryptococcal meningitis.[33,35,93,94] Patients with mild isolated pulmonary disease without cryptococcoma have been successfully treated with fluconazole monotherapy of

Figure 4: **Antifungal treatment recommendations for cryptococcal meningitis**
(A) Recommendations for people with HIV. (B) Recommendations for SOT recipients and patients without HIV or SOT. SOT=solid organ transplant. CSF=cerebrospinal fluid. TDM=therapeutic drug monitoring.

A People with HIV

First-line therapies

Induction (2 weeks)
(AIIt) Liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day (preferred in high-income settings); or
(AI) Single dose liposomal amphotericin B 10 mg/kg and 14 days of flucytosine 25 mg/kg four times a day and fluconazole 1200 mg daily (recommended in low-income settings)

→

Consolidation (8 weeks)
(AI) Fluconazole 400–800 mg daily (800 mg preferred in low-income settings)

→

Maintenance (12 months or until immune restoration)
(AIIt) Fluconazole 200 mg daily

Alternative therapies

If liposomal amphotericin B is not available:
(BIIt) Amphotericin B lipid complex 5 mg/kg daily plus flucytosine 25 mg/kg four times a day

If liposomal amphotericin B and amphotericin B lipid complex are not available:
(BI) Amphotericin B 0·7–1·0 mg/kg daily plus flucytosine 25 mg/kg four times a day; or
(BI) Amphotericin B 1 mg/kg daily and 5-flucytosine 25 mg/kg four times a day for 1 week, followed by fluconazole 1200 mg daily for 1 week

If flucytosine is not available:
(BIII) Liposomal amphotericin B 3–4 mg/kg daily plus fluconazole 800–1200 mg daily;
(BIII) Amphotericin B lipid complex 5 mg/kg daily plus fluconazole 800–1200 mg daily; or
(BI) Amphotericin B 0·7–1 mg/kg daily plus fluconazole 800–1200 mg daily

If amphotericin B-based therapies are not available:
(BI) Flucytosine 25 mg/kg four times a day and fluconazole 800–1200 mg daily

If only fluconazole is available:
(CI) Fluconazole 800–1200 mg daily

(BIII) Voriconazole 200 mg twice a day (with TDM)

(BIII) Posaconazole 300 mg daily (with TDM)

(BIII) Isavuconazole 200 mg daily

(CIIt) Itraconazole 200 mg twice a day (with TDM)

Comments:
- **(AIIu)** Opening pressure should be measured at every lumbar puncture in patients with cryptococcal meningitis
- **(AIIu)** The use of amphotericin B and liposomal amphotericin B should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy
- **(AIIu)** In-hospital care for the first 1–2 weeks is encouraged to manage the early complications of cryptococcal meningitis therapy
- **(BIII)** Monitoring of flucytosine drug concentration is recommended, where available and if timely; particularly with renal dysfunction
- **(AIIu)** Check for drug–drug interactions and adjust doses as necessary
- **(CIIu)** Consider performing a lumbar puncture at the end of the first or second week of induction therapy to check for CSF sterility before antiretroviral therapy (ART) commencement
- **(CIIu)** Consider prolonging induction therapy if CSF is persistently culture positive at 2 weeks
- **(CIIt)** Adjunctive recombinant interferon-γ might be considered for persistently positive CSF yeast cultures in people with HIV-associated cryptococcal meningitis who have evidence of poor inflammatory responses or persistently positive cryptococcal CSF culture after prolonged antifungal therapy
- **(DI)** The routine use of high-dose dexamethasone in cryptococcal meningitis is not recommended
- **(CIII)** A short course of dexamethasone can be considered for specific indications such as symptomatic space-occupying lesions in the CNS with surrounding oedema or mass effect and cerebral vasculitis
- **(BIIu)** Cease maintenance therapy after 12 months of antifungal therapy in patients aviraemic on ART with a CD4 count more than 100 cells per mm^3
- **(AIII)** Restart maintenance therapy if CD4 count drops to less than 100 cells per mm^3

B SOT recipients and people without HIV or SOT

First-line therapies

Induction (minimum 2 weeks)
(AIIt) Liposomal amphotericin B 3–4 mg/kg daily plus flucytosine 25 mg/kg four times a day

→

Consolidation (8 weeks)
(AIIt) Fluconazole 400–800 mg daily

→

Maintenance (12 months)
(AIIt) Fluconazole 200 mg daily

Alternative therapies

If liposomal amphotericin B is not available:
(BIIt) Amphotericin B lipid complex 5 mg/kg daily plus flucytosine 25 mg/kg four times a day

If liposomal amphotericin B and amphotericin B lipid complex are not available:
(BIIt) Amphotericin B 0·7–1·0 mg/kg daily plus flucytosine 25 mg/kg four times a day

If amphotericin B-based therapies are not able to be used:
(CIIt) flucytosine 25 mg/kg four times a day plus fluconazole 800–1200 mg daily

(BIII) Voriconazole 200 mg twice a day (with TDM)

(BIII) Posaconazole 300 mg daily (with TDM)

(BIII) Isavuconazole 200 mg daily

(CIIt) Itraconazole 200 mg twice a day (with TDM)

Comments:
- Recommendations in HIV patient population are also applicable
- **(AIII)** Induction therapy with liposomal amphotericin B and flucytosine should be considered for any disseminated disease or isolation from a sterile site (even in the absence of CNS manifestations)
- **(AIII)** Close monitoring of tacrolimus, cyclosporine, and sirolimus concentrations (TDM) and dose reduction of these agents are recommended when azoles are co-administered[73,74]
- **(BIII)** Immunosuppressant doses need to be carefully adjusted to allow effective killing of yeasts but should be reduced slowly to avoid precipitating cryptococcosis-associated immune reconstitution inflammatory syndrome; consider a sequential or stepwise reduction of immunosuppressants with careful lowering of corticosteroids early and eliminating mycophenolate before considering reduction of the calcineurin inhibitors because of their direct anticryptococcal activity
- **(CIII)** In a patient treated for cryptococcosis, retransplantation or a new organ transplant can be considered, provided viable yeasts have been cleared from CSF and the patient is asymptomatic after receiving 12 months of anticryptococcal treatment

Grades of recommendation

- A. Strongly recommended
- B. Moderately recommended
- C. Marginally recommended

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See Online for appendix

***Panel 6:* Recommendations for pulmonary cryptococcosis**

- Stratify treatment by disease severity and presence of pulmonary cryptococcoma (appendix pp 22, 67, 84)
- Isolated pulmonary cryptococcosis in immunocompetent or immunocompromised host:
  - (AIIu) Severe disease: as for CNS disease
  - (BIIu) Mild disease: fluconazole 400 mg daily for 6–12 months (range guided by symptom resolution)
- Pulmonary cryptococcosis with CNS manifestations or other evidence of dissemination (eg, cryptococcaemia or skin lesions)
  - (AIIt) as for CNS disease
- (AIII) If the presence of *Cryptococcus* spp in respiratory specimen is deemed as airway colonisation after careful evaluation and no treatment is elected, regular follow-up is recommended, especially in the setting of future immunosuppression
- Pulmonary cryptococcoma (see cryptococcoma section)

***Panel 7:* Recommendations for non-pulmonary non-CNS disease**

- (AIIu) The recommendation for cryptococcaemia is to treat the same as for CNS disease
- (AIII) The recommendation for primary cutaneous (skin) cryptococcosis, attributed to direct inoculation without evidence of dissemination, is fluconazole 400 mg daily for 3–6 months or until healed
- (BIIu) For all other non-CNS non-pulmonary disseminated disease treat the same as CNS disease
- (BIIu) Cryptococcal eye disease should be managed in collaboration with an ophthalmologist

400 mg daily.[93,95,96] Some clinicians consider watchful-waiting and elect not to treat asymptomatic immuno-competent people who incidentally culture any *Cryptococcus* spp in their sputum and have no radiological features of pulmonary cryptococcosis, as they consider this presentation to be airway colonisation.[97] Criteria for distinguishing colonisation from infection is uncertain (panel 6; appendix pp 22, 67).

Non-pulmonary non-CNS disease

Cryptococcosis can affect any organ following haematogenous dissemination. Clinical presentation of non-CNS non-pulmonary disease without fungaemia is rare, but possible. The absence of documented fungaemia does not exclude dissemination. Barring direct inoculation into the skin following trauma, extra-pulmonary disease is by definition disseminated disease and generally requires consideration for aggressive induction therapy. There are no clinical treatment trials for non-pulmonary non-CNS cryptococcosis.

Importantly, visual changes noted in cryptococcal meningitis are frequently related to raised intracranial pressure and do not necessarily indicate direct eye involvement. Ocular cryptococcosis can occur[98,99] but is unusual and requires formal ophthalmological documentation and management. Isolated skeletal osteomyelitis is rare and often requires a combined surgical and medical approach.[100–102] Skin lesions might be polymorphic (panel 7; appendix p 67).

Specific management issues

Raised intracranial pressure

Increased intracranial pressure has been associated with a high burden of cryptococci, leading to both acute and chronic symptoms and signs (eg, visual and hearing loss) and decreased short-term survival. Clinical experience has shown that CSF outflow obstruction can be improved by removal of CSF; observational studies suggested that scheduled therapeutic lumbar punctures result in substantial improvement in survival, regardless of opening pressure.[103,104] For prolonged control of acute increased intracranial pressure, use of lumbar drains in cases without hydrocephaly or ventriculostomies in cases with hydrocephaly might be required.[105–107] Medical therapies including acetazolamide, mannitol, and corticosteroids can be detrimental (panel 8; appendix p 79).[108,109]

Timing of ART commencement

The optimal time to commence ART for HIV infection during cryptococcosis is controversial. Four randomised trials[3,110–112] to find out the optimal timing of ART initiation in HIV-cryptococcal meningitis co-infection have been done in low-income settings, using induction regimens that are not currently preferred, including fluconazole (800 mg daily) monotherapy,[110] amphotericin B 0·7 mg/kg daily,[111] and amphotericin B 0·7–1 mg/kg daily and fluconazole 800 mg daily for 2 weeks. These data seem to suggest that initiating ART within 2 weeks of cryptococcal meningitis presentation is too early in the setting of suboptimal antifungal therapy, and that delaying ART initiation for 4–6 weeks reduces the incidence of C-IRIS and death. CSF sterility before ART commencement might be another factor.[45] A retrospective analysis of combined cohorts in high-income settings did not show higher mortality in those receiving early ART in the first two weeks of antifungal therapy compared with those with delayed therapy.[113] Early ART in high-income settings will need careful justification and close monitoring; further randomised studies might be helpful.[114]

There are no studies for timing ART initiation in other forms of cryptococcosis, those with cryptococcal antigenemia, or those recommencing ART after a period of interruption. Early concerns that potent integrase inhibitors pose an increased risk of C-IRIS have been disproven.[115] Whether those presenting with cryptococcal meningitis within 2 weeks of starting ART require withholding of ART is uncertain (panel 9; appendix p 70).[116–118]

Panel 8: Recommendations for raised intracranial pressure

- (AIIu) Opening pressure should be measured at every lumbar puncture in patients with cryptococcal meningitis
- (AIII) A brain CT should be done (if CNS imaging not already done) to exclude CNS outflow obstruction
- (AIIt) Acute symptomatic elevation of the intracranial pressure (≥20 cm of CSF) should be managed by daily therapeutic lumbar punctures (ie, removal of sufficient CSF, usually around 20–30 mL), to reduce the pressure to 50% of opening pressure or to a normal pressure of ≤20 cm of CSF (documented as a closing pressure)
- (BIIu) Perform a scheduled therapeutic lumbar puncture 48–72 h after initial lumbar puncture or 7 days, regardless of initial opening pressure
- (AIIt) Persistent raised symptomatic intracranial pressure, despite therapeutic lumbar punctures, should be managed by surgical decompression via temporary lumbar drainage, shunting, or ventriculostomy, depending on local expertise and resources
- (BIII) Consider ventriculoperitoneal (preferential) and lumboperitoneal shunts (alternative) to control both acute and chronic hydrocephalus if temporary measures are not successful. Ideally, insert shunts after institution of effective antifungal therapy

Panel 9: Recommendations for the timing of ART commencement

- (DI) Immediate or very early commencement of ART is not recommended.
- (AI) If suboptimal antifungal induction therapy is used, delay ART for 4–6 weeks.
- (BIIu) If optimal antifungal induction therapy was used, consider further individualisation, taking into consideration resolution of symptoms and signs of cryptococcal meningitis, intracranial pressure (including normalisation of opening pressure), attainment of CSF cryptococcal sterility, successful identification, management of concurrent co-infections and other AIDS-defining illnesses, the patient's readiness for ART, and local experience of cryptococcal meningitis and C-IRIS management (usual range is 4–6 weeks).
- (CIIt) If possible, ensure CSF is cryptococcal culture negative before ART commencement.
- (BIII) For people who have had ART who develop cryptococcal meningitis and might need to switch to second-line ART or recommence ART, a delay of 4–6 weeks is recommended.
- (CIII) Pending further studies, consider withholding ART and restarting at 4–6 weeks in those presenting with cryptococcal meningitis within 2 weeks of starting ART.
- (BIII) Patients with isolated pulmonary cryptococcosis or those with asymptomatic cryptococcal antigenemia can commence ART earlier (eg, at 2 weeks).

Panel 10: Recommendations for antifungal resistance

For those with fluconazole resistance or emerging fluconazole resistance:

- (BIII) Consider a long (eg, 4 weeks) course of induction treatment with amphotericin B (1 mg/kg daily) or high dose of liposomal amphotericin B (3–6 mg/kg daily) together with flucytosine
- (BIII) Consider amphotericin B 1 mg/kg weekly or liposomal amphotericin B 3–6 mg/kg weekly as consolidation or maintenance therapy. Consider daily voriconazole, posaconazole, isavuconazole, or itraconazole for isolates without evidence of pan-azole resistance, as guided by antifungal susceptibility testing
- (CIII) If amphotericin B or liposomal amphotericin B are not available, adding flucytosine to high-dose fluconazole (1200 mg daily) could be considered

Resistance to antifungals

Developing secondary resistance to flucytosine is common when given as monotherapy, necessitating its use with a partner drug in cryptococcosis. Acquired resistance to polyenes, such as amphotericin B, is rare, but the emergence of fluconazole resistance is concerning.[48,49,119] Fluconazole monotherapy as induction therapy has been associated with secondary resistance.[120–123]

There are no clinical MIC breakpoints for fluconazole against *Cryptococcus* spp and insufficient data to suggest that high MICs imply worse outcomes. Interpretation of epidemiological cutoff values with the Clinical and Laboratory Standards Institute (CLSI) method for fluconazole requires accurate species identification. The epidemiological cutoff values is 8 ug/mL for *C neoformans* VNI, 16 ug/mL for *C gattii* VGI, and 32 ug/mL for *Cryptococcus deuterogattii* VGII.[124] In principle, a higher than two-fold increase in MIC during treatment could suggest development of resistance and the need for closer clinical monitoring. There are no European Committee on Antimicrobial Susceptibility Testing (EUCAST) epidemiological cutoff values available for fluconazole (panel 10; appendix p 72).

Cryptococcal persistence, clinical relapse, and culture-positive (microbiological) relapse

Distinguishing clinical relapse from persistent cryptococcal infection is challenging. Clinical relapse can be due to a microbiological relapse, C-IRIS, raised intracranial pressure (whether related to C-IRIS or not), or other infective and non-infective (CNS and non-CNS) causes (figure 2). Cryptococcal antigen persists in the CSF and blood, thus it has little clinical utility in distinguishing clinical responders from non-responders.[125] Most cases of culture-positive (microbiological) relapse occur early and result from inadequate or suboptimal induction therapy or early discontinuation of consolidation or maintenance therapy (figure 2; panel 11; appendix p 74).

Panel 11: **Recommendations for cryptococcal persistence, clinical relapse, and culture-positive (microbiological) relapse**

- (AIIt) Think broadly and investigate thoroughly for causality (CNS or non-CNS and infective or non-infective) in cases of apparent clinical relapse; investigations should include brain CT or MRI, lumbar puncture for opening pressure, and CSF analyses, including microscopy and culture
- (AIIu) Review adherence to antifungal therapy, ART, immunosuppressants, and other medications and consider drug–drug interactions; perform therapeutic drug monitoring if applicable. Optimise control of underlying diseases
- (CIII) Consider escalating antifungal therapy while awaiting CSF results (and de-escalate if culture-negative)
- (DIIu) The use of follow-up blood or CSF cryptococcal antigen (including monitoring of titres) for clinical decision making is discouraged
- (DIIu) Do not escalate antifungal therapy for persistent blood antigenemia, persistently positive CSF cryptococcal antigen, visible cryptococci in CSF (without culture positivity), or abnormal CSF microscopy or biochemistry; these are not necessarily indicators of microbiological failure

For culture-positive (microbiological) persistent or relapsed infection (figure 1):

- (BIII) Antifungal susceptibility testing should be done concurrently on all initial and relapse isolates (if stored and available); an increase in fluconazole MIC of more than two dilutions is considered concerning for the potential development of drug resistance
- (BIII) Consider reinduction with a more optimal regimen (guided by antifungal susceptibility testing)

Panel 12: **Recommendations for C-IRIS**

- (AIIt) For patients with suspected paradoxical C-IRIS, carefully exclude recurrent cryptococcal disease or new infective or non-infective conditions before attributing symptoms and signs to C-IRIS; perform a brain MRI and lumbar puncture to measure opening pressure and get CSF for microbiological and biochemical analyses
- (AIIu) Treatment of C-IRIS should include therapeutic lumbar puncture and symptomatic therapy, such as analgesia, antiemetics, and antiepileptics, if appropriate
- (AIII) Continue antifungal therapy
- (BIII) High-dose prednisolone or prednisone (usually 0·5–1·0 mg/kg daily) or dexamethasone (usually 0·2–0·3 mg/kg daily), weaned over 4–6 weeks, can be considered in those with persistent symptoms who are unresponsive to therapeutic lumbar punctures; rarely a second steroid course with taper is needed
- (DIII) Do not stop ART
- (BIII) Cases of steroid-refractory or recurrent C-IRIS should be discussed with experts in the field
- (BIIu) Steroids could be considered for PIIRS

Panel 13: **Recommendations for *C gatti***

In *C gattii* CNS disease:

- (AIII) Treat the same as *C neoformans* CNS infection
- (BIII) In non-HIV patients, consider extending induction therapy to 4–6 weeks
- (AIII) Early CSF shunting is indicated for obstructive chronic hydrocephalus

Treatment of *C gattii* lung disease is summarised in the appendix (p 17).

C-IRIS

C-IRIS has been described in people with HIV usually between 2 weeks and 3 months after commencement of ART. Patients develop exaggerated symptoms and signs or atypical inflammation, reminiscent of a paradoxical recurrence,[126,127] but C-IRIS can also occur in the setting of immune recovery or withdrawal of immunosuppressants. It has also been observed in seemingly immunocompetent individuals, including in *C gattii* infections, as a post-infectious inflammatory immune response syndrome (PIIRS).[90,128] There is no diagnostic biomarker for C-IRIS. It is diagnosed by diagnosis of exclusion (figure 2).

There have been no therapeutic trials in C-IRIS. Management strategies include therapeutic lumbar puncture and symptomatic therapies. In severe C-IRIS, corticosteroids are commonly used to dampen inflammation, although their efficacy has not been rigorously examined in clinical trials. In steroid-refractory C-IRIS, there are case reports on the use of tumour necrosis factor-α blockers, such as adalimumab[129–132] or thalidomide,[133–135] with mixed success. Corticosteroids can also be beneficial in PIIRS (panel 12; appendix p 76).[136]

C gattii

About 50–70% of *C gattii* infections occur in putatively immunocompetent hosts,[137–139] compared with 2–30% in people with HIV.[140–144] Autoantibodies to granulocyte-macrophage colony-stimulating factor and idiopathic CD4 lymphopenia are reported risk factors.[137,145–147] Notably, not all commercial lateral flow assays are able to detect *C gattii* disease.[148] Antifungal agents used for treatment are the same as for *C neoformans*.[30,32,141] However, 4–6 weeks of induction therapy might be required in some cases of non-HIV-associated meningitis with *C gattii* (panel 13; appendix p 81).[149]

Cryptococcomas

Cryptococcomas occur predominantly in the lungs and brain and are more frequent in *C gattii* infection.[140,150] CNS cryptococcomas can manifest as neurological deficits or raised intracranial pressure,[140] which requires urgent management. Corticosteroids and surgical resection can be of value.[149,151,152] Radiological lesions can persist indefinitely despite clinical and microbiological cure (panel 14; appendix p 84).[32,153] Recommendations for cryptococcomas are in.

Non-*C neoformans* and non-*C gattii* strains of cryptococcus

There are individual case reports and small case series of non-*C neoformans* and non-*C gattii* cryptococcus infections, predominantly in immunosuppressed patients. *Papiliotrema laurentii* (previously *Cryptococcus laurentii*)[154] and *Naganishia albida* (previously *Cryptococcus albidus*)[155] account for about 80% of the invasive infections in this group and usually involve the skin, lungs, bloodstream, or CNS.[156] Colonisation, especially of the skin, respiratory, and

Panel 14: Recommendations for cryptococcomas

- (AIII) Perform a biopsy or aspirate to exclude a secondary pathogen or an underlying tumour in non-responding cryptococcomas (particularly in immunosuppressed patients)
- (BIII) Consider surgical resection for accessible brain lesions more than 3 cm, lesions at risk of compressing critical structures, or large lesions not responding to therapy
- (DIII) During follow-up, do not prolong or escalate therapy for persistent radiological findings in the absence of new or worsening symptoms or signs

For CNS cryptococcoma:

- (BIII) Consider prolonging CNS antifungal induction therapy to 4–6 weeks
- (BIII) Consider corticosteroids for large cryptococcomas with surrounding mass effect or if neurological symptoms and cerebral imaging signs worsen despite a good microbiological response

The appendix (pp 22, 84) summarises treatment of lung cryptococcoma.

gastrointestinal tracts must be distinguished from true disease. In some cases, the laboratory might misidentify another yeast as *P laurentii* or *N albida* on the basis of non-definitive commercial identification methods.[157] Elevated MICs against flucytosine, fluconazole, and other azoles for some isolates have been documented but are of uncertain clinical significance (panel 15; appendix p 85).[158,159]

Pregnancy

The majority of cases of cryptococcosis in pregnancy occur in the third trimester or postpartum.[160,161] Maternal mortality from disseminated cryptococcosis is approximately 25%, and less than 50% of women carry their pregnancy to term.[161] Extensive clinical experience suggests that amphotericin B and liposomal amphotericin B are safe during pregnancy (Category B drug), and thus are the cornerstone of treatment.[161,162] Flucytosine is rated by the USA Food and Drug Administration as a Category C drug because of its direct effects on RNA and DNA metabolism. Fluconazole is a Category D drug due to its increased risk of musculoskeletal malformations, tetralogy of Fallot, and spontaneous abortions (panel 16; appendix p 86).[163–167]

Paediatrics

There is a clear need for paediatric-specific studies in cryptococcosis. CNS disease seems to predominate in paediatrics, but non-CNS disease is probably under-reported.[168–174] Clinical efficacy trials and studies to validate diagnostic tests and therapies for cryptococcosis in children are scarce. Recommendations are extrapolated from studies in adult populations. Dosing of antifungal agents needs particular attention for the paediatric patient (panel 17; appendix p 87).

Panel 15: Recommendations for non-*C neoformans* and non-*C gattii* strains of cryptococcus

- (AIII) As non-*C neoformans* and non-*C gattii Cryptococcus* spp are rarely pathogenic, careful assessment of the laboratory identification and clinical context is required to ascertain clinical significance
- (CIII) For CNS or disseminated disease, treat the same as *C neoformans* CNS infection

Panel 16: Recommendations for cryptococcosis in pregnancy

- (AIII) Use liposomal amphotericin B or amphotericin B in induction, consolidation, and maintenance therapy and for the treatment of isolated cryptococcal antigenemia
- (DII) Avoid the use of flucytosine and fluconazole in pregnancy, particularly in the first trimester; their use in the second and third trimester requires careful individualised risk–benefit assessment
- (BIII) Fluconazole can be used after delivery despite its excretion into breastmilk
- (AIII) Apply clinical judgement when considering initiation of antifungal therapy and duration of therapy, factoring in trimester of pregnancy and severity of illness
- (CIII) For asymptomatic cryptococcal antigen in pregnancy, consider intermittent polyene therapy, especially in the first trimester

Panel 17: Recommendations for paediatric cryptococcosis

For the treatment of CNS or disseminated disease:

- (AIIt) Induction: amphotericin B 1 mg/kg daily or liposomal amphotericin B 3–4 mg/kg daily plus flucytosine (100–150 mg/kg daily in 4 divided doses) for 2 weeks
- (AIIt) Consolidation: fluconazole 12 mg/kg (maximum 800 mg) daily for 8 weeks
- Maintenance: fluconazole 6 mg/kg daily (maximum 800 mg) for 6–12 months
  - (AIIt) Should be provided for people who live with HIV and are immunocompromised
  - (BIIt) Can be provided for people who are immunocompetent
- (AIII) For the treatment of severe isolated pulmonary diseases: treat the same as CNS disease
- (AIII) Treatment of mild isolated pulmonary disease: fluconazole 12 mg/kg daily (maximum 800 mg) for 6–12 months
- (AIII) Screening is recommended for children older than 10 years living with HIV in high disease prevalence areas

Conclusions

Cryptococcosis and its management is complex and challenging. Adherence to clinical practice guidelines can improve outcomes.[44,175] Although there has been substantial development of evidence from randomised controlled trials over the past 20 years, there are considerable unmet needs (appendix pp 23, 91). Addressing these challenges is particularly crucial in low-income settings, where the burden of disease is high and access to antifungal therapy is inadequate. Equally, more clinical research needs to be done in high-income settings, where host risk profiles are changing and an increasing array of presentations of cryptococcosis are being recognised, necessitating more nuanced and individualised treatment plans.

Contributors

JRP guided the structure, content, and development of the guideline. OAC contributed to the conceptual planning, management, and supervision of the project. CCC and JRP coordinated the work of the

authors. TAB, CCC, MC, FH, TSH, OL, RO, JRP, TCS, AS, and AW contributed to the coordination of data collection, data visualisation, and participants' contributions and communication and wrote the first manuscript draft. All authors contributed towards the literature review, collection and preparation of data, creation of tabled recommendations, and critical review of the manuscript.

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Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Chang CC, Harrison TS, Bicanic TA, et al. Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM. *Lancet Infect Dis* 2024; published online Feb 9. https://doi.org/10.1016/S1473-3099(23)00731-4.

Appendix

This appendix provides supplementary tables, figures, panels and supplementary text to the main manuscript, Chang et al., Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM. Please refer to the main manuscript for key figures and panels.

Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM

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Supplementary Tables

Supp. Table 1: Common risk factors and rare Immunodeficiencies to consider in the setting of cryptococcosis.

Underlying immunodeficiencies to consider	Steps/ Investigations to consider	Recommendations
Common underlying risk factors: • HIV[1-4] • SOT recipients[5-7] • Other immunosuppressive conditions such as haematological malignancies[8-10], cirrhosis[11-13], rheumatoid arthritis[14], sarcoidosis[15] • Receipt of immunosuppresant drugs such as corticosteroids[16-18], ibrutinib[19], JAK inhibitors[20-22], alemtuzumab[23,24], fingolimod[25], eculizumab[26,27] • Idiopathic CD4 lymphopenia (ICL)[28-36]	• Thorough history should include risk factors for HIV, receipt of immunosuppressive therapy, personal or family history of recurrent or unusual infections, and examination for lymphadenopathy and concurrent opportunistic infections (e.g. oral candidiasis) is recommended. • Blood investigations should include a CBC and blood smear/film, HIV testing, T cell (CD4/CD8/ natural killer (NK) cell subsets by flow cytometry.	**(AIII)** Exclusion of an underlying immunodeficiency (including performing HIV serology and CD4 T-cell count) is recommended in all patients presenting with cryptococcosis.
Rare primary immunodeficiencies: • Anti-cytokine antibodies, particularly auto-antibodies to IFN-gamma and GM-CSF (Pulmonary alveolar proteinosis) [37-43] [44] • X-linked hyper-IgM syndrome associated with a CD40 ligand (CD40L) gene mutation[45-47] and in those with hypogammaglobulinemia.[48-50] (in paediatric patients) • Inborn errors of immunity including mutations in *STAT1* causing GOF or *STAT3* causing hyper-IgE (Job's) syndrome (*STAT3* HIES), GATA2 deficiency, IL-12 receptor beta-1 deficiency[51-55] • SNPs including PRR such as dectin-2,[56] PTX3,[57,58] Fc gamma receptors,[59,60] MBL,[61] and six loci	• Blood investigations to consider include: immunoglobulin levels, complement testing, and anti-cytokine antibodies. • In-born errors of immunity are best assessed by WGS, WES or targeted gene sequencing. • A clinical immunologist experienced in immunodeficiency conditions may assist in guiding and prioritising these advanced investigations. • A personalised and pragmatic approach is necessary.	**(BIII)** Individuals without a known risk factor for disseminated cryptococcosis, particularly those with a history of other atypical fungal, mycobacterial, or bacterial infections, be considered for evaluation of an undiagnosed immunodeficiency, preferably in consultation with a clinical immunologist.

upstream of the colony-stimulating factor 1 gene encoding for M-CSF[62]		

Footnote:

CBC complete blood count

GM-CSF granulocyte-macrophage colony-stimulating factor

GOF gain-of-function

HIV Human immunodeficiency virus

IFN-gamma Interferon-gamma

JAK, Janus kinase

M-CSF macrophage colony-stimulating factor

MBL mannose-binding lectin

PRR pathogen recognition receptors

PTX3 pentraxin-3

SNPs Single nucleotide polymorphisms SNP

SOT Solid organ transplant

STAT signal transducer and activator of transcription

WES whole exome sequencing

WGS whole genome sequencing

164 ## Supp. Table 2: Key outcome determinants of cryptococcosis
165

(1)	**The yeast burden of the infection:** Higher yeast burdens are associated with worse clinical outcomes. This may be indicated by any of the following: • high quantitative cerebrospinal fluid (CSF) yeast counts (e.g., ≥10^6 CFU/ml in research settings); • high blood and CSF cryptococcal antigen (CrAg) titres (>1:512 with latex agglutination or ≥1:640 with LFA) [8,63-65]; • positive blood culture; • strongly positive CSF India Ink examinations or heavy amount of yeast cells seen by gram stain or calcofluor white stain
(2)	**Cryptococcosis syndrome type:** CM is a severe and potentially fatal condition with high mortality rates while isolated pulmonary cryptococcosis is generally a milder illness.
(3)	***Cryptococcal neoformans* species complex vs. *Cryptococcal gattii* species complex:** *C. gattii* is associated frequently with pulmonary disease, and with more cryptococcomas that may be associated with mass effect or masquerade as malignancy. It occurs more commonly in putatively immunocompetent persons. There are no clinical trials specifically for *C. gattii* and less is known about its optimal management.
(4)	**Poor host immune response:** This may be suggested by any of the following: • Prolonged CD4 lymphopenia, • An acellular CSF with low/ absent CSF leucocytosis (≤5 white blood cells (WBCs) /high power field (HPF) on microscopy), • and persistent CSF hypoglycorrhachia (low glucose). Some non-HIV related cryptococcosis may have a worse outcome than HIV-related cases, due in part, to the inability to define and/or reverse underlying immunosuppression.
(5)	**Host comorbidities:** The underlying immunosuppressive condition is an important predictor of mortality by imparting different manifestations to cryptococcosis or influencing attributable mortality of the infection.
(6)	**Delays in presentation or diagnosis:** Delays of weeks to months in diagnosis are common outside of the classical high-risk groups, even in patients with typical CM symptomatology, which results in worse outcomes**.** Cryptococcosis presentation can also be subacute or chronic and non-specific further challenging prompt diagnosis. Mortality is increased in non-HIV-infected hosts with cryptococcosis.[66]
(7)	**Optimal antifungal management and development of antifungal toxicity:** Different combinations of antifungal drugs result in very different clinical efficacy, thus selecting the optimal combination is critical. Expertise in the prevention and management of adverse events related to an antifungal therapy is necessary to facilitate successful and ongoing antifungal treatment.
(8)	**Management of raised ICP:** Uncontrolled raised ICP results in loss of consciousness, coma, and death. Management of ICP in the first one to two weeks of therapy is critical to survival, regardless of the antifungal therapy received. The

	lack of personnel, time and tools required for therapeutic lumbar punctures (LP) may impede therapeutic success.
(9)	**Development of cryptococcosis-associated immune reconstitution inflammatory syndrome (C-IRIS):** C-IRIS is associated with significant morbidity including re-hospitalisation, particularly in case of CNS involvement. Poor management of C-IRIS including missed diagnosis or over-treatment (e.g., unnecessary re-induction) may result in additional negative sequelae.
(10)	**Healthcare systems:** Worse outcomes are reported in RLS due to delayed presentation, missed screening and diagnostic opportunities; limited access to health care, including intensive care; limited antifungal options (notably 5-flucytosine and L-Amb), interrupted ART supply; reduced resources to control ICP; difficulties posed by drug toxicities; and/or sub-optimal follow-up.

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168 Supp. Table 3: Recommendations for the use of diagnostic modalities in cryptococcosis

Method	Evidence	Recommendations/ Comment
Microscopy, culture and identification	• India Ink is a traditional rapid test: Diagnostic sensitivities of 84%–98% after centrifugation[67,68] are reported in experienced laboratories particularly in high prevalence settings. • Gram stain and calcofluor white staining of CSF may also detect *Cryptococcus.* • Culture as gold standard[69] with colonies appearing on solid media after incubation at 30–35°C in aerobic conditions for 48 to 72 hours. • Molecular taxonomy has revolutionised fungal nomenclature – whether these should remain as two species complexes or be divided into multiple separate clades (species) is debated.[70,71] It may be important in the future to differentiate these separate clades as the antifungal susceptibility may differ. • *C. neoformans* and *C. gattii* isolates can be distinguished by growth on CGB agar; *C. gattii* develop a blue colour in the media, whereas *C. neoformans* do not. • Accurate species identification of fungal colonies into *C. neoformans* and *C. gattii* species complexes can be achieved reliably and rapidly by MALDI-TOF MS.[72-76]. Some jurisdictions utilise API20C AUX.[77]	• **(AIIt)** Gram stain/ India ink preparation/ calcafluor white stain of centrifuged CSF can be effective in patients suspected of CM with or without CrAg availability. • **(AIII)** Fungal culture from blood, CSF, sputum, urine, BALF or other relevant clinical specimens is recommended in all patients suspected of cryptococcosis for genus and species identification. • **(AIIt)** Blood cultures should be performed using standard bacterial or fungal culture media. • **(AIIu)** Body fluids and other clinical specimens are ideally plated on Sabouraud dextrose agar without cycloheximide and incubated at 30°C in aerobic conditions for 7-10 days. • **(AIIu)** Differentiation into *C. neoformans* and *C. gattii* species complexes is recommended in standard clinical laboratories. • **(AIIt)** Species identification into *C. neoformans* and *C. gattii* species complexes by MALDI-TOF MS, is recommended. • **(BIIu)** For qualitative cultures during therapy, we recommend centrifugation of a reasonable volume (***e.g.,*** >5 ml) of CSF and plating the sediment for culture. CSF collected at CM presentation may need only 7 days of culture whereas CSF collected after antifungal commencement may require longer periods of culture (e.g., 14 days) to adequately assess for CSF sterility.
***In vitro* susceptibility testing.**	• BMD using CLSI or EUCAST protocols are reference methods to determine the MIC of *Cryptococcus* against amphotericin B, 5-flucytosine or azoles.	• **(AIIt)** All primary cryptococcal isolates should have a baseline fluconazole, and ideally flucytosine MIC

	• Commercial MIC techniques using automated systems (Vitek®2, BioMérieux) and colorimetric assays (SensiTitre™ YeastOne™, ThermoFisher Scientific) show good agreement with BMD for the three antifungal classes.[78,79] • Agreement with results using agar-based diffusion gradient systems (Etest®, BioMérieux; MTS, Liofilchem) is more variable, and is worst with amphotericin B.[78-81] • Susceptibility thresholds for *Cryptococcus* yeasts are based on epidemiological cut-off values, abbreviated as ECV (CLSI) or ECOFF (EUCAST) to distinguish non-wild type yeasts with potentially detectable acquired resistance to a particular drug. ECV/ECOFF are method-dependent and vary by *Cryptococcus* genotype.	determined or stored for future comparative testing of relapse isolates.
Latex agglutination (LA) and Enzyme Immunoassay (EIA) for CrAg	• LA kits for CrAg are reportedly 93–100% sensitive and 93–98% specific versus culture, compared with 85.2-99% and 97%, respectively for EIA in the CSF. The diagnostic performance of these commercial tests in CSF and blood vary[82], and have been extensively reviewed, including in the WHO guidelines supplement of 2018.[1] • Dilution of sera or CSF may be necessary to remove the pro-/post-zone effect (to exclude false-negatives).[82,83]	• **(BII)** Both LA and EIA tests have been generally superseded by LFA though some jurisdictions continue to use these for antigen detection.
Lateral flow assay (LFA) for CrAg	• LFA records overall sensitivities and specificities of 97%–100% and 95%–99%, compared to CSF culture respectively.[84-88] • Notably, commercial CrAg LFAs may have reduced ability to detect all pathogenic *Cryptococcus* species[65,89], although this does not apply to the IMMY CrAg® LFA test (Oakland, Oklahoma, US) which has been most thoroughly evaluated.	• **(AIIt)** Detection and quantification of CrAg in blood and CSF by LFA is recommended as a first-line rapid diagnostic test, in both RLS and RRS. Any positive result regardless of titre level should be appropriately investigated. • **(CIII)** CrAg tests can be performed on other body fluids, including BALF and urine, but their sensitivity and specificity in relation to these sample types have not been assessed rigorously.

	• False negatives due to pro-/post-zone effect are rare but have been reported to be overcome by dilution.[90]. • Not all commercial LFAs are able to detect *C. gattii* disease. The detection of *C. tetragattii* may be missed with CryptoPS (Biosynex, Illkirch-Graffenstaden, France) and cryptococcal antigen LFA (Dyanmiker Biotechnology, Tianjin, China).[89] • The diagnostic performance of these tests in CSF and blood have been extensively reviewed, including in the WHO guidelines supplement of 2018.[1]While the CrAg tests appear to have a lower sensitivity in HIV-negative patients (85% compared to 100% in PLHIV for LFA), LFA still appears to be significantly better than the EIA,[91] although there has not been a systematic evaluation of CrAg LFA in *C. gattii* disease.	• **(DIIu)** The use of follow-up blood or CSF CrAg (including monitoring of titres) for clinical decision-making is discouraged.
Cryptococcal antibody	• Not used in current clinical practice.	• **(DIII)** Cryptococcal antibody testing is not recommended for diagnosis or management of cryptococcosis at this time.
Quantitative culture[92] for fungal burden	• Used as a surrogate treatment response biomarker in Phase 2 and 3 clinical trials. Cryptococcal clearance is calculated for each patient and the mean slope for patients in a given treatment arm is calculated and termed EFA.[92] • Not routinely performed in clinical laboratories as they are labour-intensive and individual EFAs or quantitative cultures have not yet driven therapeutic decisions. Data are lacking in non-HIV settings.	• **(BIII)** We recommend quantitative yeast CSF cultures be considered for comparison in individual cases (*e.g.*, persistent/ relapsed infection).
Molecular detection	• Reference laboratories can reliably identify the implicated *Cryptococcus* species using molecular tools, such as ITS-based sequencing[93], by MLST[76] and by NGS, but these are limited to research settings • The CSF Biofire® FilmArray® (BioMérieux, Marcy-l'Etoile, France) multiplex PCR Assay for CNS pathogens, includes *C.*	• **(CIIu)** At present, molecular detection for *Cryptococcus* may be done but is less sensitive than CSF CrAg LFA. • **(AIII)** Patients suspicious for CM with a negative CSF Biofire® FilmArray® and patients who test incidentally positive for this despite a low pre-test probability should have a CSF CrAg performed concurrently.

	neoformans/ C. gattii, however, false negative and false positives have been reported.[94-97] This test is increasingly used as a generic CNS diagnostic panel but has less direct utility in those suspicious for CM. This assay is not as sensitive as CrAg in CSF and a negative test does not rule out CNS cryptococcosis. In patients with low yeast burdens, sensitivity may be improved by centrifuging the CSF and performing PCR on the cell pellet. A report suggested this may identify viable yeasts in follow-up CSF examination.[98] • Quantitative PCR allowing identification and quantification is under development[99]	
Histopathology	• Yeast cells may be pleomorphic, typically range from 4–20 µm in diameter, with a single-base narrow budding. • Rare morphological variants have been noted.[100] • Fungal-specific stains such as GMS and PAS stain the cryptococcal cell wall, and capsular stains include mucicarmine, alcian blue. • Mucicarmine is relatively specific for *Cryptococcus* but can be negative in capsule-deficient isolates (*Candida*, *Histoplasma* and *Coccidioides* are negative and *Blastomyces* is only weakly positive). • The Fontana-Masson stain converts melanin and other silver-reducing material in the cryptococcal cell wall to a dark brown to black colour.[101] • A combination of mucicarmine or alcian blue plus Fontana-Masson stains is more sensitive than each of these alone.[102,103]	• **(AIIu)** When *Cryptococcus* is provisionally identified in H&E sections, a combination of capsular stains (e.g., mucicarmine or alcian blue) and cell wall stains (e.g., Fontana-Masson for melanin) should be performed. • **(BIII)** Pan-fungal PCR/ sequencing may be useful when putative cryptococci are visualised in tissue biopsy sections but not cultured, or in the uncommon situation when phenotypic methods provide uncertain results.

	• Nucleic-acid (DNA)-based methods such as broad-range (pan-fungal) PCR/ sequencing may assist in identifying *Cryptococcus* in tissue.[104-106]	
Imaging	• Cerebral imaging by CT or MRI may aid in the diagnosis of CM and its complications, with MRI generally out-performing CT.[107,108] • Classical CNS findings include leptomeningeal and pachymeningeal lesions and enhancement; punctate hyperintensities; dilatation of perivascular spaces (also known as Virchow Robin spaces) sometimes leading to mild mass effect; mass lesions with surrounding oedema (pseudocysts, granulomas and cryptococcomas); and increased ICP with hydrocephalus (ventriculomegaly) and sulcal effacement; lacunar and cortical infarcts with or without evidence of vasculitis; and cerebral atrophy.[6,107-115] • Pulmonary cryptococcosis: Nodules are the most common radiological findings - can be single or multiple; are often peripheral or subpleural. Other findings include segmental or lobar alveolar consolidation, pulmonary infiltrates with ground glass appearance, cavitation which may occur within areas of consolidation or in nodules, hilar and mediastinal lymphadenopathy. Multiple lesions and cavitating lesions are more common in immunosuppressed individuals. Pulmonary cryptococcomas have been reported more commonly in *C. gattii* species complex infections, where they can be especially large.[116,117]	• **(AIIt)** Imaging is an important adjunct to the diagnosis and management of cryptococcosis and the choice of imaging modality depends on clinical presentation and access to the technology. We recommend MRI brain and CT chest where available since they are most sensitive and provide more information, but CT brain and chest x-ray are acceptable alternatives.

Footnote:

BALF Bronchoalveolar lavage fluid

BMD Broth microdilution

CLSI (Clinical and Laboratory Standards Institute)

CGB canavanine-glycine-bromothymol blue

CNS central nervous system

CSF cerebrospinal fluid

CT computer tomography

ECV Epidemiologic cutoff values

ECOFF Epidemiologic cutoff values

EIA Enzyme immunoassay

EFA early fungicidal activity

EUCAST European Committee on Antimicrobial Susceptibility Testing

GMS Gomori methenamine silver

H&E haematoxylin and eosin

LA Latex agglutination

LFA Lateral flow assay

ICP Intracranial pressure

MALDI-TOF MS Matrix-assisted laser desorption/ionization-time of flight

MIC Minimum inhibitory concentration

MRI magnetic resonance imaging

MLST multi-locus sequence typing

NGS next generation sequencing

PAS periodic acid Schiff

PCR Polymerase chain reaction

RLS resource-limited setting

RRS resource-rich setting

196 Supp. Table 4: Adverse drug reactions and symptoms of CM

	Headache	Nausea (N) / Vomiting (V)	Fevers	Confusion	Seizure	Loss of Vision	Loss of hearing	Loss of mobility	Fatigue	Electrolyte change	Cytopenia	Renal impairment	LFT abnormalities	Death	Comment	Prevention/ Treatment
CM	X	X	X	X	X	X	X	X	X					X	• Untreated CM leads to death	• Antifungal therapy • ICP Management • Treat underlying immunosuppression
Raised ICP	X	X	X	X	X	X	X	X						X	• Majority of CM • Early morning headache, N+V • ICP symptoms and signs usually improve with therapeutic LPs	• Proactive therapeutic LP (remove large volume of CSF) (~20-30 ml) • Acutely, may require lumbar drain or ventriculostomy. • Chronically, may require ventricular shunt
CNS C-IRIS	X	X	X	X	X	X	X	X	X					X	• Occurs in ~20-30% of CM, usually in the first 3 months post ART initiation or change in immunosuppressants with transplant recipients	• Avoid starting ART too early (< 4 weeks) • Treat symptomatically • Consider prednisone/prednisolone or dexamethasone
Amb-D		X	X							X	X	X	X		• ~32% experience renal impairment • ~30% have fevers • Significant infusion reactions • Exacerbates anaemia • thrombophlebitis with an associated risk of secondary bacteraemia, hypokalaemia, hypomagnesaemia, and anaemia[118,119]	• Regular IV fluid and K+ and Mg+ replacement • Avoid other nephrotoxic agents • Avoid in renal impairment • Regular antiemetic • Consider premedication to prevent infusion reactions • Frequent complete blood counts (CBCs) and electrolyte measurements are essential.
L-Amb		X	X							X			X		• ~15% experience renal impairment • ~10% have fevers • Exacerbates anaemia	• Regular IV fluid and K+ and Mg+ replacement • Frequent complete blood counts (CBCs) and electrolyte measurements are essential. • Consider less frequent dosing in moderate/severe renal impairment
5-Flucytosine		X	X								X		X		• Leukopenia, Thrombocytopenia, Pancytopenia and Hepatotoxicity are all dose-dependent[120 8,92,121-125] • IV formulation is only stable between 18-25°C and higher temperatures may lead to	• Dose-adjust in renal impairment • Perform TDM 3-5d after change of dose (peak 2-hours after oral dose or 30 mins after IV dose) should be <100 mg/L to minimise toxicity, particularly in renal impairment • In RLS, check FBE/CBC regularly as a surrogate of drug toxicity

															spontaneous conversation to 5-fluoruracil. • Studies in RLS suggest little relational between oral 5-flucytosine levels and fungal clearance or toxicity.	
Fluconazole		X											X		• Equivalent oral and IV dose • Inhibitor of CYP2C19, CYP2C9, CYP34A • Concurrent rifampicin may reduce fluconazole levels • Monotherapy failures seen in induction therapy • Fluconazole exhibits linear pharmacokinetics, and serum and CSF fluconazole levels are strongly correlated.[126]	• Dose-adjust to renal function • Check for drug-drug interaction (*e.g.*, with amiodarone, cyclosporin, phenytoin, rifampacin, sulphonylureas) need careful consideration and with high drug levels may result in QTc prolongation. • Fluconazole susceptibility testing should be performed concurrently on isolates collected at initial presentation and at microbiological relapse. • Fluconazole TDM is not routinely done in clinical practice[127] but may be considered in the critically ill and obese, high MICs, or in hosts with multiple drug-drug interactions. However, there are no accepted therapeutic fluconazole drug levels predictive of treatment efficacy in CM.

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198 ## Supp. Table 5: CM treatment in PLHIV, SOT and non-HIV-non-SOT

199 **Antifungal treatment recommendations for CM in adult PLHIV, SOT, non-HIV-non-SOT**
200 **populations.**

Patient population	First-line agent	SoR QoE	Alternative agents	SoR QoE	Other recommendations
PLHIV	**<u>INDUCTION</u> (2w):** ***L-Amb 3-4 mg/kg daily plus 5-FC 25 mg/kg four times a day** **<u>OR:</u>** **#Single dose L-Amb 10 mg/kg and** **14 days of 5-FC 25 mg/kg four times a day and Fluconazole 1200 mg daily**	**AIIt** **AI**	*Where L-Amb is not available:* ABLC 5mg/kg daily plus 5-FC 25 mg/kg four times a day	BIIt	• **(AIIu)** In-hospital care for the first 1-2 weeks is encouraged to manage the significant early complications seen with CM therapy. • **(AIIu)** The use of Amb-D and L-Amb should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy. • **(AIIu)** Frequent (at least every alternate day) complete blood counts, renal function tests and electrolyte measurements, are recommended to assess for therapy-related bone marrow, nephrotoxicity, and fluid and electrolyte changes. Liver function tests at baseline and at least weekly are recommended. • **(BIII)** Monitoring of 5-flucytosine drug levels, where available and if timely, is recommended. particularly with renal dysfunction. • **(AIIu)** Check for drug-drug interactions and dose-adjust as necessary. • **(CIIu)** Consider performing an LP at the end of the first or second week of induction therapy to check for CSF sterility prior to ART commencement. • **(CIIu)** Consider prolonging
			Where L-Amb and ABLC are not available: Amb-D 0.7-1.0 mg/kg daily plus 5-FC 25 mg/kg four times a day; *<u>OR</u>* Amb-D 1 mg/kg daily and 5-flucytosine 25 mg/kg four times a day for 1 week, followed by fluconazole 1200 mg daily for 1 week.	BI BI	
			Where 5-FC is not available: L-Amb 3–4 mg/kg daily plus fluconazole 800–1200 mg daily <u>OR:</u> ABLC 5 mg/kg daily plus	BIII BIII BI	

			fluconazole 800–1200 mg daily <u>OR:</u> Amb-D 0.7-1 mg/kg daily plus fluconazole 800-1200 mg daily		induction therapy if CSF is persistently culture positive at 2 weeks. • **(CIIt)** Adjunctive recombinant IFN-gamma might be considered for persistently positive CSF yeast cultures in HIV-associated CM for those who have evidence of very poor inflammatory responses or persistently positive cryptococcal CSF culture after prolonged antifungal therapy. • **(DI)** The routine use of high-dose dexamethasone in CM is not recommended. • **(CIII)** A short course of dexamethasone may be considered for specific indications such as symptomatic space-occupying lesions in the CNS with surrounding oedema or mass effect and cerebral vasculitis. • **(DI)** We recommend against the use of adjunctive sertraline and tamoxifen in CM. • **(AIII)** Ongoing advocacy is needed to ensure populations with cryptococcosis have access to the most effective drug regimen(s). Focus on screening, early diagnosis, and effective management of CM in all healthcare systems is essential to reduce mortality, morbidity, and costs. • **(BIIu)** Cease maintenance therapy after 12 months of antifungal therapy in patients aviremic on ART
			Where amphotericin-based therapies are not available: 5-FC 25 mg/kg four times a day and fluconazole 800mg-1200 mg daily	BI	
			Where only fluconazole is available: Fluconazole 800-1200 mg daily	CI	
	<u>CONSOLIDATION</u> (8w): **[%]Fluconazole 400-800 mg daily** *(Note: 800 mg preferred in RLS)*	**AI**	Voriconazole 200 mg twice a day (with TDM)	BIII	
			Posaconazole 300 mg daily (with TDM)	BIII	
			Isavuconazole 200 mg daily	BIII	
			Itraconazole 200 mg twice a day (with TDM)	CIIt	
	<u>MAINTENANCE</u> (12m/ immune restoration):	**AIIt**	Voriconazole 200 mg twice a day (with TDM)	BIII	
			Posaconazole 300 mg daily (with TDM)	BIII	

	Fluconazole 200 mg daily		Isavuconazole 200 mg daily	BIII	with CD4 count >100 cells/mm^3. • **(AIII)** Restart maintenance therapy if CD4 drops <100 cells/mm^3.
			Itraconazole 200 mg twice a day (with TDM)	CIIt	
Solid Organ Transplant recipients	**<u>INDUCTION</u> (minimum 2w):** **L-Amb 3-4 mg/kg daily plus 5-FC 25 mg/kg four times a day**	**AIIt**	*Where L-Amb is not available:* ABLC 5 mg/kg daily plus 5-FC 25 mg/kg four times a day	BIIt	• **(AIII)** We recommend that CNS-based therapy be considered for any disseminated disease or isolation from a sterile site (even in the absence of CNS manifestations). • **(AIII)** Close monitoring of tacrolimus, cyclosporine and sirolimus (TDM) levels and dose reduction of these agents are recommended when azoles are co-administered.[174,175] • **(BIII)** Immunosuppressant doses need to be carefully adjusted to allow effective killing of yeasts but should be reduced slowly to avoid precipitating C-IRIS. Consider a sequential or stepwise reduction of immunosuppressants with careful lowering of corticosteroids early and eliminating mycophenolate before considering reduction of the calcineurin inhibitors because of their direct anticryptococcal activity. • **(CIII)** In a patient treated for cryptococcosis, re-transplantation or a new organ transplant may be considered provided viable yeasts have been cleared from CSF and the patient is asymptomatic after
			Where L-Amb nor ABLC are available: Amb-D 0.7-1.0 mg/kg daily plus 5-FC 25 mg/kg four times a day	BIIt	
			Where amphotericin B-based therapies are not able to be used: 5-FC 25 mg/kg four times a day plus Fluconazole 800-1200 mg daily	CIIt	
	<u>CONSOLIDATION</u> (8w): **Fluconazole 400–800 mg daily**	**AIIt**	Voriconazole 200 mg twice a day (with TDM)	BIII	
			Posaconazole 300 mg daily (with TDM)	BIII	

			Isavuconazole 200 mg daily	BIII	receiving 12 months of anticryptococcal treatment.
			Itraconazole 4200 mg twice a day (with TDM)	CIIt	
	<u>MAINTENANCE (12m):</u> **Fluconazole 200 mg daily**	**AIIt**	Voriconazole 200 mg twice a day (with TDM)	BIII	
			Posaconazole 300 mg daily (with TDM)	BIII	
			Isavuconazole 200 mg daily	BIII	
			Itraconazole 200 mg twice a day (with TDM)	CIIt	
Non-HIV non-transplant	**Induction (minimum of 2w), Consolidation and Maintenance:** **Same options as for CNS/CM treatment in SOT**	**AIIt**	Same as for CNS/CM treatment in SOT	BIIt	• **(BIII)** Consider extending induction therapy for *C. gattii* CNS infection to 4-6 weeks in non-HIV patients.

Footnote:

* preferred in resource-rich settings (RRS)

#: recommended in resource-limited settings (RLS)

% consolidation dose of Fluconazole 800 mg daily is preferred in RLS

TDM: therapeutic drug monitoring

Supp. Table 6: Treatment of pulmonary cryptococcosis and pulmonary cryptococcoma

		Severity of pulmonary cryptococcosis?	
		SEVERE pulmonary cryptococcosis (i.e., multiple lesions, large lesions (≥2 cm), lobar consolidation, cavitation, multi-lobar involvement, hypoxaemic).	**MILD pulmonary cryptococcosis** (i.e., asymptomatic or mildly symptomatic patients or with a solitary small nodule (<2 cm))
Presence of CNS and/or disseminated cryptococcosis?	**Presence of pulmonary cryptococcoma?**	**Preferred therapy/ Comment**	**Preferred therapy/ Comment**
Yes	Regardless	2w induction[&]	2w[&]
No	No	2w induction **(AIIu)**	**(BIIu)** Fluconazole 400-800 mg daily for 6-12 months^**
	Multiple (all <2 cm)	2w induction **(AIIu)**	N/A
	Multiple (and if any are ≥2 cm)	Consider 4-6w induction **(BIIu)**	N/A
	Single – Large (≥2 cm)	2w induction **(AIIu)**	N/A
	Single - Small (<2cm)	2w induction **(AIIu)**	**(BIIu)** Fluconazole 400-800 mg daily for 6-12 months^**
	Yes, but has been completely excised (and is asymptomatic)	N/A	**(BIII)** Fluconazole 400 mg daily for 3-6 months^**

Footnote:
& Consider prolonging induction in select hosts (e.g., SOT recipients, *C. gattii* infection, in non-HIV populations or those with CNS cryptococcomas.
(Single dose 10mg/kg L-Amb with 14d of 5-FC 25.kg 4x a day and Fluconazole 1200 mg daily) has only been trialled in HIV-CM.
^ May consider a shorter duration (e.g. 3 months) in immunocompetent individuals
** If the presence of *Cryptococcus spp.* in respiratory specimens is deemed as airway colonisation after careful evaluation and no treatment is elected, regular follow-up is recommended especially in the setting of future immunosuppression.

Yellow box:

- **Induction:**

  (AIIt) L-Amb 3-4 mg/kg daily and 5-flucytosine 25 mg/kg four times daily

  OR

  (AI) [#]Single dose L-Amb 10 mg/kg and 14 days of 5-FC 25 mg/kg four times a day and Fluconazole 1200 mg daily
- **Consolidation:**

  Fluconazole 400-800 mg daily for 8 weeks,
- **Maintenance:**

  Fluconazole 200 mg daily for 12 months

Supp. Table 7: Urgent, and unmet needs in cryptococcosis

1. Determination of the best strategy for pre-emptive therapy in isolated cryptococcal antigenemia.
2. New, more potent oral anticryptococcal drugs that lead to efficient (rapid) killing of yeasts in CSF as induction alternatives.
3. In-depth understanding of genetic and immunological factors associated with non-HIV CM.
4. Utilisation of quantitative CSF yeast counts or other more practical markers of fungal load, EFA and quantitative PCR calculations into clinical practice, including validation of these in diverse hosts with low and high cryptococcal burden.
5. Interventions to properly diagnose and manage C-IRIS. and optimal management of altered mental status.
6. Clinical trials targeting non-CNS disease, particularly pulmonary cryptococcosis and clinical trials to shorten the length of consolidation and maintenance therapy in CNS cryptococcosis.
7. Clinical trials in resource-rich settings (RRS), particularly in SOT, and non-HIV, non-SOT populations and successful integration with knowledge gained from trials in RLS.
8. Continued advocacy to improve access to optimal antifungal agents and healthcare in RLS, and continued surveillance for antifungal resistance.
9. Improved understanding of the pathogenesis of raised ICP and the critical importance of therapeutic LPs (including access to manometers).
10. The development of clinical breakpoints and ECV/ECOFF for *Cryptococcus spp.,* and rapid diagnostic tools to identify genotypes.
11. Exploring, managing, and improving the substantial mid- and long-term morbidity of cryptococcosis and its impact on patients, their families, and communities.

Supplementary Panels

Supp. Panel 1: Guideline development and processes

Panel 1: Guideline development and processes

This global guideline on cryptococcosis is issued by the European Confederation of Medical Mycology (ECMM) and International Society for Human and Animal Mycoses (ISHAM) in cooperation with the American Society for Microbiology (ASM). This guideline has been endorsed by 75 societies or institutions from 63 countries including Infectious diseases Society of America (IDSA). (see **s32**)

Experts were identified based on their publication activity in the field of cryptococcosis in the previous 5 years, their involvement in patient management, and their distribution over world regions defined by the United Nations. These include physicians and scientists representing the fields of clinical infectious diseases, microbiology, pathology, radiology, research, surgery, paediatrics, haematology, intensive care, dermatology, and pharmacy.

This guideline follows the structure and definitions of previous guidelines on invasive fungal infections which are in accordance with the Grading of Recommendations Assessment, Development and Evaluation (GRADE) and Appraisal of Guidelines for Research & Evaluation (AGREE) systems, as previously described. Strength of recommendation (SOR) and quality of evidence (QOE) are provided **(Panel 1)**.

Keyword search included “cryptococcosis” OR “cryptococcal meningitis” OR “cryptococcal meningitides” OR “cryptococcal meningoencephalitis” OR “pulmonary cryptococcosis” OR “cryptococcal lung disease” OR “cryptococcus” OR “*Cryptococcus neoformans*” OR “*Cryptococcus gattii*”.

As with other ECMM ISHAM guidelines, experts were invited to review a video tutorial and slide deck discussing the scope, objective, and methodology of the guidelines. Assistance to, and supervision of the group were provided by the coordinators (OAC, SR). Documents were shared, updated, and agreed upon as previously described.[128,129] Once all tables were finalised, a core writing group (JP, TB, CC, MC, FH, TH, RO, AS, TS, OL) produced the first draft. Recommendations were consensus-based. If no consensus was found, majority vote was used. Multiple iterations were drafted and reviewed by the core writing group, and a mature manuscript draft was circulated to all the writing group participants for comments and editing in December 2022.

In February 2023, a public consultation phase ensued. Comments received were evaluated, and either dismissed or used to change the manuscript, resulting in a final review by the authors in August 2023. A total of 75 scientific societies from 63 countries across Europe, North, South and Central America, Asia and Africa reviewed and endorsed the guidance document in February through August 2023. See **sFig1** for map and details of endorsing societies (**s32**).

233 ## Supp. Panel 2: Abbreviations

ABLC Amphotericin B lipid complex
AFST Antifungal susceptibility testing
AIDS Acquired Immunodeficiency Syndrome
Amb-D amphotericin B deoxycholate (conventional amphotericin)
ART Antiretroviral therapy
ASM American Society of Microbiology
BALF Bronchoalveolar lavage fluid
BMD Broth microdilution
CBC complete blood count
CLSI (Clinical and Laboratory Standards Institute)
CGB canavanine-glycine-bromothymol blue
CM Cryptococcal meningitis or cryptococcal meningoencephalitis
CNS Central nervous system
CrAg Cryptococcal Antigen
CSF cerebrospinal fluid
CT computer tomography
DNA deoxyribonucleic acid
ECV Epidemiologic cutoff values
ECOFF Epidemiologic cutoff values
ECMM European Confederation of Medical Mycology
EIA Enzyme Immunoassay
EFA early fungicidal activity
EUCAST European Committee on Antimicrobial Susceptibility Testing
FBC full blood count
FDA U.S. Food and Drug administration
GOF gain-of-function
GM-CSF granulocyte-macrophage colony-stimulating factor
GMS Gomori methenamine silver
H&E haematoxylin and eosin
HIV Human immunodeficiency virus
HPF high power field
ICP Intracranial pressure
IFN-gamma Interferon-gamma
IRIS Immune reconstitution inflammatory syndrome
ISHAM International Society of Human and Animal mycology
IV Intravenous
JAK Janus kinase
K+ Potassium
L-Amb Liposomal amphotericin
LA Latex agglutination
LFA Lateral flow assay
LFT Liver function test
M-CSF macrophage colony-stimulating factor
MALDI-TOF MS Matrix-assisted laser desorption/ionization-time of flight
MBL mannose-binding lectin
Mg+ Magnesium
MIC Minimum inhibitory concentration
MRI magnetic resonance imaging
MLST multi-locus sequence typing
NGS next generation sequencing
OI opportunistic infection
OP Opening pressure
PAS periodic acid Schiff
PCR Polymerase chain reaction
PIIRS post-infectious inflammatory immune response syndrome
PLHIV Persons living with HIV
PRR pathogen recognition receptors
PTX3 pentraxin-3
QTc corrected QT interval
RCT randomised controlled trial

RLS resource-limited setting
RNA ribonucleic acid
RRS resource-rich setting
SNPs Single nucleotide polymorphisms SNP
SOT Solid organ transplant
STAT signal transducer and activator of transcription
TDM: therapeutic drug monitoring
TNF-alpha Tumour necrosis factor-alpha
U&E Urea and electrolytes
USA United States of America
VP ventriculoperitoneal
W weeks
WBC white blood cells
WES whole exome sequencing
WGS whole genome sequencing
WHO World health organization

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Supp. Panel 3: Definitions

Asymptomatic cryptococcal antigenemia: Persons who test positive for blood CrAg LFA but apparently have no clinically overt symptoms or signs of cryptococcal meningoencephalitis (CM). It is important to note that ideally, an LP with CSF CrAg and CSF analysis are needed to thoroughly exclude CM.

Blood cryptococcal antigen: In this document, we use this combined term to refer to the detection and quantification of cryptococcal antigen in whole blood, serum, or plasma. Cryptococcal antigen was previously detected by latex agglutination (LA) but in many settings, the most commonly performed method is with the lateral flow assay (LFA), of which the IMMY test has been best studied, and the most robust. The comparative quantification of LA to LFA ~10.

Clinical relapse: Recrudescence of clinical symptoms after a period of quiescence. This is usually characterised by a recurrence of symptoms and signs at the original site of infection (e.g., headache(s), seizures or decreased conscious level) and/or positive cultures after initial normalisation of signs, symptoms, and negative cultures.

Central nervous system (CNS) cryptococcosis includes CM (see below) and any CNS involvement including CNS cryptococcomas, CNS mucinous or gelatinous pseudocysts, vasculitis or infarcts.

Colonisation: presence of a microorganism on/in a host, with growth and multiplication of the organism, but without interaction between host and organism. Practically, this may be difficult to distinguish from cryptococcal infection.

Cryptococcal meningitis or cryptococcal meningoencephalitis (CM) is the most common and most severe of the cryptococcal syndromes. CM presents with a constellation of severe headaches, fevers, nausea, vomiting, visual or hearing loss, confusion, seizures and altered consciousness. Reduced visual acuity, double vision, hearing impairment and bilateral sixth nerve palsies are common false localising signs related to raised intracranial pressure (ICP). Patients with CM may also have minimal or no symptoms.

Cryptococcoma is a mass lesion attributed to any *Cryptococcus* species, usually diagnosed by radiological imaging (most commonly in the brain or lungs) but can also occur in the skin or other organs. It can appear in any size and can occur as a single or as multiple lesions. Histologically this is usually a walled-off granuloma with inflammatory infiltrates and yeast resembling cryptococci. Cryptococcomas are more commonly seen with *C. gattii* species complex infections[117,130] where they may be large.

Cryptococcosis or cryptococcal disease includes all forms of clinical syndromes and disease caused by the *C. neoformans / C. gattii* species complexes.

Cryptococcosis-associated IRIS (C-IRIS), involving the CNS is one of the more frequent and severe forms of IRIS. Paradoxical CNS C-IRIS is typically diagnosed in a PLHIV co-infected with CM who despite an initial period of improvement with antifungal therapy and appropriate therapeutic LPs, develops recurrent symptoms following commencement of ART, usually without evidence of microbiological (culture-positive) relapse **(Figure 2).** In contrast, unmasking C-IRIS occurs in a PLHIV with no known symptoms of cryptococcosis at time of ART commencement, but who later develops new symptoms and signs of cryptococcosis. C-IRIS is not limited to PLHIV. There are no diagnostic biomarkers, thus this is a diagnosis of exclusion. Typically, CSF examination at presentation of C-IRIS reveals pleocytosis consistent with an inflammatory state, CSF cultures are usually negative (but need not be) and

the opening pressure (OP) may be raised. Brain imaging may reveal T2 hyperintensity of the leptomeninges and cerebral oedema. Worsening pneumonia or pulmonary infiltrates may be noted on chest imaging.

Culture-positive (microbiological) relapse: CSF cultures are positive for *Cryptococcus* after previously documented negative CSF cultures.

Disseminated cryptococcosis can involve any organ, following haematogenous spread. This is usually associated with fungaemia (blood culture positive) but need not be.

Non-pulmonary, non-CNS cryptococcosis. This may arise in one of two situations: (a) Haematogenous dissemination to a particular organ (*e.g.*, kidney or skin) without CNS and pulmonary involvement (b) Primary cutaneous (skin) cryptococcosis may be due to direct inoculation.

Persistent cryptococcal infection is characterised by ongoing symptoms and signs, AND positive cryptococcal cultures from the disease site after 4 weeks of optimal (polyene-based) antifungal therapy at an established effective antifungal dose. As fluconazole monotherapy results is associated with poorer CSF clearance and slower clinical improvement, the period before which persistence is established is extended.

Pre-emptive therapy is primarily defined as treatment of asymptomatic cryptococcal antigenemia.

Primary prophylaxis refers to the use of fluconazole in persons who are at risk for but with no history or current clinical evidence of cryptococcosis. Current evidence is limited to PLHIV.

Pulmonary cryptococcosis refers to cryptococcal involvement of the lungs and/or pleura. This is often subclinical and is commonly diagnosed incidentally on imaging. Disease severity may be classified as mild or severe although these criteria are based on clinical judgment and are not validated. In principle, asymptomatic or mildly symptomatic patients, or those with a solitary small nodule (<2 cm) are considered to have mild disease while those with multiple lesions, large lesions (≥2 cm), lobar consolidation, cavitation, multi-lobar involvement, and hypoxemia are considered to have severe disease.

Raised intracranial pressure (ICP) is common in CM and is usually defined by an opening pressure of ≥20 cm of CSF. Symptoms and signs include morning headaches, nausea, vomiting, double vision, decreasing conscious levels, seizures, and coma. Some may be asymptomatic.

Therapeutic lumbar puncture (LP). Therapeutic LPs are key to the management of raised ICP and an important arbiter of successful CM management. This involves measurement of the opening pressure and removal of sufficient CSF (usually 20-30 ml) to reduce the pressure to 50% of OP and/or to a normal pressure of ≤20 cm of CSF (recorded as a closing pressure). Patients with raised ICP usually report symptomatic relief post-procedure (e.g., headache subsides) and conscious levels may also improve. Generally, LPs should be repeated daily until symptoms resolve and OP has normalised. Daily consideration of symptoms, signs and ICP should determine need for and duration of decompressive measures.

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Supp. Panel 4: Navigating the treatment sections of these guidelines

The treatment sections form the core of these guidelines. CM is the most severe and most commonly encountered form of cryptococcosis and is the syndrome for which we have the best, albeit incomplete understanding – management of CM thus forms the "spine" to the treatment recommendations. The treatment section includes with a discussion on antifungal agents, a table of core management principles in CM, a detailed description of CM in HIV, followed by CM in SOT, and CM in non-HIV, non-transplant hosts, and adjunctive therapies trialed in CM. Brief sections on the treatment for pulmonary and non-pulmonary/non-CNS cryptococcosis follow. Common abbreviations and important definitions are found in **sPanel 2 (s25)** and **sPanel 3 (s27)**, respectively.

There is a myriad of important specific management issues encountered in CM and cryptococcosis – these are discussed in the following sections, and include management of ICP, timing of ART, cryptococcal resistance, clinical relapse and IRIS, biomarkers. Underlying immunodeficiencies in unexpected cryptococcosis, treatment of *C. gattii*, management of cryptococcomas, significance or non-*neoformans*, non-*gattii Cryptococcus*, and management of cryptococcosis in pregnancy and paediatrics.

Therapy is addressed initially by summarising 10 management principles of CM and our key recommendations in **Panel 5.** These include (1) screening and risk stratification; (2) utilisation of optimal fungicidal drugs in CM for induction therapy; (3) minimisation of drug toxicity; (4) management of raised ICP; (5) excluding and managing underlying immunosuppression; (6) utilisation of appropriate consolidation and maintenance therapy; (7) optimal timing for commencement/recommencement of ART; (8) remaining vigilant for clinical relapse; (9) management of culture-positive persistence/ relapse; and (10) careful diagnosis and management of C-IRIS. While much of the data and recommendations are derived from PLHIV, many of these principles are applicable to CM in non-HIV settings. These principles and the strategy for CM management (including clinical relapse) are incorporated into a clinical algorithm, in **Figure 2**.

Historical and contemporary clinical treatment trials in HIV-associated CM including the unique challenges and the divergent clinical practices in RRS and RLS are discussed, as means of justification and rationale for our treatment recommendations. The choice of induction therapy is evolving and is particularly complex. We chose to provide the recommended induction antifungal therapy for HIV-associated CM based on the availability of antifungal agents available, recognising that drug-access is variable and dynamic – this is summarised in **Figure 3.** A list of preferred and alternative antifungal therapy for CM in PLHIV, SOT and non-HIV-non-SOT patients is found in **Figure 4** and listed **sTable 5 (s18)**. Both **Figures 3 and 4** are best read in conjunction with the accompanying text in the relevant CM sections to appreciate the evidence and discourse leading to these key recommendations.

The many overlapping symptoms and signs of the complications seen in CM including adverse drug reactions associated with antifungal therapy are shown in **sTable 4 (s16).** A brief description of the antifungal agents used in cryptococcosis can be found in **sPanel 5 (s30).** There is a complex interplay of pulmonary cryptococcosis, cryptococcoma and *C. gattii* disease, requires careful consideration. We summarise the treatment recommendations for these often overlapping conditions in **sTable6 (s22)**.

Importantly, **Figure 1 summarises the entire suite of first-line recommendations for all the cryptococcosis syndromes in adults** discussed in this guideline. This figure illustrates the critical importance of determining the extent of disseminated disease (be it CNS or non-CNS) as some may require aggressive induction therapy. Only mild pulmonary disease and skin cryptococcosis due to direct skin inoculation are suitable for oral fluconazole therapy. Subtleties in the management of *C. gattii* infection and cryptococcomas are also presented. Finally, detailed text is included in the appendix **(s36** onwards**)** to allow a more complete understanding of a complex infection and some illustrative images are found in **sFig 2 (s35)**

Supp. Panel 5: Antifungal drugs in cryptococcosis

Amphotericin B-based agents, 5-flucytosine and fluconazole are the cornerstones to successful cryptococcosis therapy. *Cryptococcus* species are intrinsically resistant to echinocandins.[131,132] Conventional deoxycholate amphotericin B (Amb-D) and 5-flucytosine were developed in the 1950s, while fluconazole was developed in the early 1990s. Development of lipid-associated formulations of amphotericin B, began just over 30 years ago, largely driven by the dose-limiting nephrotoxicity seen with **Amb-D** and its high rates of infusion reactions.[133,134] (See **sTable4** (**s16)** for common adverse drug reactions). Unfortunately, cost factors continue to limit access to the full suite of essential antifungal agents. Further complicating the construction of universal guidelines is: (1) the defining CM studies in RRS were performed more than 30 years ago; and (2) more recent studies have been performed in RLS settings where, until recently Amb-D is the only polyene option available. (See discussion on drug equity).

Generally, lipid-associated amphotericin B formulations are less nephrotoxic than Amb-D.[133,134] The most widely used lipid-associated formulations is **liposomal amphotericin** (**L-Amb**, AmBisome®, Gilead, Foster City, CA, USA). Other formulations include the less commonly used **amphotericin B lipid complex** (**ABLC**, Abelcet®, Enzon Pharmaceuticals, Bridgewater, New Jersey, USA) and the now rarely used **amphotericin B colloidal dispersion** (**ABCD**, Amphocil®/Amphotec®, Three River Pharmaceuticals, Cranberry Township, Pennsylvania, USA) due to its high rates of infusion reactions.[135,136] Other liposomal amphotericin B products exist but each exhibits different pharmacokinetics, tissue dispersion and toxicity, due to differences in physiochemical characteristics.[134,137] Indeed, mere production differences can result in significant differences in pharmacokinetics, toxicity, and efficacy,[137,138] thus, data from one agent cannot be directly translated to another.

Of the lipid-formulations, L-Amb is the smallest in size and has a negative charge which facilitates its escape from the mononuclear-phagocyte system, resulting in a high maximum concentration and much larger area-under-the-curve compared with Amb-D. It has a long terminal half-life of approximately 152 hours.[134] L-Amb is the least nephrotoxic and as such the most widely available and used lipid-based formulation in RRS.[134,139,140] A new oral lipid nanocrystal formulation of Amphotericin (MAT2203, Matinas Biopharma) was recently explored in a small open-label study in CM-HIV with normal conscious level.[141]

5-Flucytosine is rapidly absorbed, and has good bioavailability and CSF penetration, but is also principally excreted in the kidneys, thus requires renal-dose adjustment.[125] It may be administered orally or intravenously agent, with equal effect.[124] The daily weight-based dose of 100 mg/kg is divided into 3-4 doses – in this guideline we recommend 25 mg/kg four times daily. 5-Flucytosine is generally well tolerated as part of the two-week induction therapy but can lead to cytopenia and hepatotoxicity.[120 8,92,121-125] There are general guidelines for dose reduction with reduced renal function.[142,143] Therapeutic drug monitoring (TDM) should be considered in those with renal dysfunction, if available and timely to minimise toxicity,[127,144] but many will use surrogate measurements for toxicity with frequent CBCs to detect cytopenias (see **sTable 4, s16**).

Fluconazole has excellent oral bioavailability and good CSF penetration; it is also renally excreted, and requires renal dose-adjustment.[145] Fixed doses are usually used, though for children, obese or critically ill patients, weight-based dosing of 12 mg/kg loading followed by 6 mg/kg daily, accompanied by TDM is advised.[146] Fluconazole is a potent inhibitor of cytochrome P450 thus drug-drug interactions (e.g., with amiodarone, cyclosporin, phenytoin, rifampicin, sulphonylureas) need careful consideration and with high drug levels may result in QT prolongation (see **sTable 4, s16**),

Alternative antifungals such as triazoles (e.g., itraconazole[120,147-151], voriconazole[152-159], posaconazole[160-162] and isavuconazole[163,164]) have been used sporadically in cases of cryptococcosis with some success. Indications include fluconazole intolerance, resistance, or drug interactions or toxicities. TDM is critical with itraconazole, posaconazole, or voriconazole. Importantly, it is not known whether any of these newer triazoles are more efficacious than fluconazole for cryptococcosis and none are readily accessible in RLS. Where an alternative agent is required, voriconazole, posaconazole and isavuconazole are recommended over itraconazole for reasons of drug-drug interactions and CNS penetration.

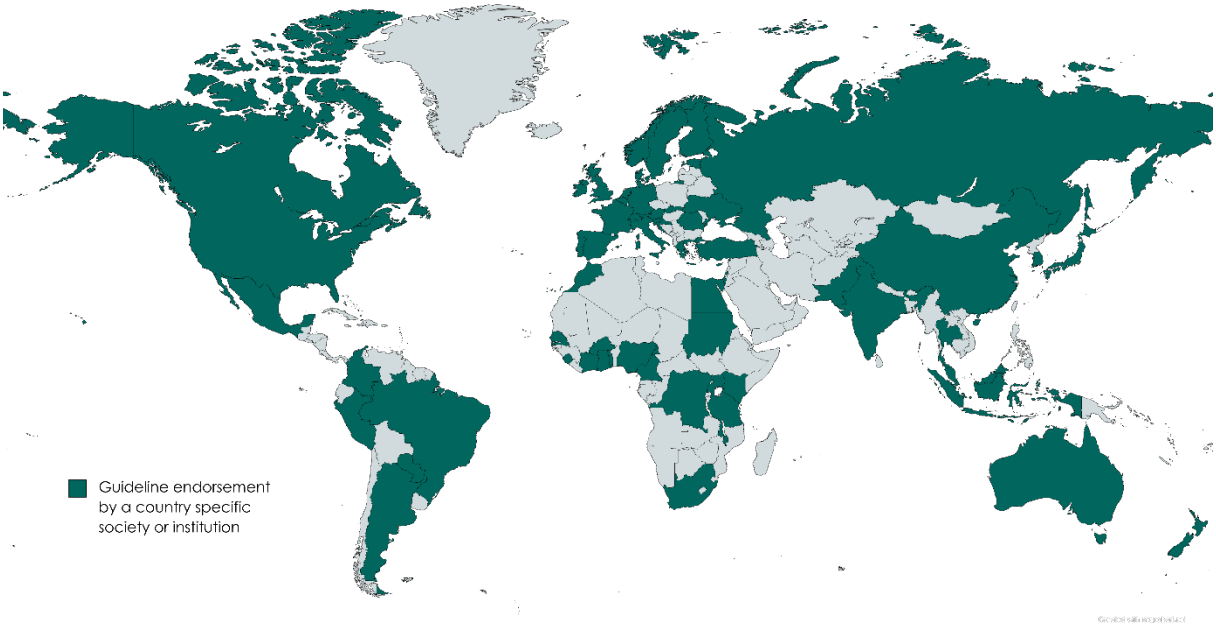
Looking into the future, of the **newer antifungals** in the pipeline, manogepix and fosmanogepix (a small-molecule inhibitor of the conserved fungal Gwt1 protein) show promising *in vitro* activity against *Cryptococcus*. but clinical experience is lacking.[165-167] Another Gwt1 inhibitor APX2039 showed exceptional activity in a rabbit model.[168] Neither olorofim, an orotidine (dihydroorotate

dehydrogenase inhibitor)[169] nor ibrexafungerp, an oral triterpenoid glucan synthetase inhibitor[170] have any activity against *Cryptococcus in vitro.*

Supplementary Figures

Supp. Figure 1: Map of societies and institutions endorsing the ECMM cryptococcosis guidelines.

Countries with scientific societies or institutions endorsing the ECMM-ISHAM-ASM Guideline for Cryptococcosis.



Intercontinental (6)

- GAFFI Global Action Fund for Fungal Infections
- ICHS International Immunocompromised Host Society
- ISAC International Society of Antimicrobial Chemotherapy
- MMSA Medical Mycology Society of the Americas

Africa (4)

- FIDSSA Federation of Infectious Diseases Societies of Southern Africa
- MMSN Medical Mycology Society of Nigeria
- PAMWG Pan Africa Mycology Working Group
- SAHCS Southern African HIV Clinicians Society

Asia (19)

- ISHAM Asia Fungal Working Group
- CMEA Medical mycology society of Chinese Medical Education Association
- Mycology Committee of Chinese Society for Microbiology (CMS)
- IDAT Infectious Diseases Society of Thailand
- Indonesian Neurological Association
- IDST Infectious Disease Society of Taiwan
- INSHAM Indonesia Society for Human and Animal Mycology
- IPO-EASID International Public Organization "Euro-Asian Society for Infectious Diseases"
- ISMM Indian Society of Medical Mycologist

- Japanese Association for Infectious Diseases
- KSCM The Korean Society of Clinical Microbiology
- KSID Korean Society of Infectious Diseases
- MMIDSP Medical Microbiology & Infectious Diseases Society of Pakistan
- Mycological Society of China
- MyICID Malaysian Society of Infection Control and Infectious Diseases
- Pediatric Infectious Society of Thailand
- Singapore Society of Infectious Disease
- TAS Thai AIDS Society
- TMMF Thai Medical Mycology Forum

Europe (30)

- AEM Asociación Española de Micología, Sección de Micología Médica (Spanish Society of Mycology)
- AGIHO Arbeitsgemeinschaft Infektionen in der Hämatologie und Onkologie (Working Group on Infections in Haematology and Oncology of the German Society of Haematology and Oncology)
- ANRS Emerging Infectious Diseases; L'ANRS Maladies infectieuses émergentes
- ASPOMM Portuguese Association of Medical Mycology
- Belgian Hematology Society
- BSMM British Society for Medical Mycology
- DSI Danish Society of Infectious Diseases
- DZIF German Center for Infection Research
- EFISG-ESCMID Fungal Infection Study Group from the European Society of Clinical Microbiology and Infectious Diseases
- EPMyN European Pediatric Mycology Network
- Estonian Society for Laboratory Medicine
- FIMUA Federazione Italiana Micologia Umana e Animale
- HSMM Hellenic Society of Medical Mycology
- IACMAC Russian Interregional Association for Clinical Microbiology and Antimicrobial Chemotherapy
- IFS Irish Fungal Society
- Medical School of the European University
- Nordic Society for Medical Mycology
- NVMy Dutch Society for Medical Mycology
- ÖGMM Austrian Society for Medical Mycology
- PENTA Paediatric European Network for Treatment of AIDS Child Health Research (Fungal Infection Working Group)
- RSMMM Romanian Society of Medical Mycology and Mycotoxicology
- Section for Mycology of Croatian Microbiological Society
- SEIFEM sorveglianza Epidemiologica Infezioni nelle Emopatie
- SITA Società Italiana Terapia Antiinfettiva
- Slovak Society of Chemotherapy
- Slovenian Society for Medical Mycology

- Swiss Society of Microbiology
- Turkish Society of Medical Mycology

Middle East (1)

- ISMM Israel Society for Medical Mycology

Central/South America, Caribbean (9)

- ACIN Colombian Association of Infectious Diseases
- API Asociación Panamericana de Infectología
- Asociación Mexicana de Infectología y Microbiología clínica
- Asociación Mexicana de Micología Médica
- Mexican Academy of Dermatology
- SADI Sociedad Argentina de Infectología
- SBI Sociedade Brasileira de Infectologia (Brazilian Society of Infectious Diseases)
- SPEIT Peruvian Society of Infectious Diseases
- SPM Paraguayan Society of Microbiology

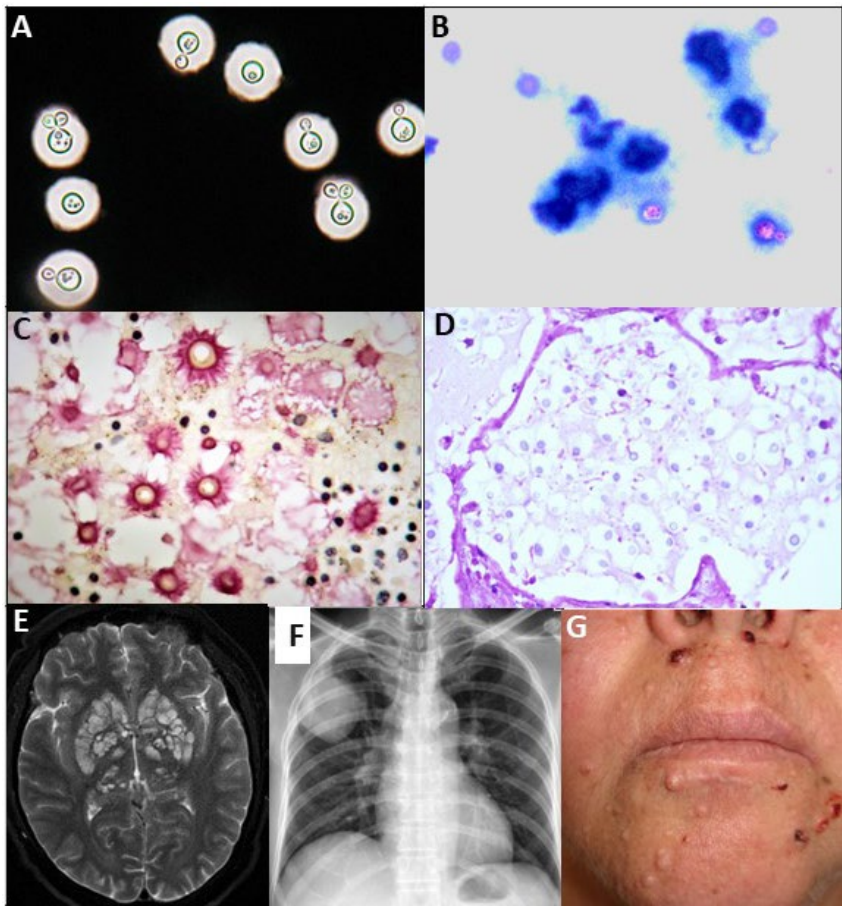
North America (3)

- AMMI Association of Medical Microbiology and Infectious Disease Canada
- IDSA Infectious Diseases Society of America
- American Society of Microbiology

Oceania (4)

- ANZMIG Australia and New Zealand Mycology Interest Group
- ASHM Australasian Society for HIV, Viral Hepatitis and Sexual health Medicine
- ASID Australasian Society for Infectious Diseases
- Sydney-ID University of Sydney Infectious Disease Institute

Supp. Figure 2: Illustrations of diagnostic and clinical features of cryptococcosis



(From top, left to right) (A). India Ink stain of cerebrospinal fluid (CSF) showing encapsulated budding yeast. (B) Giemsa stain of blood smears collected from a febrile patient with underlying liver cirrhosis showing budding yeast. (C) Mucicarmine stain of brain tissue showing multiple yeast organisms. (D) H&E stain of lung tissue obtained by transbronchial biopsy from a renal transplant recipient who presented with fever and shortness of beath, showing numerous encapsulated budding yeasts packed in the alveoli. Fungal cultures from the lung tissue grew *C. neoformans.* (E) T2-weighted scan of MRI brain of an apparently immunocompetent man who presented with headaches and mental confusion showing multiple hyperintense cyst-like lesions in bilateral basal ganglia suggestive of gelatinous pseudocysts seen in cerebral cryptococcosis. CSF analysis revealed encapsulated budding yeast, cryptococcal antigen lateral flow assay was positive, and cultures grew *C. gattii.* (F) Chest X-ray of a previously well man who presented with low grade fevers and productive cough, demonstrating a large right upper lobe mass later. Transbronchial biopsy of the lung tissue confirmed this as a pulmonary cryptococcoma and tissue cultures grew *C. gattii*. (G) Multiple skin nodules in a patient with biopsy proven disseminated skin cryptococcosis.

Introduction

The encapsulated yeasts *Cryptococcus neoformans* and *Cryptococcus gattii* are heterobasidiomycetous yeasts now known to be represented by species complexes. The first reported patient with cryptococcosis dates back to 1894.[171] Until the late 1970s, reports of human infections were rare, but with a rising immunocompromised host population, associated in particular with the HIV/AIDS pandemic, *Cryptococcus* yeasts have become a commonly encountered pathogen. The World Health Organization fungal pathogen list in 2022 ranked *Cryptococcus neoformans* as one of its top priority pathogen. [172]

Cryptococcus is mainly acquired in childhood (>80% seroprevalence at 5 years old), leading to asymptomatic pneumonia. Some dormant yeasts can persist in the lung parenchyma and lymph nodes. The immune system can control the dormant yeasts for a long period of time (latency), but finally reactivation and dissemination can occur upon immunosuppression leading to classical symptoms. In parallel, in immunocompromised patients cryptococcosis can occur following recent exposure to cryptococci without the latency phase.[173]

Due to a unique predilection for the central nervous system (CNS), *Cryptococcus* is the commonest cause of fungal CNS disease in the world today. The highest burden of disease is encountered in low- and middle-income countries (LMIC), especially sub-Saharan Africa[2], where HIV/AIDS remains the dominant driving factor. In resource-rich settings (RRS), cryptococcosis is increasingly seen in solid organ transplant (SOT) recipients, in non-HIV/ non-SOT populations, and less so in persons living with HIV (PLHIV). In France, PLHIV and non-HIV now bear equal burden of cryptococcosis.[174] Cryptococcosis most often involves the lung and/or CNS but disseminated disease may impact any organ. Importantly, *Cryptococcus* can also cause disease in apparently healthy hosts.

Morbidity from cryptococcosis remains substantial ranging from 10-70% depending on the disease syndrome and severity, underlying predisposing conditions of the host, and the healthcare system in which the patient is managed.[2,175-177] Only three classes of drugs are available for therapy: polyenes (conventional deoxycholate Amphotericin B (Amb-D), and various lipid-associated amphotericin B products of which liposomal amphotericin (L-Amb) is most commonly used), azoles (particularly fluconazole) and the pyrimidine analogue, 5-flucytosine. Multiple carefully controlled interventional trials, conducted primarily in resource-limited health care settings of late, have strengthened the evidence base for treatment and has led to improved outcomes.[92,121,123,178-180]

Several published diagnostic and management guidelines for cryptococcosis have framed its pertinent clinical issues.[1,7,181-189] With this background, the following is a comprehensive synthesis and update of cryptococcosis management guidelines issued by The European Confederation of Medical Mycology (ECMM), endorsed by 75 societies and institutions globally. Details on the formation of the guideline group and processes in writing the guidelines are described (see **sPanel 1, s24**). We utilised the Grading of Recommendations Assessment, Development and Evaluation (GRADE) and Appraisal of Guidelines for Research & Evaluation (AGREE) systems, as previously described (see **Panel 1**).[128,129]

This guidance document is intended to facilitate clinical decision making, while simultaneously providing an overview of areas of uncertainty in the field of cryptococcosis and challenges that we expect to face over the coming decades. While aspects of cryptococcosis management may evolve over time, the principles in these guidelines provide a comprehensive framework supported by the latest evidence. This document has been prepared by physicians and scientists representing the fields of clinical infectious diseases, microbiology, pathology, radiology, research, surgery, paediatrics, haematology, intensive care, pulmonology, dermatology, and pharmacy. Drawn from experts from all regions of the world, the guideline writing group gives voice to expertise and challenges from both resource-rich settings (RRS) and resource-limited settings (RLS) in a globally relevant document.

Epidemiology of cryptococcosis

Populations at risk

Evidence:

Cryptococcosis accounts for significant morbidity and mortality globally. In 2014, an estimated 223,100 cases of CM occurred globally, with 73% of these in sub-Saharan Africa.[2] The estimated pooled prevalence of cryptococcal antigenaemia calculated from studies performed in PLHIV (majority with CD4 < 100 cells/mm^3) in RLS from 2007-2018 was 8% (ranging from 1.7-33%).[190] In resource-rich settings, this ranged from 3.9-5%.[191,192] While classical at-risk patient populations include PLHIV, SOT recipients and individuals with other immunosuppressive conditions or receiving immunosuppressant drugs, apparently immunocompetent hosts also experience cryptococcosis.[177,193-195] Further risk factors include chronic pulmonary disease, use

of the novel oral tyrosine kinase inhibitor such as ibrutinib[19], fingolimod (a sphingosine1-phosphate receptor modulator used in multiple sclerosis[25]), idiopathic CD4 lymphopenia[29,30], cirrhosis[11-13] and the presence of anti-cytokine autoantibodies[41,196,197]. Other rare primary immunodeficiencies have been recognised in seemingly immunocompetent persons with cryptococcosis (discussed under exploration for an underlying immunodeficiency).

Recommendations:

- **(AIII)** Cryptococcosis should be considered in any patient presenting with suspicious symptomatology, and microbiology, regardless of their immune status.
- **(AIII)** Exclusion of an underlying immunodeficiency (including performing HIV serology and CD4 T-cell count) is recommended in all patients presenting with cryptococcosis.

Yeasts causing cryptococcosis

Evidence:

Discovered in the environment and as a cause of human disease in the 1890s, the basidiomycetous yeast initially known as *Saccharomyces neoformans*[198] was designated as *Cryptococcus neoformans* in 1935.[199] Four serotypes, A, B, C, D, were originally identified, based on cryptococcal capsular antigens. Serotypes A and D were named *Cryptococcus neoformans* (later separated into *C. neoformans* var. *grubii* and *C. neoformans* var. *neoformans,* respectively), and serotypes B and C were named *C. neoformans* var. *gattii.*[200,201] In the mid-1970s, Kwon-Chung identified the sexual stage of these heterothallic basidiomycetes, consisting of two teleomorphs: *Filobasidiella neoformans* (A, D) and *Filobasidiella bacillispora* (B, C).[202]

C. neoformans and *C. gattii* are now well established as species complexes. *C. neoformans* is the predominant causative agent of cryptococcosis in PLHIV, while *C. gattii* more commonly causes disease in apparently immunocompetent hosts. While both can cause a similarly wide repertoire of cryptococcosis syndromes, *C. neoformans* has a predilection for CNS disease while *C. gattii* is often associated with pulmonary disease and can frequently cause cryptococcomas both in the CNS and lungs.[116,203,204]

Molecular taxonomy has revolutionised fungal nomenclature. Recently, it was proposed that *C. gattii* (serotype B and C) be split into sibling species: *C. gattii* sensu stricto (genotype VGI), *C. bacillisporus* (genotype VGIII), *C. deuterogattii* (genotype VGII), *C. tetragattii* (genotype VGIV) and *C. decagattii* (genotype VGVI), and some recent isolates (genotype VGV) were proposed

as a new sibling species.[205] Multiple genotypes have also been identified within *C. neoformans* serotype A, namely VNI, VNII, VNBI/VNBII and VNIII (a hybrid of VNI and VNIV) and serotype D (genotype VNIV)[206] with the proposed name of *C. deneoformans*. Thus, *C. bacillisporus/C. tetragattii/C. decagattii* are mostly associated with PLHIV while *C. gattii s.s./C. deuterogattii* are more often encountered in apparently immunocompetent hosts.[207,208] There is ongoing debate within the scientific community as to whether *C. neoformans* and *C. gattii* should be considered as species complexes or divided into separate clades (species) with corresponding changes in nomenclature for clinical use.[70,71] It may be important in the future to differentiate these separate clades as the antifungal susceptibility may differ. For the purposes of this guideline, we primarily refer to *C. neoformans* and *C. gattii* as species complexes.

Furthermore, nomenclature within the genus *Cryptococcus* has been extensively revised as molecular relationships have become better understood. This taxonomic revision placed the environmental yeasts, *Cryptococcus adeliensis*, *C. albidus, C. diffluens*, *C. humicola*, *C. laurentii*, *C. liquefaciens*, *C. luteolus, C. macerans,* and *C. uniguttulatus,* into the new genera *Papiliotrema, Naganishia,* and *Filobasicium.*[209] These fungi are only occasionally pathogenic in humans.

While host immunity clearly influences the clinical outcomes of cryptococcosis, the impact of cryptococcal strain virulence is only just emerging.[210-216] There have been associations found in linking "burst" groups, sequence type (ST) groups, and even genome wide association studies (GWAS) to mortality. For example, *C. neoformans* genotype VNB has been associated with higher mortality in PLHIV with CM.[217] Conversely, genotype VNII strains were associated with a favourable outcome in a cohort of renal transplant recipients with CM.[218]

Recommendations:

- **(AIIu)** Differentiation into *C. neoformans* and *C. gattii* species complexes is recommended in standard clinical laboratories.

Protean presentations and variable outcomes

Evidence

Cryptococcosis most often presents as cryptococcal meningoencephalitis (CM), with one or more symptoms of severe headaches, fevers, nausea, vomiting, double vision, visual or hearing loss, confusion, and seizures. Patients with severe CM may manifest with reduced levels of

consciousness or coma, which constitute a medical emergency. Reduced visual acuity, double vision, hearing impairment and bilateral sixth nerve palsies, are common false localising signs related to raised intracranial pressure (ICP). Rarely, patients with CM may have minimal or no symptoms. Pulmonary cryptococcosis, in contrast, is most often subclinical and found incidentally on imaging of the lung. Occasionally, it presents with respiratory symptoms such as cough, chest pain, shortness of breath and fever. Rarely, *C. neoformans* can colonize and be isolated from respiratory secretions of patients with chronic obstructive lung disease or asthma without disease.[219,220] Disseminated cryptococcosis can involve any organ of the body such as skin, liver, spleen, kidney, and the prostate gland, thus a detailed examination is required. Rarely, osteoarticular involvement or pyomyositis may be presenting features – these require appropriate histological and microbiological tests. (See next section on diagnostics).

Prompt, optimised treatment of CM and management of its complications is critical to avoid morbidity and a potentially fatal outcome. Mortality from CM remains unacceptably high even in recent treatment trials performed in RLS, ranging from 24-47% at 10 weeks[121,123,179,221]. Contemporary studies show a 2-year mortality of 56-60%.[222,223] There remains a significant gap in treatment outcomes between RLS and resource-rich settings (RRS).

The ten key outcome determinants of cryptococcosis are discussed in **sTable 2 (s8)** and include: (1) The yeast burden of the infection, (2) Cryptococcosis syndrome type, (3) *C. neoformans* complex vs. *C. gattii* complex, (4) Poor host immune response, (5) Host comorbidities, (6) Delays in presentation or diagnosis, (7) Optimal antifungal management, (8) Management of raised ICP, (9) Development of cryptococcosis-associated immune reconstitution inflammatory syndrome (C-IRIS) and (10) Healthcare systems in which the patient is managed.

Recommendations

- **(AIIt)** All patients with suspected or confirmed cryptococcosis (including cryptococcal antigenemia) require careful clinical assessment for CNS, pulmonary and other body site involvement. Investigations for disseminated disease should include: (1) lumbar puncture (LP) with measurement of CSF opening pressure (OP), glucose, protein, cell counts, microscopy, culture, and quantification of CSF CrAg; (2) quantification of blood CrAg , and cultures of blood, sputum (or other respiratory specimens) and/or of other affected sites; (3) ideally, brain imaging (preferably MRI) and chest imaging (preferably CT).

- **(AIII)** Clinicians need to carefully interface prognosis and cryptococcal guidelines into a bedside interpretation and plan for the individual.

Diagnostic methods

Here we discuss the complementary diagnostic modalities used to establish the diagnosis, extent, severity, and prognosis of cryptococcosis. See **sTable 3 (s10)** for a summary of this section.

Microscopy, culture, and antifungal susceptibility testing

Evidence:

Traditionally, visualisation of India Ink positive encapsulated yeasts on microscopy of CSF provided a rapid and useful and specific diagnostic test. However, this has largely been superseded by the cheap, and more sensitive and accurate lateral flow antigen assay (LFA). On **microscopy**, cryptococci appear as globular encapsulated yeast cells, ranging in size from 4–20 µm in diameter, with narrow-based single budding. They can be identified rapidly in a colloidal medium of India ink mixed with CSF when present at concentrations of at least 10^3-10^4 colony forming units (CFU) of yeasts per millilitre of CSF. Diagnostic sensitivities of 84–98% have been reported in high prevalence settings with expert laboratory experience[67,68] but in patients with low yeast burdens, they may be as low as 42%. The sensitivity of Gram stain is similar to India ink when performed by an experienced technologist[68], and Calcafluor white staining is often used.

Culture from a body site such as blood, CSF, sputum, urine, skin, or (BALF) bronchoalveolar lavage fluid, is considered as the gold standard for the diagnosis of cryptococcosis[69] but may be slow (48 hours to a week) for growth. *Cryptococcus* is easily grown on standard fungal culture media (e.g., Sabouraud's dextrose agar) and bacterial culture media. Colonies can usually be observed after incubation at 30–35°C in aerobic conditions for 48 to 72 hours. Antibacterial agents, preferably chloramphenicol, can be added to the media to minimise bacterial contamination. However, *Cryptococcus* does not grow in the presence of cycloheximide at the concentration used in certain selective fungal media. *Cryptococcus* colonies typically appear as opaque, white to creamy colonies with variable degree of mucoid appearance which is related to the polysaccharide capsule. Chromagar or Bird Seed Agar (Staib's Medium) may be used for sputum and urine culture.

C. neoformans and *C. gattii* isolates can be distinguished by growth on canavanine-glycine-bromothymol blue (CGB) agar, though not always performed in routine diagnostics. Colonies of *C. gattii*, which is resistant to canavanine, assimilate glycine and develop a blue colour whereas those of *C. neoformans* do not although reports of accuracy vary.[224-227] Delayed or very weak CGB positive tests may result from rare interspecies hybrids[228,229]. Rare non-pathogenic cryptococcal species may test positive on CGB agar.[225] Accurate species identification of fungal colonies into *C. neoformans* and *C. gattii* species complexes can be achieved reliably and rapidly by MALDI-TOF MS.[72-76]

***In vitro* susceptibility testing**.

Broth microdilution (BMD) using CLSI (Clinical and Laboratory Standards Institute) or EUCAST (European Committee on Antimicrobial Susceptibility Testing) protocols are reference methods to determine the minimum inhibitory concentration (MIC) of *Cryptococcus* against amphotericin B, 5-flucytosine or azoles. The method for performing BMD varies between CLSI and EUCAST with different microdilution trays, media constituents, inoculum size, and time and method of MIC determination, and BMDs are not usually practical in clinical laboratories.[230]

There has been no consistent relationship between pre-treatment MICs for amphotericin B, 5-flucytosine or fluconazole and the mycological or clinical outcomes.[81,231,232] *Cryptococcus* species are considered intrinsically resistant to echinocandins.[131,132]

Susceptibility thresholds for *Cryptococcus* yeasts are based on epidemiological cut-off values, abbreviated as ECV (CLSI) or ECOFF (EUCAST) to distinguish non-wild type yeasts with potentially detectable acquired resistance to a particular drug. ECV/ECOFF are method-dependent and vary by *Cryptococcus* molecular type. Importantly, ECV/ECOFF do not necessarily predict clinical success. Clinical breakpoints in cryptococcosis are extremely hard to generate as it requires treatment outcome data, pharmacokinetic and pharmacodynamic data in addition to MIC distribution – this is only available for Amphotericin B (1 ug/mL by EUCAST).

EUCAST (v3, 2022-01-18) provides ECOFFs for *C. neoformans:* Amphotericin B (tentatively 1 ug/mL), Posaconazole 0.5 ug/mL, Voriconazole 0.5 ug/mL. For *C gattii*, the ECOFF for Amphotericin B is tentatively 0.5 ug/mL, Posaconazole 1 ug/mL. There is no ECOFF for fluconazole, flucytosine, or isavuconazole using EUCAST reference method.

CLSI endorsed ECVs for fluconazole are available for a few genotypes of *Cryptococcus.* This is 8 mcg/mL for *C. neoformans VNI,* 16 ug/mL for *C. gattii VGI* and 32 ug/mL for *C. deuterogattii*).[230] Species identification must be done by molecular typing or MALDI-TOF MS for valid use.

Commercial MIC techniques using automated systems (Vitek®2, BioMérieux) and colorimetric assays (SensiTitre™ YeastOne™, ThermoFisher Scientific) show good agreement with BMD for the three antifungal classes, but none have clinical breakpoints or method-specific ECVs. Agreement with results using agar-based diffusion gradient systems (Etest®, BioMérieux; MTS, Liofilchem) is more variable, and worst with amphotericin B.[78-81]

Recommendations:

- **(AIIt)** Gram stain/ India ink preparation/ calcafluor white stain of centrifuged CSF may be effective in patients suspected of CM with or without CrAg availability.
- **(AIII)** Fungal culture from blood, CSF, sputum, urine, BALF, or other relevant clinical specimens is recommended in all patients suspected of cryptococcosis for genus and species identification.
- **(AIIt)** Blood cultures should be performed using standard bacterial or fungal culture media.
- **(AIIu)** Body fluids and other clinical specimens are ideally plated on Sabouraud dextrose agar without cycloheximide and incubated at 30°C in aerobic conditions for 7-10 days.
- **(AIIt)** Species identification into *C. neoformans* and *C. gattii* species complexes by MALDI-TOF MS is recommended.

  (AIIt) All primary cryptococcal isolates should have a baseline fluconazole, and ideally flucytosine MIC determined or stored for future comparative testing of relapse isolates.

Serologies (Antigen and Antibody)

Evidence:

The development of serological tests for cryptococcal polysaccharide antigens (CrAg) has improved access to rapid diagnosis globally. This was the first fungal antigen test used widely in clinical practice and remains the most robust serological test in mycology. Serum/ plasma and CSF CrAg can be detected using latex agglutination test (LA), enzyme immunoassay (EIA) or the more convenient and now the most widely used lateral flow assay (LFA). The diagnostic performance of these commercial tests in CSF and blood vary[82], and have been extensively reviewed, including in the WHO guidelines supplement of 2018.[1] LA kits for CrAg are reportedly 93–100% sensitive and 93–98% specific, compared with 85.2-99% and 97%, respectively for EIA.[233] Dilution of sera or CSF may be necessary to remove the pro-/post-zone effect leading to false-negatives.[82,83]

LFA has the advantage of functioning as a point-of-care assay with overall sensitivities and specificities of 97–100% and 95–99%, respectively.[84-88] Furthermore, the LFA CrAg is accurate, simple to use, and cheap. This test forms an important part of the new screening and diagnostic algorithms. Notably, commercial CrAg LFAs may have reduced ability to detect all pathogenic *Cryptococcus* species[65,89], although this does not apply to the IMMY CrAg® LFA test (Oakland, Oklahoma, US) which has been most thoroughly evaluated. False negatives due to pro-/post-zone effect are rare but have been reported to be overcome by dilution.[90]. Not all commercial LFAs are able to detect *C. gattii* disease. The detection of *C. tetragattii* may be missed with CryptoPS (Biosynex, Illkirch-Graffenstaden, France) and cryptococcal antigen LFA (Dyanmiker Biotechnology, Tianjin, China).[89] While the CrAg tests appear to have a lower sensitivity in HIV-negative patients (85% compared to 100% in PLHIV for LFA), LFA still appears to be significantly better than the EIA,[91] although there has not been a systematic evaluation of CrAg LFA in *C. gattii* disease.. The diagnostic performance of these tests in CSF and blood have been extensively reviewed, including in the WHO guidelines supplement of 2018.[1]

Cryptococcal antibodies are not used in current clinical practice and its application in research is further limited by its lack of standardisation both in terms of target protein, isotype, and assay methodology. Old studies performed in paediatric populations without known cryptococcal infection suggests its presence increases with increasing age[234], which led to idea that cryptococcal antibodies are positive in most people by the time they reach adulthood though this

has not been vigorously tested. A pre-HIV era study suggested the presence of cryptococcal antibodies at CM presentation as a good prognostic indicator[235], and a small unmatched cohort study suggested that patients who later developed cryptococcosis compared to those who did not, had a lower IgM to glucuronoxylomannan pre-transplant[236]. Both are likely confounded by the presence of immunosuppressive conditions, where antibodies are less likely to be detectable.[237,238] Post-infection, cryptococcal antibodies are known to persist for at least a few months and there is no correlation to antigen level or to clinical course.[239] Despite the lack of clinical utility, a recent retrospective study of cryptococcal diagnostics ordered in a large national commercial laboratory in the USA showed an unexpectedly large number of cryptococcal antibody testing was ordered, perhaps mistakenly, with a surprisingly low (0.1%) positivity rate.[240] However, there is ongoing work on the value of humoral immunity in cryptococcosis pathogenesis research.[241]

Recommendations:

- **(AIIt)** Detection and quantification of CrAg in blood and CSF by LFA is recommended as a first-line rapid diagnostic test, in both RLS and RRS. Any positive result regardless of titre level should be appropriately investigated.
- **(CIII)** CrAg tests can be performed on other body fluids, including BALF and urine, but their sensitivity and specificity in relation to these sample types have not been assessed rigorously.
- **(DIII)** Cryptococcal antibody testing is not recommended for diagnosis or management of cryptococcosis at this time.

Molecular-based Testing

Evidence:

Publicly available, quality-controlled, reference sequences for gene sequence-based identification can be found in the ISHAM Barcoding Database (www.its.mycologylab.org) and the WI-KNAW database (Westerdijk Fungal Biodiversity Institute, https://www.wi.knaw.nl/). Furthermore, reference laboratories can reliably identify the implicated *Cryptococcus* species using molecular tools, such as ITS-based sequencing[93] from tissue biopsies and by multi-locus sequence typing (MLST).[76] Next generation sequencing (NGS) offers a method for species identification, detection of drug resistance mutations and epidemiological information. Although

its use is currently limited by cost, complexity of the data and lack of a comprehensive database, future implementation into clinical practice, particularly in outbreak settings may be feasible.[242-246]

Molecular-based studies of cryptococcosis have accelerated in the last decade. At present, whole genome sequences (WGS) are available for hundreds of cryptococcal strains. These have improved precision in taxonomy, and epidemiology and in understanding of pathobiology.[210,212,214,217] In the future, understanding the genetic make-up of a fungal isolate might contribute to determining outcomes of infection. On the other hand, host genomic sequencing efforts continue with the aim of identifying cryptococcus-associated susceptibility genes.[62,247]

A molecular-based diagnostic PCR test for CNS infections, the Biofire® FilmArray® (BioMérieux, Marcy-l'Etoile, France) multiplex PCR Assay for CNS pathogens, includes *C. neoformans/ C. gattii*. In patients with low yeast burdens, sensitivity may be improved by centrifuging the CSF and performing PCR on the cell pellet. False negative results have been reported.[94-97]This assay is likely not as sensitive as CrAg in CSF and has not been reliably adopted into clinical practice. Quantitative PCR allowing identification and quantification is under development.[99]

Recommendations:

- **(CIIu)** At present, molecular detection for *Cryptococcus* may be done but is not required in routine clinical practice.
- **(AIII)** Patients suspicious for CM with a negative CSF Biofire® FilmArray® and patients who test incidentally test positive for this despite a low pre-test probabllity should have a CSF CrAg performed concurrently.

Imaging

Evidence:

Cerebral imaging by computer tomography (CT) or magnetic resonance imaging (MRI) may aid in the diagnosis of CM and its complications, with MRI generally out-performing CT.[107,108] Importantly, CT and even MRI brain may be normal in patients with confirmed CM.[110] Thus, normal CNS imaging does not exclude the diagnosis of CM. The range of radiological findings likely reflects the following: yeast burden; disease onset and progression of cryptococcosis in individual patients; the phase of illness and treatment; the host's age; the presence and degree of immunodeficiency (particularly HIV, which has its own CNS radiological manifestations); comorbidities; and lastly advances in radiology.

Classical findings include leptomeningeal and pachymeningeal lesions and enhancement; punctate hyperintensities; dilatation of perivascular spaces (also known as Virchow Robin spaces) sometimes leading to mild mass effect; mass lesions with surrounding oedema (pseudocysts, granulomas and cryptococcomas); and increased ICP with hydrocephalus (ventriculomegaly) and sulcal effacement; lacunar and cortical infarcts with or without evidence of vasculitis; and cerebral atrophy.[6,107-115] A few case series have also focused on non-HIV CM-related radiology findings: most report a similar spectrum of findings. One study that described ependymitis and choroid plexitis as additional non-HIV-associated CM findings.[248] MRI findings can worsen during successful and unsuccessful therapy and evidence of new or persistent inflammation may be a sensitive marker for IRIS.

Nodules are the most common radiological findings in pulmonary cryptococcosis; they can be single or multiple and are often peripheral or subpleural. Other findings include segmental or lobar alveolar consolidation, pulmonary opacities with ground glass appearance, cavitation which may occur within areas of consolidation or in nodules, and hilar and mediastinal lymphadenopathy. Rarely, a halo sign, a reverse halo sign and occasionally, pleural effusion(s) or thickening have been reported.[249-260] Multiple lesions and cavitating lesions are more common in immunosuppressed individuals. Pulmonary cryptococcomas have been reported more commonly in *C. gattii* infections, where they can be especially large and abut on critical structures.[116,117] Case series describing PET/CT changes in pulmonary cryptococcosis support the presence of solitary and multiple nodules,[261] with varying standardised uptake values (SUV) in early images,[261,262] and no differences in SUV noted between immunocompetent and immunocompromised individuals.[261] Spiculation, convergence of peripheral vessels, pleural indentation and lymphadenitis are other features noted on PET/CT.[262]

Involvement of other organs may be seen on imaging as masses. Skeletal cryptococcosis is rare but can occur in any bone, particularly the vertebrae, skull and femur, and is often diagnosed radiologically as osteomyelitis.[263] MRI of the spine can reveal spinal arachnoiditis manifesting with nerve root clumping and enhancement.[264] Cryptococcosis has been noted as an incidental finding on PET/CT scans, with hypermetabolic lesions seen particularly in rare locations such as the duodenum, adrenal glands and within muscles,[265-269] or as a late infective complication in those who are under PET/CT surveillance for malignancy.[270]

Recommendation:

- **(AIIt)** Imaging is an important adjunct to the diagnosis and management of cryptococcosis and the choice of imaging modality depends on clinical presentation and access to the technology. We recommend MRI brain and CT chest where available since they are most sensitive and provide more information, but CT scan of brain and chest x-ray are acceptable alternatives.

Pathology

Evidence:

Cryptococcosis is occasionally diagnosed by histopathology alone[67], particularly when a cryptococcoma masquerades as a malignancy in lung, brain, bone, or other organs. Multiple, multi-morphic skin lesions,[271,272] which occur predominantly in immunosuppressed patients are typical of disseminated cryptococcosis and are readily amenable to diagnostic biopsy or culture.

Tissue sections stained with haematoxylin and eosin (H&E) reveal individual cryptococci or large numbers of cryptococcal cells with pale blue nuclei, surrounded by empty spaces which originally contained capsular material and may provide the first clue to diagnosis. Yeast cells typically range from 4–20 μm in diameter and can vary in size within a single body site. Rare morphological variants have been noted in research settings, such as very large titan cells (~50–100 μm in diameter), with thick capsules and cell walls[273], pseudohyphae, yeasts with multiple budding, chains of budding yeasts and acapsular cryptococci[100]

The associated host inflammatory response varies from scattered lymphocytes in the presence of many extracellular cryptococci, to an intense granulomatous reaction with giant cells with or without intracellular cryptococci and/or fibrosis, or sometimes, no visible cryptococci.[274,275] Minimal inflammatory responses are typical in both cerebral cryptococcosis and in pulmonary disease in advanced HIV.[274,276,277]

Fungal-specific stains such as Gomori methenamine silver (GMS) stain cryptococcal cell wall and the typical narrow necks associated with budding are easily seen. The fungal capsular stains mucicarmine, alcian blue and periodic acid Schiff (PAS) also stain the cryptococcal capsule in tissue sections. Mucicarmine is relatively specific for *Cryptococcus* (*Candida*, *Histoplasma* and *Coccidioides* are negative and *Blastomyces* is only weakly positive). The

Fontana-Masson stain converts melanin and other silver-reducing material in the cryptococcal cell wall to a dark brown to black colour.[101] Both Fontana-Masson and GMS have the advantage of staining acapsular cryptococci.[100,278] Individually, mucicarmine, alcian blue or Fontana--Masson stains identify most cryptococci in tissue sections but a combination of mucicarmine or alcian blue plus Fontana-Masson stains is more sensitive than each of these alone.[102,103]

Immunohistochemical methods using anticapsular monoclonal antibodies can distinguish between *C. neoformans* and *C. gattii* with 100% accuracy[279] but have not been developed for commercial use. Nucleic-acid (DNA)-based methods such as broad-range (pan-fungal) PCR, typically targeted to regions of the (multicopy) rRNA gene, or probe hybridisation, can definitively identify *Cryptococcus* both in fresh tissue and, with reduced sensitivity but high specificity, in formalin-fixed paraffin-embedded tissue (FFPE)[104,105] (reviewed by Rickerts et. al.[106]).

Recommendations:

- **(AIIu)** When *Cryptococcus* is provisionally identified in H&E sections, a combination of capsular stains (e.g., mucicarmine or alcian blue) and cell wall stains (e.g., Fontana-Masson) should be performed.
- **(BIII)** Panfungal PCR/ sequencing may be useful when putative cryptococci are visualised in tissue biopsy sections but not cultured, or in the uncommon situation when phenotypic methods provide uncertain results.

Screening/Primary Prophylaxis/Pre-emptive Therapy

Evidence:

Screening may be performed on serum or plasma from patients at risk of cryptococcosis to presumptively identify unrecognised infection in apparently well (or unrecognised symptomatic or pre-symptomatic) persons. This is currently performed using LFA CrAg tests. Its intended outcome is to identify and successfully refer those strongly suspected of CM predominantly (and less so other cryptococcosis syndromes) for physician-directed therapy or risk-reduction strategies. Existing supportive evidence for screening individuals for cryptococcosis is limited to PLHIV. **Primary prophylaxis** refers to the use of fluconazole in persons who are at risk for but with no history of or prior evidence of cryptococcosis. **Pre-emptive therapy** is defined as

834 treatment of antigenemic, asymptomatic, high-risk patients, usually with fluconazole, and is
835 sometimes termed “early treatment”.

836 The role of **fluconazole primary prophylaxis** was established in the 1990s.[280-283] Of late, a
837 prospective, randomised, double-blinded study in Uganda enrolled HIV-infected, ART-naïve
838 adults with CD4 of <200 cells/mm^3, randomising to 200 mg fluconazole three times a week
839 versus placebo, and stopping when the CD4 count reached at least 200 cells/mm^3.[284] Of 1,519
840 persons enrolled, 19 developed cryptococcosis (one in the fluconazole group and 18 in the
841 placebo group; (p=0.0001); adjusted HR (aHR) 18.7 (95% CI 2.5 -140.7). This study suggests
842 that one case of cryptococcal disease may be prevented by treating 45 patients with baseline
843 CD4 <200 cells/mm^3. The REALITY trial was an open-label, factorial-design trial of children and
844 adults with CD4 <100 cells/mm^3, which demonstrated that an enhanced package of
845 antimicrobial prophylaxis, including trimethoprim-sulfamethoxazole, isoniazid-pyridoxine,
846 fluconazole, azithromycin, albendazole at varying durations, reduced overall death rates at 24
847 weeks and 48 weeks. Rates of tuberculosis and cryptococcal infection were significantly
848 reduced[285], as were deaths from cryptococcosis.[286] A sub-study of the REALITY trial measured
849 serum CrAg retrospectively on stored specimens at baseline and week 4 and, as expected,
850 found the incidence of new cryptococcal disease and death from cryptococcosis to be
851 significantly higher in those with cryptococcal antigenaemia.[287] A 2018 Cochrane systematic
852 review of nine clinical trials, with a total 5,426 participants, examined the role of primary
853 antifungal prophylaxis for cryptococcal disease in HIV-infected adults and children. The
854 conclusion of this review was that prophylaxis probably reduces the risk of developing
855 cryptococcosis as well as of deaths due to cryptococcal disease (RR 0.29, 95% CI 0.11-0.72).[288]
856 However, with widespread availability of CrAg testing, primary prophylaxis is likely limited only in
857 settings without access to CrAg.

858 In RLS, however, access to healthcare for ensuing investigations, the cost of antifungal therapy,
859 and cost of LFAs, often necessitate rationalisation of resources. Strategies used include limiting
860 screening to an asymptomatic subpopulation of PLHIV usually by a CD4 threshold (e.g. CD4
861 <200 cells/mm^3); not quantifying the degree of antigenaemia (i.e. only positive or negative
862 CrAg), or possibly performing further investigations such as LP, in only a subset of those who
863 are antigenaemic above a certain titre (e.g. titre >1:80) based on data that these patients are
864 more likely to have CNS involvement.[63] Resource constraints usually also limit the
865 investigations to LP and CSF examination and chest XR, leaving a significant subset of
866 antigenaemic persons without definable disease but who still require pre-emptive antifungal

therapy. Importantly, asymptomatic cryptococcal antigenaemia in PLHIV is an independent predictor of death during the first 12 weeks of ART[289].[290]

Trials examining the dose and/or duration of pre-emptive therapy continue and are generally supportive. A meta-analysis on the impact of pre-emptive fluconazole reported on observational studies and most were of small cohorts.[291] Studies using pre-emptive fluconazole for asymptomatic serum CrAg positivity in PLHIV continue to report failures.[292] A trial on adjunctive sertraline in asymptomatic antigenemia was stopped due to harm.[293] Prospective studies examining other pre-emptive therapeutic strategies for asymptomatic antigenemia are presently ongoing (such as the addition of single dose L-Amb 10 mg/kg to fluconazole 800 mg daily (ACACIA NCT03945448) and addition of 5-flucytosine to fluconazole (EFFECT)).

Recommendation:

- For adult PLHIV with CD4 <200 cells/mm^3 (ART-naive or after a period of ART discontinuation):
  - **(AI)** Perform a blood CrAg by LFA for the screening of cryptococcosis, and determine the CrAg titre if positive.
  - **(AIIt)** All patients with cryptococcal antigenaemia should be carefully assessed and investigated for cryptococcosis (see section above) and treated as appropriate.
  - **(AIIu)** In asymptomatic cryptococcal antigenaemic PLHIV without clinical cryptococcosis after thorough investigation (including at least a LP), fluconazole 1200 mg daily for 2 weeks (when ART may be initiated), followed by 800 mg daily for 8 weeks, and 200 mg thereafter for about 6 months is recommended. (Guidance may be updated contingent on results of prospective trials.)
  - **(BI)** In clinical settings where CrAg LFA screening is not available (despite WHO's strong recommendations), universal primary prophylaxis with fluconazole 100 mg daily in PLHIV with CD4 count <200 cells/mm^3 is recommended.
- Other (non-HIV) populations:
  - **(DIIu)** Routine blood CrAg screening, primary prophylaxis and pre-emptive therapy are not currently recommended in non-HIV populations.

Principles of treatment and navigating the guidelines

The bulk of our understanding of cryptococcosis is in the management of CM. This is complex and requires clinical expertise and committed personnel. The following section focuses firstly on the main agents used in cryptococcosis, followed by the evidence and recommendations on antifungal therapy in CM in different populations (*i.e.*, PLHIV, SOT and non-HIV, non-transplant), adjunctive therapy and antifungal therapy for pulmonary and non-pulmonary, non-CNS cryptococcosis. The subsequent section addresses the many complications of cryptococcosis.

We summarise ten management principles of CM and our key recommendations in **Panel 5.** These include (1) screening and risk stratification; (2) utilisation of optimal fungicidal drugs in CM for induction therapy; (3) minimisation of drug toxicity; (4) management of raised ICP; (5) excluding and managing underlying immunosuppression; (6) utilisation of appropriate consolidation and maintenance therapy; (7) optimal timing for commencement/ recommencement of ART; (8) remaining vigilant for clinical relapse; (9) management of culture-positive persistence/ relapse; and (10) careful diagnosis and management of C-IRIS. While much of the data and recommendations are derived from PLHIV, many of these principles are applicable to CM in non-HIV settings. These principles and the strategy for CM management (including clinical relapse) are incorporated into a clinical algorithm, in **Figure 2**.

Historical and contemporary clinical treatment trials in HIV-associated CM including the unique challenges and the divergent clinical practices in RRS and RLS are discussed, as means of justification and rationale for our treatment recommendations. The choice of induction therapy is evolving and is particularly complex. We chose to provide the recommended induction antifungal therapy for HIV-associated CM based on the availability of antifungal agents, recognising that drug-access is dynamic – this is summarised in **Figure 3.** Accompanying this is **Figure 4,** which provides a list of preferred and alternative antifungal therapy for CM in PLHIV, SOT and non-HIV-non-SOT patients. Both **Figures 3 and 4** are best read in conjunction with the accompanying text in the relevant sections to appreciate the evidence and discourse leading to these key recommendations. **sTable 4 (s16)** demonstrates the many overlapping symptoms and signs of the complications seen in CM including adverse drug events associated with antifungal therapy.

sTable 6 (s22) details the treatment recommendations for the often overlapping conditions of pulmonary cryptococcosis, pulmonary cryptococcoma and *C. gattii* disease. Importantly, **Figure 1** summarises the entire suite of first-line recommendations for all the cryptococcosis syndromes in adults discussed in this guideline. This figure illustrates the critical importance of determining the extent of disseminated disease (be it CNS or non-CNS) as these may require aggressive induction therapy. Only mild pulmonary disease and localised skin cryptococcosis due to direct skin inoculation are suitable for initial oral fluconazole therapy. Subtleties in the management of *C. gattii* infection and cryptococcomas are also outlined.

Antifungal agents

The evidence for choice and duration of antifungal treatment for cryptococcosis is predominantly based on data from HIV-CM co-infected populations in RLS. Amphotericin B-based agents, 5-flucytosine and fluconazole are the cornerstones to successful cryptococcosis therapy. **Conventional deoxycholate amphotericin B (Amb-D)** and 5-flucytosine were developed in the 1950s, while fluconazole was developed in the early 1990s. Development of lipid-associated formulations of amphotericin B, began just over 30 years ago, largely driven by the dose-limiting nephrotoxicity seen with Amb-D and its high rates of infusion reactions.[133,134] Unfortunately, cost factors continue to severely limit access to these agents and the majority of CM prospective clinical treatment trials are based on Amb-D. Further complicating the construction of universal guidelines is: (1) the defining CM studies in RRS were performed more than 30 years ago; and (2) more recent studies have been performed in RLS settings where Amb-D is the only polyene option available. Other significant toxicities seen with Amb-D include thrombophlebitis with an associated risk of secondary bactaeremia, hypokalaemia, hypomagnesaemia, and anaemia with an average fall in haemoglobin of 2.3 g/dL over 2 weeks.[118,119] Caution is advised with use of Amb-D in the presence of renal impairment, or those at high risk of renal impairment. Pre-hydration and regular potassium and magnesium replacement therapy are recommended to prevent renal toxicity, and frequent complete blood counts (CBCs) and electrolyte measurements are essential.

The most common lipid-associated formulation is **liposomal amphotericin** (**L-Amb**, AmBisome®, Gilead, Foster City, CA, USA). Other formulations include **amphotericin B lipid complex** (**ABLC**, Abelcet®, Enzon Pharmaceuticals, Bridgewater, New Jersey, USA) and **amphotericin B colloidal dispersion**/ cholesteryl sulphate complex (**ABCD**, Amphocil®/Amphotec®, Three River Pharmaceuticals (Kadmon Corporation), Cranberry

Township, Pennsylvania, USA).[135,136] There is a long list of other liposomal amphotericin B products (reviewed in Adler-Moore et al.[137]). Importantly each lipid-associated compound exhibits different pharmacokinetics, tissue dispersion and toxicity, due to differences in their physiochemical characteristics such as particle size, lipid formulation, and electrostatic charge.[134,137] Indeed, even between liposomal agents of the same composition, mere production differences can result in significant differences in pharmacokinetics, toxicity, and efficacy.[137,138] Thus, data from one agent cannot be directly translated to another.

Generally, lipid-associated amphotericin B formulations are less nephrotoxic than Amb-D.[133,134] ABLC is the largest of these compounds, is recognised by macrophages and sequestered into the spleen and liver, with a high volume of distribution and a lower circulating serum level compared to Amb-D.[134] ABCD has been associated with more infusion reactions than Amb-D, thus its usage is limited. L-Amb is the smallest in size and has a negative charge. This facilitates its relative escape from the mononuclear-phagocyte system, with a high maximum concentration and much larger area-under-the-curve compared with Amb-D. It has a long terminal half-life of approximately 152 hours.[134] L-Amb is the least nephrotoxic and as such the most widely available and used lipid-based formulation, with less than 10% of infused drug being excreted into urine and faeces.[134,139,140] A new oral lipid nanocrystal formulation of Amphotericin (MAT2203, Matinas Biopharma) was recently explored in a small open-label study in CM-HIV, limited to those with normal Glascow coma scale.[141]

5-flucytosine is rapidly absorbed, has good bioavailability, and penetrates CSF and multiple organs, and is also principally excreted in the kidneys.[125] It may be used as an oral or intravenous agent.[124] The daily weight-based dose of 100 mg/kg is divided into 3-4 doses – in this guideline we recommend 25 mg/kg four times daily, but this requires renal dose adjustment.

5-flucytosine is generally well tolerated as part of the two-week induction therapy but can lead to cytopenia and hepatotoxicity.[120 8,92,121-125] Of note, the intravenous formulation is only stable between 18 and 25°C and higher temperatures may lead to spontaneous conversation to 5-fluoruracil. Therapeutic drug monitoring (TDM) should be considered especially in those with renal dysfunction if available, preferably on days 3-5 of therapy or 3-5 days after any alteration in dose[127,144]. Peak concentration (2-hours after oral dose or 30 minutes after IV dose) should be < 100 µg/ml, to minimise toxicity. Of note, all recent studies of 5-flucytosine (using the oral formulation) in HIV-associated CM have been conducted in RLS where TDM was not available. Here, studies examining 5-flucytosine levels retrospectively suggest little relationship between 5-flucytosine levels and fungal clearance or toxicity.[124,294] In fact, many clinicians will perform

frequent blood counts, using development of cytopenias as a surrogate for drug toxicity due to unavailability of drug levels or simply turnaround time on drug levels is too long for clinical decisions. In the event of severe cytopenia, 5-flucytosine should be withheld. It should be noted that many patients of African descent have underlying neutropenia and recover spontaneously with repeat testing.[295]

Fluconazole has excellent oral bioavailability and good CSF penetration; it is also renally excreted, and requires renal dose-adjustment.[145] It is usually given orally but intravenous formulations exist (with equivalent dosing). Fixed doses are usually used, though for children, obese or critically ill patients, weight-based dosing of 12 mg/kg loading followed by 6 mg/kg daily, accompanied by TDM is advised.[146] Fluconazole is a potent inhibitor of cytochrome P450 thus drug-drug interactions (*e.g.*, with amiodarone, cyclosporin, phenytoin, rifampacin, sulphonylureas) need careful consideration and high drug levels may result in QTc prolongation.

Alternative antifungals such as triazoles (e.g. itraconazole[120,147-151], voriconazole[152-159], posaconazole[160-162] and isavuconazole[163,164]) have been used sporadically in cases of cryptococcosis with some success. Indications include fluconazole intolerance, resistance, or drug interactions or toxicities. TDM is critical if itraconazole, posaconazole, or voriconazole are used. There is more historical experience with itraconazole in cryptococcosis compared with the newer triazoles.[147] However, the significant intra- and inter-patient variability in pharmacokinetics, drug-drug interactions, and the difficulty in predicting steady-state plasma concentrations from the dosage, makes consistent drug exposure challenging for intraconazole[296], thus other triazoles are preferred. Importantly, it is not known whether any of these newer triazoles are more efficacious than fluconazole for cryptococcosis, and none are readily accessible in RLS.

Looking into the future of the **newer antifungals** in the pipeline, manogepix and fosmanogepix (a small-molecule inhibitor of the conserved fungal Gwt1 protein) show promising *in vitro* activity against *Cryptococcus*, but clinical experience is lacking.[165-167] Another Gwt1 inhibitor APX2039 showed exceptional activity in a rabbit model.[168] Neither olorofim, an orotidine (dihydroorotate dehydrogenase inhibitor)[169] nor ibrexafungerp, an oral triterpenoid glucan synthetase inhibitor[170] have any activity against *Cryptococcus in vitro.*

HIV-associated cryptococcal meningoencephalitis (CM)

Evidence:

Induction treatment. Multiple studies support the combination of Amb-D plus 5-flucytosine as the induction treatment of choice in HIV-associated CM. This approach was first trialled in PLHIV by van der Horst and colleagues, who found addition of 5-flucytosine to Amb-D, showed a strong trend to improved CSF sterility at 2 weeks (60% vs. 51%, p=0.006).[120] Use of 5-flucytosine was also associated with a decreased frequency of relapse.[149] Subsequently, Brouwer et al., reported that this combination cleared cryptococci (measured as early fungicidal activity, EFA), more rapidly from the CSF than either Amb-D alone or Amb-D plus fluconazole.[92] An older, prospective cohort study in France, found that this combination was also associated with the lowest rate of mycological treatment failure or death.[8] Importantly, a survival advantage at day 70 of the combination of **Amb-D 1 mg/kg daily plus 5-flucytosine 25 mg/kg four times a day** over Amb-D alone was demonstrated in a large prospective study conducted in Vietnam.[121]

In **RLS**, the side effects and difficulty of monitoring and safely managing 2 weeks of Amb-D induction treatment led to phase 2 studies, exploring alternative regimens. Fluconazole monotherapy, even at doses up to 1200 mg daily, was associated with a greater than 50% mortality at 10-weeks, and with one year mortality over 75%.[297-299] It is not recommended unless there is no alternative and referral for more effective treatment is not possible. An oral combination of fluconazole 1200 mg daily plus 5-flucytosine 25 mg/kg four times a day was associated with a significant improvement in EFA compared with fluconazole alone.[300] Not surprisingly, the addition of a short, 5-7 day course of Amb-D at 1 mg/kg daily to oral fluconazole or oral fluconazole and 5-flucytosine showed improved rates of cryptococcal clearance[301,302], similar to rates seen with 14 days of Amb-D. Notably there was no decrease in rate of clearance in the second week, perhaps due to the long half-life of Amb-D or to its post antibiotic effect in brain tissues.[301,302] On the other hand, a retrospective study in China suggest that short-term usage of Amb-D of less than 14 days was associated with increased mortality in CM.[303] These results set the stage for comparative studies in reduced polyene exposure to reduce toxicity but have acceptable efficacy.

In the phase 3 ACTA trial conducted in centres in sub-Saharan Africa, the oral combination of fluconazole 1200 mg daily and 5-flucytosine 25 mg/kg four times a day for 2 weeks was compared with 1-week of Amb-D 1 mg/kg daily or 2 weeks of Amb-D 1 mg/kg daily as induction therapy, with the latter two arms randomly being partnered with fluconazole 1200 mg daily or 5-

flucytosine 25 mg/kg four times a day.[123] One-week Amb-D plus 5-flucytosine followed by fluconazole 1200 mg daily given in the second week was the best performing induction arm, with a 24% 10-week mortality. There were fewer side effects and the same EFA, comparing one week with two weeks of Amb-D (plus 5-flucytosine). The oral combination was non-inferior to the 2-week regimens containing Amb-D. Although the EFA was less rapid than with the Amb-D regimens, the oral combination was associated with the fewest side effects. Amb-D when combined with 5-flucytosine was associated with improved EFA and survival, compared with fluconazole as the partner drug. Following the ACTA trial, the WHO and Southern African HIV Clinicians Society Guidelines were updated to recommend 1-week Amb-D 1 mg/kg daily plus 5-flucytosine as the preferred induction regimen.[1,185]

Subsequently, in light of the long half-life of L-Amb especially in brain tissue[304], and support of intermittent dosing success in animals for CM[304] and precedent for using a single, high 10 mg/kg dose of L-Amb in visceral leishmaniasis[305], one or more high doses of L-Amb were compared to L-Amb 3 mg/kg daily for 2 weeks in a phase II study in Botswana and Tanzania. In the AMBITION phase II trial, a single 10 mg/kg dose of L-Amb resulted in an EFA similar to daily doses of 3 mg/kg, and side effects were markedly less than in prior trials using Amb-D.[306] In a subsequent phase III study of 844 participants across sites in Southern and East Africa, a single initial 10 mg/kg dose of L-Amb, on the oral backbone of fluconazole 1200 mg daily plus 5-flucytosine 25 mg/kg four times a day for 2 weeks was compared with the existing WHO recommendation of 1-week Amb-D 1 mg/kg daily plus 5-flucytosine followed by 1 week of fluconazole 1200 mg daily.[180] The trial demonstrated that the single dose L-Amb-based regimen was non-inferior to the WHO recommendation (10 week mortality 24.8% vs 28.7%, respectively, risk difference -3.9%, 90%CI -9.0 - 1.2), with results consistent across intention to treat, per-protocol, adjusted and unadjusted analyses. The EFA of the regimens was very similar (-0.40, vs -0.42 $\log_{10}$ CFU/ ml CSF/d, respectively), while the L-Amb regimen was significantly better tolerated: Grade 4 anaemia, mean change in haemoglobin over the first week, and proportion of participants receiving blood transfusion were 3%, -0.3 g/dl, and 8% in the L-Amb arm, compared with 15%, -1.9 g/dl, and 18% with the WHO regimen. There was also a greater rise in creatinine, more hypokalaemia, and more thrombophlebitis requiring antibiotics in the 1-week Amb-D arm. It is important to note that these results cannot be extrapolated to other lipid-associated products. Following the AMBITION trial, the WHO guidelines now recommend the single dose L-Amb-based regimen combined with fluconazole 1200 mg and flucytosine as the preferred antifungal therapy in PLHIV with CM.[188]

In RLS, the extra cost of L-Amb may be offset by a reduced need for monitoring, and fluid and electrolyte supplements and reduced intravenous administration costs, and by the ability to discharge some patients earlier (although most patients will need at least 1-2 weeks of inpatient care to monitor and manage complications, including raised ICP). Given the difficulties of delivering routine care in RLS, of ensuring close monitoring and management of Amb-D-associated side effects and of providing transfusions, it is also possible that the relative benefit of the L-Amb regimen may be greater in "real-life" than was demonstrated in the trial setting. On the other hand, sequential non-inferiority trials may miss the step-wise erosion of therapeutic efficacy.

The data from the ACTA and AMBITION trials may not be directly applicable to **resource-rich settings** where the current standard of treatment is **L-Amb 3-4 mg/kg daily plus 5-flucytosine 25 mg/kg four times a day for 2 weeks**. This is different from the comparators used in ACTA (of 1 or 2 weeks of Amb-D 1 mg/kg) and in AMBITION (the previous WHO recommendation, which was derived from ACTA, of Amb-D 1 mg/kg daily and 5-flucytosine 100 mg/kg daily for 1 week followed by 1 week of fluconazole 1200 mg daily). The lesser nephrotoxicity of L-Amb compared with Amb-D and the importance of the addition of 5-flucytosine are both long-established. The use of high-dose fluconazole is not widely accepted in patients managed in resource-rich settings where comorbidities and potential drug-drug interactions need to be carefully considered and hepatotoxicity is less well tolerated. In a retrospective review of cryptococcosis treatment and adherence to treatment guidelines in a centre in USA, only a third of patients completed the 14-days of 5-flucytosine.[307] Thus, caution is advised against the over-reliance on both these oral agents as induction therapy. Furthermore, complications of CM during induction therapy such as increased ICP require close clinical monitoring and generally require hospital care for at least one, if not two or more weeks.

Unfortunately, there have been few comparative studies in management of CM in resource-rich healthcare systems since the 1997 clinical trial by Van der Horst and colleagues.[120] A small, randomised study showed improved culture conversion, similar clinical efficacy and less nephrotoxicity with L-Amb at 4 mg/kg daily compared to Amb-D 0.7 mg/kg daily, (both arms without 5-flucytosine, followed by fluconazole 400 mg daily) .[308] Hamill et al., found similar survival and response rates with L-Amb at 3 mg/kg daily and 6 mg/kg daily compared to Amb-D at 0.7 mg/kg daily.[309] However, review of the data by the US Food and Drug administration (FDA) supported 6 mg/kg daily.[310] L-Amb at 3 mg/kg daily was associated with the fewest side

effects. Significant clinical experience has now accumulated in RRS where L-Amb 3-4 mg/kg daily has become the standard of care.

The use of ABLC in CM is supported by a few smaller randomised trials and retrospective cohorts[311,312] but much less rigorous than L-AmB. There are no clinical trial data on ABCD in CM. A low and high dose study of a generic liposomal amphotericin B product – Fungisome® (Lifecare Innovations, Gurguram, Haryana, India) was reported with no comparator, in a small number of CM patients in India.[313] Despite the evidence that Amb-D 1 mg/kg daily compared with 0.7 mg/kg daily has improved rate of fungal clearance, some institutions persist with the lower dose Amb-D mostly due to nephrotoxicity issues and/ or insurance or reimbursement limitations.

Overall, there has been a shift towards 2 weeks of L-Amb as the first-line polyene due to its reduced toxicity in RRS. This standard 2-week induction course of L-Amb and 5-flucytosine has shown better tolerability and a low early mortality of 6% in one nationwide observational study.[314], and stands as the recommended regimen in RRS hospitalisation during these first one to two weeks and thus allows for careful monitoring and management of the many complications of CM.

Role of CSF sterility post-induction therapy

Some treatment guidelines advocate performing a LP 2-weeks after initiation of therapy (prior to changing induction therapy to consolidation therapy) to assess CSF culture sterility as a marker of successful induction.[1,181,184,189] Other guidelines – particularly those focused on RLS - do not.[185,188] Mycological success has been associated with improved outcomes and reduced clinical relapse.[315] In PLHIV with CM, CSF sterility prior to ART commencement has been shown to be associated with reduced occurrence of neurological deterioration, microbiological relapse and C-IRIS[315] (discussed in later sections).

The sensitivity of the clearance LP is reduced when very small volumes of CSF are used. Other concerns include delay in adjustment of the treatment regimen while waiting on culture results. The preciseness of the follow-up LP and its necessity are controversial, but we encourage regular microbiological culture assessment of CSF for induction treatment success.

Consolidation and maintenance treatment

There are no recent trials of consolidation and maintenance therapy. Two early studies showed that fluconazole (400 or 600 mg daily) was superior to itraconazole for consolidation therapy.

[120,148] With current extensive published data on the safety of an 800 mg daily dose, evidence of a fluconazole dose response effect,[297,316] and with nevirapine (levels of which may increase with concomitant fluconazole) no longer used for initial antiretroviral therapy, the consolidation dose of 800 mg daily of fluconazole is typically preferred in RLS.[1,185] Concerningly, a gradual rise in median fluconazole MICs in cryptococcal isolates collected during initial CM presentation[317,318] may support a higher consolidation dose.

Maintenance therapy with fluconazole 200 mg daily has been shown to be highly effective at preventing relapse, and is superior to weekly Amb-D and itraconazole[149,319,320] Rarely, triazoles such as voriconazole[152-159], posaconazole[160,161,321] or isavuconazole[163,164] are used as alternatives to fluconazole due to concerns of fluconazole resistance, drug toxicity or drug-drug interactions (See section on cryptococcal resistance). These triazoles have *in vitro* and animal data to support their use in cryptococcosis. However, none of these alternative drugs have been formally trialled but there are relatively more reports and experience with voriconazole.

After a minimum of 1 year of antifungal therapy in patients established on ART, who have a suppressed viral load and/ or a CD4 > 100 cells/mm^3, a very low incidence of subsequent relapse has been reported.[322-328]

Recommendations:

Acknowledging the complexity of CM management, and the additional resource-related challenges, we provide treatment strategies for CM treatment in PLHIV based on availability of antifungal drugs; preferred and alternative strategies are offered **(see Figures 3 and 4, sTable 5)**. Clinical practice is rapidly evolving with the read-out of recent trials, changes in international guidance and increasing access to antifungal drugs in RLS.

Induction (2 weeks):

Preferred:

- **(AII) L-Amb 3–4 mg/kg daily plus 5-flucytosine 25 mg/kg four times a day** (preferred in RRS)**;**

  OR

- **(AI) a single dose of L-Amb 10 mg/kg with 14 days of (5-flucytosine 25 mg/kg four times a day and fluconazole 1200 mg daily)** (preferred in RLS)**.**

Alternatives:

- *Where L-Amb is not available:*
    - **(BIIt)** ABLC 5mg/kg daily plus 5-flucytosine 25 mg/kg four times a day.
- *Where L-Amb and ABLC are not available:*
    - **(BI)** Amb-D 0.7-1 mg/kg daily plus 5-flucytosine 25 mg/kg four times a day; OR
    - **(BI)** Amb-D 1 mg/kg daily and 5-flucytosine 25 mg/kg four times a day for 1 week, followed by fluconazole 1200 mg daily for 1 week.
- *Where 5-flucytosine is not available:*
    - **(BIII)** L-Amb 3–4 mg/kg daily plus fluconazole 800–1200 mg daily.
    - **(BIII)** ABLC 5 mg/kg daily plus fluconazole 800–1200 mg daily.
    - **(BI)** Amb-D 0.7-1 mg/kg daily plus fluconazole 800–1200 mg daily.
- *Where amphotericin-B based therapies are not available:*
    - **(BI)** 5-flucytosine 25 mg/kg four times a day and fluconazole 800-1200 mg daily.
- *Where only fluconazole is available:*
    - **(CI)** Fluconazole 800-1200 mg daily.

Consolidation (8 weeks):

Preferred:

- **(AI) Fluconazole 400-800 mg daily**
    - (Note 800 mg daily is preferred in RLS)

Alternatives *(from most to least preferred)*:

- **(BIII)** Voriconazole 200 mg twice a day (with TDM)
- **(BIII)** Posaconazole 300 mg daily (with TDM)
- **(BIII)** Isavuconazole 200 mg daily
- **(CIIt)** Itraconazole 200 mg twice a day (with TDM)

Maintenance (12 months/ immune restoration):

Preferred:

- **(AIIt) Fluconazole 200 mg daily**

Alternatives *(from most to least preferred)*:

- **(BIII)** Voriconazole 200 mg twice a day (with TDM)
- **(BIII)** Posaconazole 300 mg daily (with TDM)
- **(BIII)** Isavuconazole 200 mg daily
- **(CIIt)** Itraconazole 200 mg daily twice a day (with TDM)

Other recommendations:

- **(AIIu)** In-hospital care for the first 1-2 weeks is encouraged to manage the significant early complications seen with CM management.
- **(AIIu)** The use of Amb-D and L-Amb should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy.
- **(AIIu)** Frequent (at least every alternate day) complete blood counts, renal function tests and electrolyte measurements, are recommended to assess for therapy-related bone marrow, nephrotoxicity, and fluid and electrolyte changes. Liver function tests at baseline and at least weekly are recommended.
- **(BIII)** Monitoring of 5-flucytosine drug levels, where available and if timely, is recommended, particularly with underlying renal dysfunction.
- **(AIIu)** Check for drug-drug interactions and dose-adjust as necessary.
- **(CIIu)** Consider performing an LP at the end of the second week of induction therapy to check for CSF sterility prior to ART commencement.
- **(CIIu)** Consider prolonging induction therapy if CSF is persistently culture positive at 2 weeks.

Solid organ transplant (SOT) recipients

Evidence:

Cryptococcosis is the third most common invasive fungal infection in SOT recipients, with an incidence of 4.5-33.8%[175-177] and significant mortality.[7] SOT recipients comprise a third of non-HIV related cryptococcosis (33.8%) in the United States compared with a much lower incidence of 1.3% in haematopoietic stem cell transplant recipients.[329] The majority of cryptococcosis in SOT occurs late and is due to reactivation disease. However, acute infection in donors with unrecognized cryptococcosis or primary infection has been described. Notably, organs from donors with infection at any site may transmit cryptococcosis.[183,330] Clinical clues in donor-

derived infection include cryptococcosis detected within 30 days of transplantation or in more than one recipient from a single donor or infection in the graft or surgical site.[183 331]

Understanding the cause of death in the case of deceased donor or heightened awareness in donors from endemic areas or history of environmental exposure may aid early recognition of donor-derived infections. The optimal timing for transplantation or re-transplantation in a patient with prior cryptococcosis is unknown and is thus a clinical decision.

Renal transplant recipients have long been known to be at risk for cryptococcosis[332-334] and the nephrotoxicity of Amb-D in RLS settings is especially problematic. Of late, heart and small bowel recipients are thought to be at highest risk, likely due to their more advanced immunosuppression.[335] Glucocorticoids as part of anti-rejection therapy, cytotoxic therapies and calcineurin inhibitors are significant risk factors for non-HIV related cryptococcosis[329] but the risk varies with different immunosuppressive therapy regimens. For example, calcineurin inhibitors (tacrolimus and cyclosporine), which possess some direct anticryptococcal activity, are associated with less aggressive disease [336-338] On the other hand, T-cell depleting antibodies such as alemtuzumab and anti-thymoglobulin when used to treat graft rejection, cause severe and prolonged CD4-cell depletion and pose a particularly high risk for severe cryptococcosis. Cryptococcosis in transplant recipients commonly presents as CNS or pulmonary disease but may also present with unusual manifestations including cutaneous disease[272,339] and pericarditis.[340] A multicentre study of SOTs suggested that CNS disease is associated with late onset disease, fungemia and/or serum CrAg >1:64.[339] Notably, serum CrAg may be negative in SOT recipients, particularly those with single lung nodules or in lung transplant recipients.[341]

There are no randomised treatment trials solely in SOT recipients and hence no high-level evidence to guide antifungal choice or duration of therapy. Recommendations are based on data from management of PLHIV, though the high dose fluconazole used in the ACTA and AMBITION trials are not generally recommended in SOTs, due to significant drug-drug interaction and potential toxicity. Not surprisingly, the use of Amb-D for cryptococcosis in renal transplant recipients is associated with significant graft dysfunction.[218] Compared with Amb-D, lipid formulations of amphotericin B are associated with a significant reduction in mortality.[342] Increased mortality was seen in those with altered mental status, absence of headache, liver failure[335], and persistently positive CSF cultures at 2 weeks.[343] Rapid reduction in dosing of immunosuppressants, particularly calcineurin inhibitors, may lead to IRIS.[344] Importantly, serum

chemistries and complete blood counts should be followed closely to detect and avoid serious drug-related toxicities during induction therapy.

Recommendations:

- **(AIIt)** In SOT with CNS cryptococcal disease, we recommend induction therapy with L-Amb 3-4 mg/kg daily and 5-flucytosine 25 mg/kg four times daily, for a minimum of 2 weeks, followed by fluconazole 400–800 mg daily as consolidation therapy (usually for 8 weeks), and 200 mg daily as maintenance therapy (generally for 12 months).
- **(AIII)** In view of the immunosuppressed state of SOT recipients, we recommend that CNS-based therapy be considered for any disseminated disease or isolation from a sterile site (even in the absence of CNS manifestations).
- **(BIII)** In SOT recipients with isolated mild pulmonary cryptococcosis, fluconazole 400 mg daily for 6-12 months may be considered If CNS disease is excluded.
- **(AIII)** Close monitoring of tacrolimus, cyclosporine and sirolimus (TDM) levels and dose reduction of these agents are recommended when azoles are co-administered.[345,346]
- **(BIII)** Immunosuppressant doses need to be carefully adjusted to allow effective killing of yeasts but should be reduced slowly to avoid precipitating C-IRIS. Consider a sequential or stepwise reduction of immunosuppressants with careful lowering of corticosteroids early and eliminating mycophenolate before considering reduction of the calcineurin inhibitors because of their direct anticryptococcal activity.
- **(CIII)** In a patient treated for cryptococcosis, re-transplantation or a new organ transplant may be considered provided viable yeasts have been cleared from CSF and the patient is asymptomatic after receiving 12 months of anticryptococcal treatment.
- **(DIIu)** Routine screening, primary prophylaxis, or pre-emptive therapy for cryptococcosis in transplant recipients are not recommended, except if donor organ is found to contain cryptococci which will require pre-emptive therapy.
- See **Figure 4** and **sTable 5 (s22)**

Treatment of CM in Non-HIV, Non-Transplant Hosts

Evidence

The non-HIV, non-SOT group is heterogenous, ranging from apparently normal hosts to those with haematological malignancies, long-term use of corticosteroids or severe liver disease. There is no single therapeutic regimen that meets all patient needs. The two major studies prior to the era of AIDS investigated a low dose polyene and high dose 5-flucytosine regimen.[347,348]

These studies emphasised the value of combination therapy, but mortality rates still approached 25% and there have been no recent RCTs in these populations. A retrospective study of 517 non-HIV, non-transplant patients with CM using the Japanese national database showed that the addition of 5-flucytosine to L-Amb resulted in a trend towards reduced mortality at day 14.[314]

Today, mortality remains substantial in this group likely due to the heterogenous nature of their underlying diseases including some with a non-reversible immunodeficiency and diagnostic delays due to lack of suspicion.[307] The mortality appears to be extremely high amongst patients with cirrhosis.[11-13]

The preferred regimen in this heterogeneous group of patients mirrors that for CM in PLHIV or SOT recipients in RRS. Induction therapy may be extended in those with persistently positive cultures and/or symptoms.

Recommendations

- **(AIIt)** We recommend L-Amb 3–4 mg/kg daily and 5-flucytosine 25 mg/kg four times a day, for a minimum of 2 weeks as induction therapy, followed by fluconazole 400–800 mg daily as consolidation therapy (8 weeks), and 200 mg daily as maintenance therapy (generally for 12 months).
- **(AIIu)** These patients should all be investigated for HIV infection and have CD4 counts determined.
- See **Figure 4** and **sTable 5 (s22)**

Adjunctive therapies in CM-HIV

Evidence:

Adjunctive therapy trials in CM are limited to PLHIV. Exogenous IFN-gamma has been studied in two randomised trials. In 2004, a phase 2, double-blind, placebo-controlled study of adjunctive IFN-gamma-1b (100 or 200 μg thrice weekly for 10 weeks) or placebo, on the background of 0.7 mg/kg daily of Amb-D with or without 5-flucytosine 25 mg/kg four times a day for 14 days was trialled in CM-HIV co-infected individuals (n=70). There was a trend towards increased CSF clearance of cryptococci in the IFN- treatment group although adverse events including fever, rigors, headache, malaise, and fatigue were significantly increased.[349] In a subsequent randomised, open-label study, much shorter courses (two or six doses of 100 μg IFN-gamma-1b) were added to standard antifungal therapy (Amb-D 0.7 mg/kg daily with 5-flucytosine 25 mg/kg four times a day) (n=88).[350] The two IFN-gamma-containing arms demonstrated faster rates of CSF fungal clearance compared with standard antifungal therapy

alone. It was not powered to detect differences in mortality. The greatest benefit in yeast clearance was observed in patients with minimal endogenous IFN-gamma and TNF-alpha responses. Further studies are needed and targeted to such patients, before IFN-gamma can be routinely recommended in clinical practice. Some clinicians reserve exogenous IFN-gamma for those with prolonged CSF culture positivity despite optimal antifungal drug regimen but this indication has not been tested in clinical trials.

In the Crypto-Dex study in PLHIV in Asia and Africa, adjunctive high-dose dexamethasone (0.3 mg/kg/day, weaning weekly over 6 weeks to 1 mg/day) added to Amb-D 1 mg/kg daily and fluconazole 800mg daily was terminated early by the data safety and monitoring board (DSMB) due to harmful effects.[351] These negative effects included increased rates of disability, slower fungal clearance, more adverse events, and a trend towards increased mortality at 10 weeks (47% *vs*. 41%).[351] However, corticosteroids have been useful in select cases such as cerebral cryptococcomas with significant cerebral oedema in *C. gattii* disease[352] and considered in cryptococcal induced cerebral vasculitis. The role of corticosteroids in the management of C-IRIS is discussed in later sections.

Sertraline, a selective serotonin reuptake inhibitor, was investigated as adjunctive therapy in Astro-CM, a large phase III double-blind, randomised placebo-controlled study conducted in Uganda in HIV-CM co-infected patients[353] and in a smaller study in Mexico.[354] Both trials were futile. A randomised open-label trial of adjunctive tamoxifen, an oestrogen receptor antagonist and an antineoplastic agent, taken at high doses of 300 mg daily for 14 days was also ineffective[355] Another adjunctive study included the use of an antibody to heat shock protein (*HSP90*, Mycograb®); two concurrent trials **(**NCT00324025 and NCT00847678) were terminated early without results**.**

Recommendation:

- **(CIIt)** Adjunctive recombinant IFN-gamma might be considered in HIV-associated CM for those who have evidence of very poor inflammatory responses or persistently positive cryptococcal CSF culture after prolonged antifungal therapy.
- **(DI)** The routine use of high-dose dexamethasone in CM is not recommended.
- **(CIII)** A short course of dexamethasone may be considered for specific indications such as symptomatic space-occupying lesions in the CNS with surrounding oedema or mass effect and cerebral vasculitis.
- **(DI)** We recommend against the use of adjunctive sertraline and tamoxifen in CM.

Treatment of Pulmonary cryptococcosis

Evidence

There are no randomised treatment studies in pulmonary cryptococcosis. Recommendations are based on clinical experience, case reports and extrapolation from CNS disease studies. Disease severity may be classified as mild or severe though the criteria for distinguishing the two are based on clinical judgment and are not validated. In principle, asymptomatic disease, or mildly symptomatic patients or with a solitary small nodule (<2 cm) are considered to have mild disease while those with multiple lesions, large lesions (≥2 cm), lobar consolidation, cavitation, multi-lobar involvement, and hypoxemia are considered to have severe disease. Patients with cryptococcaemia, evidence of CNS involvement, serum CrAg ≥ 1:512 (by latex agglutination or 10-fold higher by LFA[65]), or severe pulmonary disease, should be treated as CM.[107,120,356,357]. Patients without cryptococcoma and with mild isolated pulmonary disease may be treated with azole monotherapy.[356 358 359]

Recommendations

- Stratify treatment by disease severity and presence of pulmonary cryptococcoma **(sTable 6, s22)**
- Isolated pulmonary cryptococcosis in immunocompetent or immunocompromised host:
  - Severe disease: **(AIIu)** as for CNS disease.
  - Mild disease: **(BIIu)** Fluconazole 400 mg daily for 6-12 months (range guided by symptom resolution).
- Pulmonary cryptococcosis with CNS manifestations or other evidence of dissemination (eg. cryptococcaemia or skin lesions).
  - **(AIIt)** as for CNS disease.
- Pulmonary cryptococcoma (see cryptococcoma section).

Treatment of Non-pulmonary/ Non-CNS Disease

Evidence:

While cryptococcosis can affect any organ following haematogenous dissemination, clinical presentation of non-CNS/non-pulmonary disease without fungemia is rare. Barring direct inoculation into the skin following trauma, extrapulmonary disease is by definition “disseminated

disease" and generally requires consideration for CNS-based therapy. There are no clinical treatment trials for non-pulmonary, non-CNS cryptococcosis.

Importantly, visual changes noted in CM are frequently related to raised ICP and do not necessarily indicate direct eye involvement. Ocular cryptococcosis can occur[360,361] but is unusual and requires formal ophthalmological documentation and management. Isolated skeletal osteomyelitis has been reported in both immunosuppressed and immunocompetent patients, but is rare, and often requires a combined surgical and medical approach.[263,362,363] Particularly, new granulomas in patients with sarcoidosis can pose a diagnostic dilemma and require careful consideration for new onset cryptococcosis.[15] Skin lesions may be polymorphic appearing as nodular, papulopustular, abscesses, and ulcers and can occur in both immunocompetent and immunocompromised hosts.

Recommendations:

- Cryptococcaemia:
  - **(AIIu)** Treat as for CNS disease.
- Primary cutaneous (skin) cryptococcosis attributed to direct inoculation without evidence of dissemination:
  - **(AIII)** Fluconazole 400 mg daily for 3-6 months or until cicatrisation.
- All other non-CNS/ non-pulmonary disseminated disease:
  - **(BIIu)** Treat as for CNS disease.
- **(BIIu)** Cryptococcal eye disease should be managed in collaboration with an ophthalmologist.

Specific Management Issues

Optimal antifungal therapy is but one aspect of the complex management requirements in CM and cryptococcosis. Here we discuss key management issues encountered in CM and cryptococcosis (principles summarised in **Panel 5**). Raised ICP is common in CM and its management is key to therapeutic success. The treatment of cryptococcosis is long, and complications of drug therapy, drug resistance, clinical relapse and C-IRIS may occur. An approach to clinical relapse or neurological deterioration including differential diagnoses is proposed; this serves to guide therapeutic decisions and to prevent unnecessary procedures

(see **Figure 2**). Optimal timing of ART initiation in PLHIV with CM, C-IRIS and fluconazole-resistant disease are discussed. Unique clinical situations such as cryptococcosis in pregnancy and in paediatric populations are discussed and nuances in the management of *C. gattii* and cryptococcomas are considered. With an increasing number of cryptococcosis cases seen in seemingly immunocompetent individuals, we summarise data on inborn errors of immunity, idiopathic CD4 lymphopenia and anti-cytokine antibodies and suggest an approach to explore for an underlying immunodeficiency. Notably, the clinical presentation of raised ICP, CNS C-IRIS and adverse drug events commonly overlap, thus clinical expertise in distinguishing these conditions is required. **sTable 4 (s16)** summarises important issues in relation to symptoms, signs, and drug toxicities.

Management of raised intracranial pressure (ICP)

Evidence:

The deleterious effect of increased ICP in patients with CM has been recognised for almost half a century. Increased ICP has been associated with high burden of yeasts demonstrating, both acute and chronic symptoms and signs (e.g., visual and hearing loss) and decreased short-term survival. For instance, a persistently elevated ICP during the 2-week antifungal induction period was associated with more frequent clinical failure.[364-366] The mechanism for acute raised ICP is not certain but is likely due to a high yeast burden causing structural blockage of the arachnoid granulations and lymphatic villi by yeasts and capsule-derived polysaccharides, resulting in a communicating hydrocephalus.[367] Raised ICP may also be exacerbated by cerebral oedema associated with the host inflammatory reaction to the yeasts within the CNS.

Clinical experience has shown that CSF outflow obstruction can be improved by removal of CSF.[366,368] For prolonged control of acute ICP, use of lumbar drains or ventriculostomies may be required.[369-371] Mechanically decompressing the subarachnoid space may improve the level of consciousness and reduce deafness and visual changes. On the other hand, medical therapy for increased ICP has not been successful. For instance, corticosteroids can acutely reduce ICP, but they have substantial adverse side effects including increased host disability and mortality and thus are not routinely recommended unless there is a component of IRIS or in the presence of cerebral cryptococcomas with surrounding mass effect. Acetazolamide and mannitol do not reduce ICP and are associated with substantial side effects.[372,373]

Although raised ICP is a significant problem in acute CM, it also can occur during the development of C-IRIS (where it is corticosteroid-responsive). Classical obstructive hydrocephalus secondary to scarring/chronic inflammation is a later complication, diagnosed by clinical presentation and cerebral imaging, and is responsive to a standard ventriculoperitoneal (VP) shunt placement.

Recommendations:

- **(AIIu)** Opening pressure (OP) should be measured at every LP in patients with CM.
- **(AIII)** Perform a CT brain (if CNS imaging not previously done) to exclude CNS outflow obstruction.
- **(AIIt)** Acute symptomatic elevation of the ICP (≥ 20 cm of CSF) should be managed by daily therapeutic LPs (i.e., removal of sufficient CSF (usually around 20-30 ml) to reduce the pressure to 50% of OP and/or to a normal pressure of ≤20 cm of CSF (documented as a closing pressure)).
- (**BIIu**) Perform a scheduled therapeutic LP around 48-72 hours after initial LP and/or 7 days, regardless of initial OP.
- **(AIIt)** Persistently raised symptomatic ICP despite therapeutic LPs should be managed by surgical decompression via temporary lumbar drainage, shunting, or ventriculostomy depending on local expertise and resources.
- **(BIII)** Consider ventriculoperitoneal (preferential) and lumboperitoneal shunts (alternative) to control both acute and chronic hydrocephalus if temporary measures are not successful. Ideally, insert shunts after institution of effective antifungal therapy.

Timing of antiretroviral therapy commencement

Evidence:

The optimal time to commence antiretroviral therapy (ART) during cryptococcosis is controversial. Three randomised trials to determine optimal timing of ART initiation in HIV-CM co-infection have been conducted in African and Asian countries. In a Zimbabwean study of 54 patients receiving fluconazole 800 mg daily, excess mortality was observed in the group receiving ART <72 hours after commencement of fluconazole compared with >10 weeks.[374] A smaller study in Botswana, included 27 persons receiving Amb-D 0.7 mg/kg daily; more episodes of C-IRIS were reported in patients within the early ART arm.[375] In the only adequately

powered (COAT) trial (induction therapy with Amb-D 0.7-1mg/kg daily and fluconazole 800 mg daily for 2 weeks), conducted in Uganda and South Africa, enrolment was halted by the DSMB after 177 persons were enrolled, due to excess mortality (15%) at 26-weeks in the early ART arm (1-2 weeks) versus the deferred arm (5-6 weeks).[178] These data seem to suggest that initiating ART within 2 weeks of CM presentation is too early and that delaying ART initiation for 5 weeks reduces the incidence of C-IRIS. However, the shortest acceptable time to commence ART during CM therapy is unknown. Potential risks of a delay include discontinuity in care, development of new opportunistic infections with the prolonged immunodeficient states, and ongoing HIV transmission.

In a separate, prospective study, 106 HIV-CM patients underwent routine CSF culture and analysis at a median of 15 days and commenced ART at a median of 18 days from CM diagnosis and treatment (Amb-D 1 mg/kg daily). Patients with culture-negative CSFs prior to ART commencement had significantly lower rates of neurological deterioration, microbiological relapse, and C-IRIS, with a trend to reduced mortality over 24 weeks.[315] Patients on optimal induction antifungal therapy (L-Amb 3-4 mg/kg or Amb-D 1 mg/kg with 5-flucytosine 25 mg/kg four times a day) are likely to achieve CSF sterility quicker than those treated in these ART-timing studies where combinations of 0.7-1 mg/kg Amb-D with fluconazole or fluconazole monotherapy were used. More recently, a retrospective analysis from combined cohorts of several studies in resource-rich settings did not appreciate higher mortality in those receiving early ART in the first two weeks of antifungal therapy compared to those with delayed therapy.[376] In all, early ART in RRS, will need careful justification and close monitoring; further randomised studies may be helpful.[377]

There are no studies for timing of ART initiation in other forms of cryptococcosis, those with cryptococcal antigenemia or those recommencing ART after a period of ART interruption. Initial concerns that potent integrase inhibitors (such as raltegravir and dolutegravir) pose an increased risk of C-IRIS have been disproven by a large prospective study of HIV-infected adults and children with CD4 <100 cells/mm^3 in four African countries. The addition of raltegravir to standard ART had no effect on the incidence of any IRIS, TB-IRIS, or C-IRIS.[378]

Recommendation:

- **(DI)** Immediate or very early commencement of ART is not recommended.
- Depending on antifungal induction therapy availability:

- Sub-optimal/limited access: **(AI)** Delay ART for 4-6 weeks.
- Optimal and in RRS: **(BIIu)** Consider further individualisation taking into consideration resolution of symptoms and signs of CM and ICP (including normalisation of OP), attainment of CSF cryptococcal sterility, successful identification and management of concurrent co-infections and other AIDS-defining illnesses, the patient's readiness for ART and local experience of CM and C-IRIS management. (Usual range is 4-6 weeks).

- **(CIIt)** Ideally, we encourage ensuring that the CSF is cryptococcal culture negative prior to ART commencement.
- **(BIII)** For ART-experienced persons who develop CM and may need to switch to second-line ART or recommence ART, we recommend a delay of 4-6 weeks.
- **(CIII)** Pending further studies, consider withholding ART and restarting at 4-6 weeks in those presenting with CM within 2 weeks of starting ART.
- **(BIII)** Patients with isolated pulmonary cryptococcosis or those with asymptomatic cryptococcal antigenemia may commence ART earlier (e.g., 2 weeks).

Cryptococcal resistance to antifungals

Evidence:

The predominant concern regarding cryptococcal resistance to antifungals is with fluconazole and 5-flucytosine, both pillars to cryptococcosis treatment. Secondary resistance to 5-flucytosine is common if given as monotherapy, thus it is always given with a partner drug in cryptococcosis. Attributing MICs to outcomes is particularly difficult in cryptococcosis as combination therapy is common, and older studies have usually utilised Amb-D based induction therapy. Acquired resistance to polyenes such as Amb-D is rare.

Mechanisms. Several cryptococcal virulence determinants have impacted drug resistance, including melanisation, titanisation, and replicative age of the yeast.[379,380] Resistance to amphotericin B is mediated via reduced ergosterol content.[381] 5-flucytosine resistance mutations in pyrimidine pathway genes rapidly emerge during monotherapy.[382] The clinical implication of a "high" 5-flucytosine MIC when it is used in combination with AmB-D and particularly with fluconazole, remains uncertain. Resistance to fluconazole arises from mutations or overexpression of the target *ERG11* gene as well as overexpression of efflux genes.[383] One reason for apparent discrepancies in *in vitro* MIC and clinical outcomes may be the intrinsic heteroresistance of *C. neoformans* and *C. gattii* to fluconazole, facilitating transient aberrant

chromosomal duplication of resistance genes (aneuploidy) in response to the drug[316,384-387]. This phenomenon was associated with reduced fungal clearance in azole-treated mice[388,389] and HIV-infected patients with CM, treated with fluconazole monotherapy.[390] Notably, even fluconazole doses of 1200 mg daily will not prevent emergence of heteroresistance[316], but this heteroresistance is effectively suppressed by combination therapy with 5-flucytosine.[390]

Fluconazole-resistance. Whilst population-based surveillance studies have suggested that cryptococcal resistance to antifungals is minimal[391 392], studies based in South Africa, Taiwan and Uganda have documented the emergence of fluconazole resistance.[317,318,393] Secondary resistance has been implicated as a cause of relapse in patients treated with induction fluconazole monotherapy.[390,394-396] A systematic review of 29 published studies (4,995 isolates, 5% relapse, 1988-2017) showed an overall 18.7% of nearly 5000 isolates had MICs above the epidemiological cut-off value (ECV) of 8 mg/L with 10.6% of incident isolates recording fluconazole primary resistance by this definition.[396] However, this review pooled data for various genotypes and results generated by different antifungal susceptibility testing method. Nevertheless, the emergence of fluconazole resistance is concerning particularly in light of a small older study suggesting that fluconazole MICs correlated with outcome.[397]

As discussed in in vitro susceptibility, the only available clinical breakpoint is for *C. neoformans* with Amphotericin B, using the EUCAST method. Interpretation of ECOFFs (using the CLSI method) for fluconazole requires accurate species identification. Interpretation of epidemiological cut-off values (ECVs), using the CLSI method for fluconazole requires accurate species identification. Based on CLSI, the ECV for *C. neoformans VNI* is 8 ug/mL, and 16 and 32 ug/mL, for *C gattii* VGI and *C. deuterogattii VGII, respectively.*[132] In principle, a higher than 2-fold increase in MIC during treatment may suggest development of resistance, and the need for closer clinical monitoring. There are no EUCAST ECOFFs available for fluconazole. (see sDoc1 for details).

Recommendations:

- For those with concerns of fluconazole-resistance or emerging fluconazole-resistance:
    - **(BIII)** Consider a longer course of treatment with Amb-D (1 mg/kg daily) or higher dose of L-Amb (3-6 mg/kg daily) together with 5-flucytosine (about 4 weeks) as induction therapy.

- **(BIII)** Consider Amb-D 1 mg/kg weekly or L-Amb 3-6 mg/kg weekly as consolidation/maintenance therapy. Consider daily voriconazole, posaconazole, isavuconazole or itraconazole for isolates without evidence of pan-azole resistance.
- **(CIII)** Where Amb-D or L-Amb are not available, adding 5-flucytosine to high-dose fluconazole (1200 mg daily) may be considered.

Cryptococcal persistence, clinical relapse, and culture-positive (microbiological) relapse

Evidence:

We use the term “persistence” where cultures are persistently positive after a specified period of antifungal therapy, while “relapse” implies recrudescence after a period of quiescence. We distinguish clinical relapse from microbiological relapse, where the former is based on clinical manifestations, while the latter is culture-based. **Persistent cryptococcal infection** is characterised by ongoing symptoms and signs, AND positive yeast cultures at the disease site (e.g., CSF for CNS infection) after 4 weeks of recommended antifungal therapy at an established effective dose (or longer if fluconazole induction monotherapy is used). Cryptococcal antigen persists in the CSF and blood thus have little clinical utility in distinguishing clinical responders from non-responders.[398] Importantly, persistence of antigenemia, positive CSF CrAg, visible cryptococci in CSF (without culture positivity), and abnormal CSF microscopy or biochemistry are not precise indicators of microbiological failure and do not necessitate escalation in antifungal treatment regimen. In the absence of persistently positive cultures, ongoing symptoms are most commonly due to raised ICP, which should be managed accordingly (see management of raised intracranial pressure).

Clinical relapse is characterised by a recurrence of symptoms and signs at the original site of infection (e.g., headache(s), seizures or decreased conscious level) and/or positive cultures after initial normalisation of signs and symptoms, and prior documented negative cultures. Clinical relapse may be due to microbiological relapse, C-IRIS, raised ICP (whether or not related to C-IRIS) or other infective and non-infective (CNS and non-CNS) causes. Investigations should include brain imaging, LP for OP and CSF analyses, particularly culture, and other testing as clinically indicated (see **Figure 2**). In persons with disseminated/non-meningeal cryptococcosis, comprehensive imaging is indicated to identify sanctuary sites.

Most cases of **culture-positive (i.e., microbiological) relapse** occur early and result from inadequate induction or early discontinuation of consolidation or maintenance therapy. This may be due to use of a suboptimal antifungal regimen (e.g., due to drug availability issues), subtherapeutic antifungal doses or an inadequate duration of induction therapy. This is particularly concerning in settings where high-dose fluconazole monotherapy is used as induction therapy; secondary resistance may ensue[394] (see section on cryptococcal resistance). Persons with sterile CSF cultures prior to ART commencement compared with those with positive cultures, had no microbiological relapse in 6 months (HR 0.1163 (95%CI: 0.0354-0.3818, p=0.0004),[315] and with preferred induction regimens microbiological relapse is rare.

Late microbiological failures are less common, though plausible; non-adherence to fluconazole consolidation and maintenance therapy, or suboptimal dosing particularly in the setting of drug-drug interaction, (***e.g.***, with rifampicin coadministration[399,400]) should be considered in this setting. Fluconazole exhibits linear pharmacokinetics, and serum and CSF fluconazole levels are strongly correlated.[126] Fluconazole susceptibility testing should be performed concurrently on isolates collected at initial presentation and at microbiological relapse. Fluconazole TDM is not routinely done in clinical practice[127] but may be considered in the critically ill, renally impaired and obese, when yeasts have high MICs, or in hosts with multiple drug-drug interactions. However, there are no accepted therapeutic fluconazole drug levels predictive of treatment efficacy in CM.

In episodes of symptomatic relapse following immune restoration either from commencement of ART or reduction of immunosuppression (especially in the initial 1-2 months), **C-IRIS** should be considered (see section on IRIS). A post-infectious inflammatory immune response syndrome (**PIIRS**) has also been described in previously healthy hosts. These PIIRS patients are described as having received optimal therapy with culture-conversion, who subsequently presented with deteriorating neurological status. Investigations revealed inflammation on brain MRI and evidence of immune activation in the CSF.[401] As in patients with C-IRIS, this group may benefit from steroids or other immunosuppressants,[402,403] but confirmatory evidence from randomised controlled trials is lacking.

Recommendations:

- **(AIIt):** Think broadly and investigate thoroughly to explore causality (CNS and non-CNS; infective and non-infective) in cases of apparent clinical relapse. Investigations should include CT/MRI brain, LP for OP and CSF analyses including microscopy and culture.
- **(AIIu)** Review adherence to antifungal therapy, ART/ immunosuppressants and other medications and consider drug-drug interactions. Optimise control of underlying disease(s).
- **(DIIu)** The use of follow-up blood or CSF CrAg for clinical decision-making is discouraged. Do not escalate antifungal therapy for persistent serum antigenemia, persistently positive CSF CrAg, visible cryptococci in CSF (without culture positivity), nor abnormal CSF microscopy or biochemistry. These are not necessarily indicators of microbiological failure.
- For **culture-positive** (microbiological) persistent or relapsed infection:
    - **(AIII)** Check adherence to antifungals and ART; review the possibility of; drug-drug interactions; and perform TDM (where applicable and available).
    - **(BIII)** Antifungal susceptibility testing (AFST) should be performed concurrently on all initial and relapse isolates. An increase in fluconazole MIC of ≥ 2-fold dilutions is considered significant,
    - **(BIII)** Consider recommencing induction therapy with a more optimal regimen that is guided by AFST.
- **(CIIt)** Adjunctive recombinant IFN-gamma may be considered for those failing to clear viable cryptococcus despite other antifungal measures and if there is evidence of an inadequate host immune response.
- **(BIIu)** Steroids may be considered for patients meeting PIIRS criteria.

Immune Reconstitution Inflammatory Syndrome (IRIS)

Evidence:

Immune reconstitution inflammatory syndrome (IRIS) has been classically described in HIV-infected persons after commencement of antiretroviral therapy (i.e., restoration of immune function).[404,405] Such patients develop exaggerated symptoms and signs and/or atypical

inflammation, reminiscent of a paradoxical recurrence or a newly occurring opportunistic infection (OI). Cryptococcosis-associated IRIS (C-IRIS), involving the CNS is one of the more frequent and severe forms of IRIS. Paradoxical CNS C-IRIS is typically diagnosed in a HIV-CM co-infected patient who despite an initial period of improvement with antifungal therapy and appropriate therapeutic LPs, develops recurrent symptoms following commencement of ART, usually without evidence of mycological (culture-positive) relapse **(Figure 2)**. In contrast, unmasking C-IRIS occurs in a PLHIV with no known OI at time of ART commencement, but who later develops symptoms and signs of cryptococcosis.

C-IRIS can arise in CNS and non-CNS compartments and in any immunosuppressed individual including SOT recipients and those on biologic agents, in the setting of immune recovery or withdrawal of immunosuppressants and has even been observed in seemingly immunocompetent individuals.[344]

HIV-associated C-IRIS mostly occurs between 2 weeks and 3 months after ART commencement or ART intensification but may occur earlier (within days) or as late as a year after treatment. While we have gained much knowledge in the immunopathogenesis of paradoxical CNS C-IRIS [241,315,406-413], no diagnostic biomarker for C-IRIS has been identified as suitable for use in routine clinical practice. C-IRIS thus remains a clinical diagnosis, where there is a temporal association with a change in immunosuppression and exclusion of alternative explanations (e.g., cryptococcal relapse, other infective and non-infective causes) (**Figure 2**). It is usually associated with a fall in plasma HIV viral load. A consensus clinical case definition for C-IRIS in HIV-infected patients has been proposed.[405] Careful clinical assessment and investigation are required prior to attributing symptoms and signs to C-IRIS.

In HIV infection, risk factors for C-IRIS include advanced immunosuppression (very low CD4), an acellular CSF white cell count, a high CSF protein at time of initial CM presentation and those who do not or are slow to improve their immunosuppressive state (e.g. poor CD4 recovery).[315,413] or conversely, too rapid improvement.[414] At the time of C-IRIS, serum C-reactive protein may be high, but this is a non-specific finding. Typically, CSF examination at presentation of C-IRIS reveals pleocytosis consistent with an inflammatory state, CSF cultures are usually negative (but need not be) and the opening pressure (OP) may be raised. Brain imaging may reveal T2 hyperintensity and cerebral oedema and/or worsening pneumonia or pulmonary infiltrates may be noted on chest imaging.

There have been no therapeutic trials in C-IRIS. Management strategies include therapeutic LP, and other symptomatic therapies such as analgesia, antiemetics and antiepileptics as appropriate. There is no role in intensifying antifungal therapy in C-IRIS. In severe CNS paradoxical C-IRIS, clinicians often commence corticosteroid treatment to dampen inflammation though its treatment efficacy has not been rigorously examined in clinical trials. Of note, the use of high-dose dexamethasone as adjunctive CM treatment (therefore potentially as prevention) did not show a reduction in C-IRIS.[179] Further, in steroid refractory C-IRIS, there are case reports on the use of TNF-alpha blockers such as adalimumab with mixed responses in renal transplant recipients[415,416] and in HIV-infected patients.[417,418] Thalidomide has been used in both HIV-infected patients[419,420] and in an immunocompetent patient experiencing C-IRIS.[421]

Recommendation:

- **(AIIt)** For patients with suspected paradoxical C-IRIS, carefully exclude recurrent cryptococcal disease or new infective or non-infective conditions, prior to attributing symptoms and signs to C-IRIS. Perform an MRI brain and LP to measure OP and obtain CSF for microbiological and biochemical analyses.
- **(AIIu)** Treatment of C-IRIS should include therapeutic LP, symptomatic therapy such as analgesia, antiemetics and antiepileptics where appropriate.
- **(AIII)** Continue current antifungal therapy.
- **(BIII)** High-dose prednisone/prednisone (usually 0.5-1.0 mg/kg daily) or dexamethasone (usually 0.2-0.3 mg/kg daily), weaned over 4-6 weeks may be considered in those with persistent symptoms, unresponsive to therapeutic LPs. Rarely a second steroid course with taper is needed.
- **(DIII)** Do not stop ART.
- **(BIII)** Cases of steroid-refractory or recurrent C-IRIS be discussed with experts in the field.
- **(BIIu)** Steroids may be considered for PIIRS.
- See Figure 2

Utilisation of biomarkers

Evidence:

In cryptococcosis, serum CrAg is now a well-accepted diagnostic biomarker for detection or confirmation of cryptococcal infection. High CSF titres herald a poor prognosis. In contrast, it

has little or no role in precisely monitoring the course of an individual infection, whether measured in the serum or in the CSF.[422]

Quantitative CSF cultures[92] refer to serial dilution of a specified volume of CSF and quantification of cryptococcal growth on agar plates. This parameter has been used as a treatment response biomarker in many recent phase II and III clinical trials. Cryptococcal clearance is calculated for each patient and the mean slope for patients in a given treatment arm is calculated and termed early fungicidal activity (EFA).[92] EFA has been shown to be associated with clinical outcome in a number of clinical trial cohorts.[365,423,424] It is used as a surrogate endpoint to evaluate novel drugs in phase II and lll clinical trials. Unfortunately, quantitative CSF cultures are not routinely performed in clinical laboratories as they are labour-intensive and individual EFAs have not yet driven therapeutic decisions. Data are lacking in non-HIV settings.

Future potential biomarkers include markers of innate immune response activation, measurements of cellular and humoral immunity, and transcriptomic yeast and host signatures. In particular, there is a great need for a cheap, simple to perform, sensitive and specific, predictive biomarker for C-IRIS and also to assess therapeutic success or failure.

Recommendations

- **(BIIu)** For qualitative cultures during therapy, we recommend centrifugation of a reasonable volume (*e.g.,* > 5 ml) of CSF and plating the sediment to assess for CSF sterility.
- **(BIII)** We recommend quantitative yeast CSF cultures be considered for comparison in individual cases (*e.g.*, persistent/ relapsed infection).
- **(DIIu)** The use of follow-up serum or CSF CrAg for clinical decision-making is discouraged as a persistently high serum or CSF CrAg without worsening of symptoms does not indicate clinical failure.

Exploration for an underlying immunodeficiency

Evidence:

Cryptococcosis is a well-recognised opportunistic infection in individuals with acquired immunodeficiency disorders such as HIV, SOT recipients and those taking old or new immunotherapeutic agents. Investigation for common and treatable immunodeficiency states is warranted following the diagnosis of cryptococcosis, such as, a previously unrecognised

haematological malignancy and HIV. Conversely, large radiographically apparent cryptococcomas can sometimes mimic malignant tumours.

Invasive fungal infections have been associated with a broad range of inborn errors of immunity including mutations in genes encoding for caspase recruitment domain-containing protein 9 (CARD9), Dectin-1 and 2, signal transducer and activator of transcription (STAT)-1, 3 or 4, and IL-12 receptor beta-1 (reviewed in Lionakis et al.[51]). CARD9 deficiency was identified in murine studies of cryptococcosis.[425] Cryptococcosis has also been reported in patients heterozygous for the STAT-1 "gain-of-function" gene.[52]

Single nucleotide polymorphisms within gene(s) have been sought in case-control or cohort design studies investigating disease susceptibility, particularly in immunocompetent populations. Polymorphisms associated with cryptococcosis include pathogen recognition receptors (PRR) such as dectin-2,[56] pentraxin-3 (PTX3),[57,58] Fc gamma receptors,[59,60] mannose-binding lectin (MBL),[61] and six loci upstream of the colony-stimulating factor 1 gene encoding for macrophage colony-stimulating factor (M-CSF).[62] Identification of these inborn errors of immunity with gene polymorphisms requires sophisticated immunogenomic analysis such as targeted or whole genome or exome sequencing (WGS/ WES) or a genome-wide association study (GWAS). These strategies are not readily available outside of research settings and remain undeveloped for clinical practice.

Anti-cytokine antibodies, particularly auto-antibodies to IFN-gamma and granulocyte-macrophage colony-stimulating factor (GM-CSF) have also been associated with cryptococcosis in apparently immunocompetent patients.[37-43 44] Auto-antibodies to IFN-gamma are found predominantly in East Asian patients with disseminated non-tuberculous mycobacterial (NTM) disease, salmonellosis, histoplasmosis, cryptococcosis, and talaromycosis.[197 426,427] These patients seem to have a preponderance for unusual manifestations of cryptococcosis including skeletal disease.[196] Rituximab, a chimeric monoclonal antibody against human CD20, primarily found on B cells, has been trialled for treatment in four patients with high anti-IFN-gamma antibodies and NTM with promising clinical and immunological responses.[428] Anti-GM-CSF autoantibodies have been associated with CM caused predominantly by *C. gattii.*[38,41,429] Cryptococcosis has also been described in paediatric patients with X-linked hyper-IgM syndrome associated with a CD40 ligand gene mutation[45-47] and in those with hypogammaglobulinemia, where late-onset combined immunodeficiency may be suspected.[48-50,430]

Idiopathic CD4 lymphopenia (ICL) is an increasingly recognised entity, albeit a rare, heterogeneous syndrome, diagnosed after the exclusion of conditions known to depress CD4 T cell counts including HIV.[431] This is usually identified in previously well patients with new opportunistic infections or diagnosed incidentally when lymphopenia is noted on routine blood analysis.[28] Current studies define ICL by a CD4 count of <300 cells/mm^3 or <20% of T cells. In a study of 39 patients with ICL, many had concurrent CD8 lymphopenia with low B cell or NK cell numbers.[28,36] Multiple cases of cryptococcosis have been reported in this setting.[28-36] An open-label Phase I/IIa dose-escalation study of recombinant human IL-7 in nine patients with ICL found this therapy to be safe with a resultant increase in peripheral CD4 T-cells at week 3 which peaked at week 5 and was sustained to week 12.[432] Others have prescribed exogenous IFN-gamma or IL-2 with good response.[30,433]

Practically, when faced with a patient with new cryptococcosis, a thorough history should include risk factors for HIV, receipt of immunosuppressive therapy, personal or family history of recurrent or unusual infections, and examination for lymphadenopathy and concurrent opportunistic infections (e.g. oral candidiasis) is recommended. Blood investigations should include a CBC and blood smear/film, HIV testing, T cell (CD4/CD8/ natural killer (NK) subsets by flow cytometry, immunoglobulin levels, complement testing, and anti-cytokine antibodies. In-born errors of immunity are best assessed by whole genome sequencing (WGS), whole exome sequencing (WES) or targeted gene sequencing. A clinical immunologist experienced in immunodeficiency conditions may assist in guiding and prioritising these advanced investigations. A personalised and pragmatic approach is necessary.

Recommendations:

- **(BIII)** Individuals without a known risk factor for disseminated cryptococcosis, particularly those with a history of other atypical fungal, mycobacterial, or bacterial infections, be considered for evaluation of an undiagnosed immunodeficiency, preferably in consultation with a clinical immunologist.
- See **sTable 1 (s6)**

Treatment of *Cryptococcus gattii* infection

Evidence:

Factors to consider in the investigation and management of infections by members of the *C. gattii* species complex include host status, clinicopathological presentation and antifungal

susceptibility. *C. gattii* infections occur predominantly in Australia, British Columbia, Canada, the Pacific Northwest of the USA, parts of South America and Africa.[117,434-437] Molecular type VGI, coined as *C. gattii sensu stricto*, has a global distribution but is most prevalent in Africa, Australia, and Europe. Molecular type VGII (*C. deuterogattii*) occurs also globally and is mostly reported from Australia and the Americas. Molecular type VGIII (*C. bacillisporus*) is predominantly reported from the Americas, but some cases have been reported from other continents. The relatively rare molecular type VGIV (*C. tetragattii*) is observed in southern Africa and India, while the very rare molecular type VGVI (*C. decagattii*) is exclusively reported from the Americas [76,435,438,439] Notably, not all commercial LFAs are able to detect *C. gattii* disease.[89] In a study where four commercial assays were tested against forty cryptococcal strains comprising all seven recognised species and their interspecies hybrids, incorporating serial dilutions to exclude pro-/post-zone effect, the detection of *C. tetragattii* was particularly problematic with the CryptoPS (Biosynex, Illkirch-Graffenstaden, France) and cryptococcal antigen LFA (Dyanmiker Biotechnology, Tianjin, China).[89]

There is a relationship between immune status and *C. gattii* infection, with ~50% to >70% of cases occurring in immunocompetent hosts[438,440,441] compared with 2-30% in PLHIV.[117,437,442-444] Autoantibodies to GM-CSF and idiopathic CD4 lymphopenia are known potential risk factors.[38,41,429,440]

Although global studies have demonstrated that MICs of amphotericin B, fluconazole, voriconazole, posaconazole, and 5-flucytosine against members of the *C. gattii* species complex and especially *C. deuterogattii*, may be higher compared with *C. neoformans*[445-449], However, clinical breakpoints have not been established; no evidence that isolates with higher MICs led to poorer outcomes; and outcomes with standard therapies are similar to those in *C. neoformans*.[203]

C. gattii infection presents with lung disease more often than *C. neoformans.* This may be influenced by the exact species involved. For example, *C. deuterogattii* infections caused predominant pulmonary disease from outbreaks in British Columbia and US Pacific Northwest.[440,441,450] Lung and/or brain cryptococcomas are more common in *C gattii* infections[130,352,450] although the contribution of host and species risk determinants are interdependent.[440] The likelihood of a more robust host inflammatory response in non-immunosuppressed individuals and a different cryptococcal capsular structure may explain why inflammatory disease can be more severe in *C. gattii* infections and thus treatment is more difficult or nuanced. This inflammatory response involving optic nerve tracts may contribute to

potentially steroid-responsive acute blindness.[451,452] Both *C gattii* and *C. neoformans* species complexes typically present similarly with meningitis with or without positive blood cultures in patients with AIDS.[117,437]

Management recommendations for *C. gattii* are determined by the presence of cryptococcomas, hydrocephalus and the course of the disease during antifungal therapy (see treatment of cryptococcomas). The same antifungal regimen is used as for *C. neoformans* meningitis.[116,203,437] However, 4-6 weeks of induction therapy may be required in non-HIV-associated meningitis with *C. gattii.*[352]

Because of the increased incidence of cryptococcomas and obstructive hydrocephalus, routine imaging is recommended in *C. gattii* infection.[130,352,450] The more pronounced inflammatory response can drive cerebral oedema and raised ICP. Corticosteroids have been of value in the management of large cryptococcomas with surrounding mass effect on imaging or where signs and symptoms persist or worsen despite a good microbiological response.[352,453] Clinical relapse during adequate therapy must be distinguished from C-IRIS.

Progression of disease despite adequate antifungal therapy warrants imaging and azole susceptibility testing. If fluconazole MICs are elevated, an appropriate alternative regimen should be instituted. Newer azoles may be considered for consolidation/maintenance therapy based on susceptibility testing *in vitro* but data on their clinical efficacy in *C. gattii* infections is not robust.

Recommendations:

C. gattii CNS disease

- **(AIII)** Treat *C. gattii* CNS infection with the same antifungal regimen as for *C. neoformans* CNS infection
- **(BIII)** Consider extending induction therapy to 4-6 weeks in non-HIV patients.
- **(AIII)** Early CSF shunting is indicated for chronic obstructive hydrocephalus.

C. gattii lung disease *(see sections on pulmonary cryptococcosis, pulmonary cryptococcoma and **sTable 6**).*

Treatment of cryptococcomas

Evidence:

Cryptococcomas (mass lesions) occur predominantly in the lungs and brain and appear to be more frequent in *C. gattii* infection.[117,130] Cerebral cryptococcomas have been reported in >14% of immunocompetent patients with confirmed CNS infection, 3-4% of patients with AIDS, and 10-12% of patients with other causes of immune dysfunction.[117,454] Pulmonary cryptococcomas are more common in immunocompetent patients and in *C. gattii* infection.[116,117,455]. In a Canadian series of 149 patients with *C. gattii* infection, 66% had pulmonary cryptococcomas.[130]

Cryptococcomas are diagnosed by clinical features of one or more space occupying lesions (imaging, histology and/or microbiology). CNS cryptococcomas frequently manifest as neurological deficits and/or raised ICP,[117] which requires urgent management. Adjunctive therapy with corticosteroids may be beneficial if there is surrounding oedema and mass effect,[440] noting that adverse events and debility preclude routine use of corticosteroids in HIV-associated CM.[179] Surgical resection may be considered for accessible brain lesions >3 cm, those compressing critical structures, or refractory to antifungal therapy.[456] Excision plus 6 weeks of induction antifungal therapy has been successful, even at lower amphotericin B doses than currently recommended,[116,457] but may be less effective in patients with underlying immunocompromise.[457] In the absence of surgery, radiological lesions can persist indefinitely despite clinical and microbiological cure.[116,458] The duration of consolidation/ suppression therapy varies from 12 to more than 18 months depending on physician preference.[352]

In patients with lung cryptococcomas two weeks of amphotericin B-based induction followed by consolidation with fluconazole was curative in 10 *C. gattii* patients with isolated, moderate lobar disease or large cryptococcomas.[352] Fluconazole monotherapy has been recommended for lung cryptococcomas, based on older series showing no additional benefit from amphotericin B-based induction (generally 0.5-0.8 mg/kg of Amb-D),[356] but more recently such monotherapy has been associated with treatment failure.[352] Surgical resection for diagnostic purposes, obstruction or clinical failure have been reported anecdoctally.[459-461]

Recommendations:

All cryptococcomas

- **(AIII)** Perform a brain biopsy or aspirate to exclude a secondary pathogen or an underlying tumour, in non-responding CNS cryptococcomas (particularly in immunosuppressed patients).
- **(BIII)** Consider surgical resection for accessible brain lesions >3 cm, lesions at risk of compressing critical structures, or large lesions not responding to therapy.
- **(DIII)** During follow-up, do not prolong or escalate therapy for persistent radiological findings in the absence of new or worsening symptoms or signs.

CNS cryptococcoma

- **(BIII)** Consider prolonging CNS antifungal induction therapy to 4-6 weeks.
- **(BIII)** Consider corticosteroids for large space occupying lesions with surrounding mass effect, or, where neurological symptoms and cerebral imaging signs worsen despite a good microbiological response.

Lung cryptococcoma (see sTable 6, s22)

- **(BIII)** For severe and large pulmonary cryptococcomas, consider CNS-based therapy (4-6 weeks).
- **(BIII)** For mild pulmonary cryptococcomas, consider CNS-based therapy (2 weeks induction).
- **(BIII)** If a solitary cryptococcoma has been completely excised and the patient is asymptomatic, use of fluconazole 400 mg daily for 3-6 months is still prudent.

Cryptococcosis caused by non-*neoformans* and non-*gattii* strains of *Cryptococcus*

Evidence:

There are individual case reports and case series of non-*neoformans*/non-*gattii Cryptococcus* infections. Most infections have occurred in immunosuppressed patients. *C. (Papiliotrema) laurentii*[462] and *Cryptococcus albidus* (currently *Naganishia albida*)[463] account for about 80% of the pathogenic infections in this group and most of the clinical syndromes involve skin, lungs, bloodstream, or CNS.[464] *Cryptococcus laurentii* (currently *Papiliotrema laurentii*) has also been linked to invasive device-related infections.[464] Cases of superficial or invasive cryptococcosis caused by *C. (Naganishia) liquifaciens, C. uniguttulatus, C. (Naganishia) adeliensis and C (Hanaella) luteolus* have also been reported.[465-468] Colonisation, especially of the skin,

respiratory and gastrointestinal tracts with these rare species must be distinguished from true disease.

Previous reports have highlighted diagnostic anomalies including poor growth at 35-37°C (except *N. albida, P. laurentii*); acapsular cryptococci, negative CrAg tests (LFA tests have not been evaluated) and misidentification of certain species with commercial systems (e.g. API20C, ID32C, Vitek).[468,469] In some cases, the laboratory may misidentify another yeast as *C. laurentii* or *N. albida* based on non-definitive commercial identification methods (eg. VITEK 2, VITEK MS, Bruker Biotyper).[470] Treatment of cryptococcosis caused by non-*neoformans*/non-*gattii* *Cryptococcus* is similar to that of *C. neoformans* infection. Elevated 5-flucytosine, fluconazole (especially those with prior fluconazole exposure) and other azole MICs have been documented though the clinical significance is uncertain.[469,471] Fluconazole susceptibility testing is recommended when considering its use in consolidation or maintenance therapy.[469,472]

Recommendations:

- (**AIII)** As non-*neoformans*/non-*gattii Cryptococcus* species are rarely pathogenic, careful assessment of the significance of their isolation from non-sterile and sterile sites and the clinical context is required.
- **(CIII**) For CNS or disseminated disease: Treat as for *C. neoformans* CNS infection.

Pregnancy

Evidence:

Pregnancy is associated with characteristic immunological perturbations, which impact on the presentation and natural history of cryptococcosis. Thus, the risk posed to the foetus and the risk-benefit ratio of antifungal therapy needs to be considered when managing a pregnant woman with cryptococcosis.[473] Cases of placental cryptococcosis and vertical transmission to the infant have been reported.[474-477] Maternal mortality from disseminated disease is approximately 25%; in one study of HIV-infected individuals with cryptococcosis only 44% of women carried their pregnancy to term with miscarriages and stillbirths common.[478] Cryptococcal antigenemia may first become apparent in pregnant PLHIV, in the absence of symptoms of meningitis. Cryptococcosis typically presents as limited disease of the lung or meningoencephalitis.[476,477,479-481] The majority of cases of cryptococcosis in pregnancy occur in

the third trimester or postpartum.[474,478] The rapidly changing immunological environment in the postpartum period may increase the risk of IRIS.[474,482,483]

Pregnancy also influences the choice of anticryptococcal drugs due to their potential teratogenic or other effects on the foetus. Extensive clinical experience suggests that Amb-D and L-Amb are relatively safe during pregnancy.[478,484] Thus, polyenes are the cornerstone for management of disseminated disease. 5-Flucytosine is rated by the United States FDA as a Category C drug because of its direct effects on RNA/DNA metabolism. Fluconazole use in pregnancy is associated with an increased risk of musculoskeletal malformations, tetralogy of Fallot and spontaneous abortions.[485-489] These birth defects appear with chronic and high doses (Fluconazole 400-800 mg daily) and any fluconazole effects on stillbirth rates is controversial.[490] The current FDA rating for fluconazole use in pregnancy is Category D. A relationship between other azoles and human congenital abnormalities has not been established, although voriconazole, posaconazole and isavuconazole have been associated with teratogenicity and embryotoxicity in animal studies. In one study, there was no association between high dose itraconazole use in pregnancy and birth defects.[491]

Recommendations:

- **(AIII)** Use L-Amb or Amb-D in induction, consolidation, and maintenance therapy and for treatment of isolated cryptococcal antigenemia.
- **(DII) T**he use of 5-flucytosine and fluconazole is not recommended in pregnancy. They should be avoided particularly in the first trimester and their use in the second/third trimester needs to be an individual risk-benefit assessment.
- **(BIII)** Fluconazole can be used after delivery despite its excretion into breast milk.
- **(AIII)** Apply clinical judgement when considering initiation of antifungal therapy and duration of therapy, taking into account the trimester of pregnancy and severity of illness.

Paediatric Cryptococcosis

Evidence:

The prevalence of cryptococcal disease is reportedly lower in children than adults, with infants and young children <6 years of age being less affected than children ≥6 years of age.[492-495] This low prevalence is not explained by lack of exposure, as seroprevalence studies have found that 18-–70% of children are seropositive.[234,496] LFA-based screening of HIV-infected children and

adults with CD4 < 100 cells/mm^3 in sub-Saharan Africa, revealed that serum CrAg was positive in 3.5% vs 5.8%, respectively (p=0.001).[497] In addition, CrAg positivity increased with age in the paediatric group, with the mean age of children testing positive being 14.6 years.

Among HIV-infected children, the reported prevalence of cryptococcosis is around 1–3%.[237,498-500] Amongst children presenting with clinical meningitis in Botswana with its high rates of HIV seroprevalence, *Cryptococcus* was the dominant pathogen in 13.6% of cases with clinical features of meningitis,[493] and 25.7% of those microbiologically proven.[501] Other paediatric conditions associated with an increased risk of cryptococcosis include primary immunodeficiencies, acute leukaemia, solid tumours, SOT and bone marrow transplant recipients, and receipt of immunosuppressive treatments.[54,494,502-506] In addition, sporadic cases have been described in apparently immunocompetent children.[494,504,507-515] Cryptococcosis has also been reported in neonates born to apparently well mothers without HIV.[516,517]

Cryptococcosis in children predominantly presents as CNS disease, but small case series of non-CNS disease have been reported.[494,511,518-521] Clinical efficacy trials and studies to validate diagnostics tests and therapies for cryptococcosis in children are lacking. Thus, recommendations for diagnostic and therapeutic management have been extrapolated from studies in adult populations but dosing of antifungal agents needs particular attention for the paediatric patient.

Recommendations:

- Treatment of CNS or disseminated diseases:
  - Induction: **(AIIt)** (Amb-D 1 mg/kg daily or L-Amb 3-4 mg/kg daily) plus 5-flucytosine (100–150 mg/kg daily in 4 divided doses) for 2 weeks). 5-flucytosine levels and/or CBC should be monitored closely.
  - Consolidation: **(AIIt)** Fluconazole 12 mg/kg daily for 8 weeks.
  - Maintenance: (**AIIt)** *for HIV-infected and immunocompromised*; (**BIIt)** *for immunocompetent*). Fluconazole 6 mg/kg daily for 6–12 months.
- Treatment of non-CNS, but severe isolated lung disease: **(AIII)** as for CNS disease
- Treatment of non-CNS, mild isolated lung disease: **(AIII)** Fluconazole 12 mg/kg daily for 3-6 months.
- **(AIII)** Screening is recommended for HIV-infected children in high disease prevalence areas aged ≥10 years.

Equity of treatment in resource-limited versus resource-rich healthcare systems

Evidence:

The greatest burden of cryptococcal disease is in RLS, where HIV-associated infection predominates. Resolution of issues surrounding healthcare access, management costs, including antifungal cost-effectiveness are critical to improving treatment and health outcomes. Strategies of screening and pre-emptive therapy for cryptococcosis have proven cost-effective in PLHIV in areas of high disease prevalence.[522-524] These studies were critical in the decision to implement systematic CrAg screening in South Africa, and to include screening in many other national guidelines.

Health economic analyses were built into the ACTA and AMBITION trials. In ACTA, compared with the standard of 2 weeks Amb-D plus 5-flucytosine, 1 week of Amb-D plus 5-flucytosine was found to be more effective (24% vs 38% 10-week mortality) and less costly (saving USD$424 per patient treated), and the oral combination of fluconazole plus 5-flucytosine was as effective and much less costly (saving USD$843 per patient).[123] Compared with similar cohorts treated with fluconazole alone, the fluconazole and 5-flucytosine combination was associated with a 35% mortality reduction, with an incremental cost-effectiveness ratio of USD$65 per life year saved.[525]

Trial data, together with these health economic analyses, and ongoing advocacy were instrumental in the launch of a Unitaid programme to provide care for those with advanced HIV disease, including procurement of generic 5-flucytosine and L-Amb, which are now available in seven key African countries.[526] In South Africa, implementation of routine use of 1-week Amb-D plus 5-flucytosine treatment led to a reduction in in-hospital mortality from 37% to 24%.[527] In some less well-resourced healthcare systems, where safe delivery of Amb-D is challenging, the programme facilitates use of fluconazole plus 5-flucytosine, and is now supporting roll out of the single-dose L-Amb-based regimen (being given in routine care in Uganda, Malawi, Bostwana and Zimbabwe at the time of this writing). Following on from the Unitaid initiative, countries (including those not included in the Unitaid programme) can request 5-flucytosine through the Global Fund.

The additional drug costs of the single high dose liposomal regimen in AMBITION study, may be offset, at least in part, by a reduced need for and cost of blood monitoring and the potential to

discharge some patients one or two days earlier. Gilead Sciences have committed to the supply of L-Amb at a preferential price as originally agreed to for treating visceral leishmaniasis.[528] The outcome of the AMBITION trial will allow Médecins Sans Frontières (MSF) and others who have previously accessed 1-2 weeks of L-Amb for CM treatment to switch to a regimen using only a single dose of 10 mg/kg.[180]

Still, much needs to be done to provide wider access to 5-flucytosine and L-Amb, and continued advocacy and implementation efforts are being led by the CryptoMAG group, major NGOs, and the CDC Mycology branch, who have recently launched the "Ending deaths from HIV-associated CM by 2030" initiative.[529] Certainly, wider use of more effective and safer regimens, and discontinuation of the use of fluconazole monotherapy for induction, together with screening programs and earlier diagnosis, enabled by LFA CrAg tests, hold the promise of significant reductions in HIV-associated cryptococcal mortality in LMICs.

Apart from drug cost and access issues, there are other significant challenges in RLS including access to healthcare, diagnostics such as imaging modalities and biopsies and resources such as manometers for ICP management and established C-IRIS management strategies. Consistent ART availability and long-term adherence support systems are important both for HIV and CM.

Issues of cost and access are also relevant to resource-rich settings. In particular, in the USA, without effective competition, oral 5-flucytosine costs for a 2-week course were reported to be in excess of USD $28,000 in 2015.[530] Thankfully, this has been reduced more recently with approval of new generic products.[531] The overall costs of CM initial hospitalization in a resource-rich healthcare system has been reported as USD $15,000-25,000,[532,533] and with cumulative costs of $100,000 over 5 years.[534] This does not take into account the costs of managing the substantial long-term morbidity, much of which we are only beginning to appreciate.[329,535]

Recommendations:

The reduced availability and accessibility of antifungal agents in RLS presents a daunting challenge in the management of CM. Improved access to affordable optimal antifungals and improved healthcare access and delivery systems are critical.

- **(AIII)** Ongoing advocacy is needed to ensure populations with cryptococcosis have access to the most effective drug regimen(s). Focus on screening, early diagnosis, and effective management of CM in all healthcare systems is essential to reduce mortality,

morbidity and costs.

Most Important Unmet Needs

There remain a series of important unmet needs for the management of cryptococcosis. First, all the recent studies suggest that efficient (rapid) killing of yeasts in CSF (potent EFA regimens) will lead to better clinical outcomes. We simply need new, more potent oral anticryptococcal drugs and/or regimens, including non-azole oral alternatives. Second, early identification of disease is critical and thus timing of CrAg testing and better strategies for responding to cryptococcal antigenemia in HIV and non-HIV patients will be important. Algorithms of CrAg testing and optimal integration of antigen titres and clinical variables into effective pre-emptive therapies are urgently needed. Third, management of increased ICP remains imprecise and variable, and implementation of therapeutic LPs is challenged by lack of personnel and manometers. Fourth, we require improved diagnostics for and tools to prevent and treat C-IRIS. This is currently a diagnosis of exclusion and there have been no randomised, prospective studies on its management. Fifth, validated methods of monitoring for treatment response during the induction phase and later in consolidation and maintenance phases may aid in individualising therapy for intensity and duration. Sixth, much of the research focus has been on RLS and unfortunately, resource-rich healthcare systems have not updated, studied, nor improved their antifungal strategies. How best to integrate knowledge gained from clinical trials in PLHIV in RLS into populations such as SOT and immunocompetent individuals remains debated. Seventh, despite substantial efforts undertaking clinical studies in RLS implementation, and access to optimal antifungal agents and healthcare remain discordant and elusive and thus mortality remains high. Eighth, appropriately, CNS disease has been the primary focus of most cryptococcosis studies to date, but prospective examinations on length and choice of treatment and other management studies for pulmonary and/or other non-CNS cryptococcosis is severely lacking. Ninth, we do not know how aggressively to investigate apparently normal hosts with disseminated cryptococcosis for immunogenomic deficiencies and how best to manage these deficiencies. Finally, while we grapple with the unacceptably high mortality of CM, we must also better appreciate the substantial mid- and long-term morbidity caused by CM, and its impact on patients, their families, and communities. (see **sTable7, s23**)

The list above summarises some of the important needs in cryptococcosis management as we work to improve our guidelines. There are many research directions and questions to answer. The following five research questions should be immediate priorities:

(1) Determination of the best strategy for pre-emptive therapy in cryptococcal antigenemia.

(2) Discovery and evaluation of potent and safer new antifungals agents. We are fortunate to have robust animal models to aid in these studies.[204,379,536-540]

(3) In-depth understanding of genetic immunological factors associated with non-HIV CM.

(4) Utilisation of quantitative CSF yeast counts or other more practical markers of organism load, and EFA calculations into clinical practice.

(5) Interventions to properly diagnose and manage IRIS.

Conclusions

Cryptococcosis and its management remain complex and challenging. These guidelines are very comprehensive and there is evidence that the use of and adherence to cryptococcal guidelines have a positive clinical impact.[307,541] These guidelines are designed to help clinicians manage cryptococcal diseases today. However, the range of cryptococcal disease is wide and imprecisely understood. This is particularly critical in RLS where the burden of disease is high and access to antifungal therapy remains inadequate. Equally, more needs to be learnt in resource-rich settings where host risk profiles are changing and an increasing array of presentations of cryptococcosis are being recognised, necessitating more nuanced and individualised treatment plans.

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